

QUANTUM AND CLASSICAL MACHINE LEARNING FOR ENHANCING GRAVITATIONAL WAVE SCIENCE

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Darsh Khandelwal

VITA

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ABSTRACT

Gravitational Wave (GW) astronomy is a field established almost 100 years ago by Einstein, and has undergone a remarkable transformative journey in recent years ever since the first detection of GW from Binary Black Hole (BBH) merger in 2015 by LIGO Hanford and Livingston. Due to the low amplitude of GW signals (of the order of 10^{-21}), they are embedded deep inside the detector noise of LIGO interferometric data, which generates timeseries sequences sampled at 16 KHz sampling rate. These dataset needs to be analyzed in order to detect the presence of GW. Traditional signal processing methods are effective but highly computationally inefficient and suboptimal for real-time detection.

This thesis focuses on exploring ML techniques for detecting and characterizing GW signal. We propose simplistic models of machine learning inspired by the inherent physics of the system and driven by numerical and ML techniques in the field of data-driven gravitational wave astronomy. We perform a detailed analysis of the waveforms and noise present inside the LIGO detector data through different signal processing techniques and model the amplitude of the fast fourier transforms of the waveforms obtained from numerical relativity simulations from the SXS catalog. We divide the dataset into two parts and define separate models of Spinless BBH systems and Spin BBH systems which maps fitting parameters to the binary parameters of the GW signal or vice versa. We achieved an accuracy of mean square error 7×10^{-3} on the model defined for Spinless BBH systems and 4×10^{-3} on the model defined for Spin BBH systems on the test dataset. We also propose a Long Short-Term Memory Autoencoder model to detect and denoise GW signals from the LIGO detector data which was trained on simulated noise dataset with embedded NR signals as input and NR signals as the output. The model outputs the waveforms with an mean square error of 2×10^{-3} . We benchmark the results by showing the detection and prediction of the waveform from the event GW150914. We also aimed at defining basic Quantum Neural Network models to be used in GW astronomy. We defined a simple quantum analogue of the feed-forward neural network model for the Spinless BBH systems for restricted number of features and obtained an accuracy of 2×10^{-2} using Quantum Neural Networks. These results highlight the potential of QML frameworks for scalable and efficient use of QML in GW astronomy.

LIST OF SYMBOLS OR ABBREVIATIONS

GR	General Relativity
GW	Gravitational Waves
LIGO	Laser Interferometer Gravitational-Wave Observatory
KAGRA	Kamioka Gravitational Wave Detector
CBC	Compact Binary Coalescence
BBH	Binary Black Hole
BNS	Binary Neutron Star
NSBH	Neutron Star Black Hole
PN	Post-Newtonian
NR	Numerical Relativity
EOB	Effective One Body
SXS	Simulating eXtreme Spacetime
FFT	Fast Fourier Transform
DFT	Discrete Fourier Transform
CWT	Continuous Wavelet Transform
ML	Machine Learning
AI	Artificial Intelligence
ANN	Artificial Neural Networks
NN	Neural Networks
RNN	Recurrent Neural Networks
LSTM	Long Short Term Memory Networks
CNN	Convolutional Neural Networks
MLP	Multi-Layer Perceptron
Qiskit	Quantum Information Software kit
QML	Quantum Machine Learning
QNN	Quantum Neural Networks
QBM	Quantum Boltzmann Machine
QSVM	Quantum Support Vector Machine
QCNN	Quantum Convolutional Neural Networks
QPCA	Quantum Principal Component Analysis
QLSTM	Quantum Long Short Term Memory
SNR	Signal-to-Noise Ratio

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1. INTRODUCTION

1.1 Motivation & Outline

This thesis focuses on exploring novel machine-learning techniques for enhancing Gravitational Wave (GW) astronomy. The existence of GW is derived from General Relativity (GR), which was first shown by Einstein through Einstein's field equations in the linearized theory of gravity [1, 2]. The first detection of GW by Laser Interferometer Gravitational-Wave Observatory (LIGO) in 2015 [3] established a new field of GW astronomy, which aims at detecting GW signals arising from different sources and characterizing GW signals based on their properties of the system. The experimental datasets recorded by GW observatories are sampled at a high sampling rate and are vast in nature. With the rise of Machine Learning (ML) and its proven use cases in different research domains, researchers have started employing different ML models in different domains of GW astronomy [4, 5]. This includes improving the detection rate and the accuracy of detection and analyzing the different types of glitches present in the signal and their origin. GR models are highly non-linear in nature and cannot be fully explained by the Newtonian theory of gravity. We propose machine learning techniques arising purely from the simplified physical understanding of the system and inspired by the well-established theoretical results. We explore numerical approaches to describe different system processes and extract information about the system from the available data. We aim to propose novel ML models that are inspired by physics and driven by ML models to produce reliable and robust results. We also aim to apply the quantum analogue of these models in search of better performance for a limited number of parameters. The Outline of this thesis is described below.

Chapter 1 starts with a brief introduction to GW astronomy and a brief history of the evolution of GW physics so far. Then, we discuss how these GW are detected and describe in-depth details about the detector, which is followed by the theory of gravitational physics through Newtonian approximation. We then discuss the useful techniques that are employed in our project, including signal processing techniques, ML techniques and Quantum Machine Learning (QML) techniques.

Chapter 2 describes our two novel ML models: the Neural Network (NN) model and the Long Short-Term Memory (LSTM) autoencoder model. A NN model is used to

determine the physical parameters of the system or the parameters of the GW signal. On the other hand, LSTM autoencoders are proposed to detect the presence of GW signal in LIGO detector data and predict the waveform embedded inside the detector data.

Chapter 3 illustrates a quantum analogue of the ML model proposed in the **Chapter 2** and provides a description of the model.

Chapter 4 illustrates the results obtained in the above chapters and finally discusses the conclusion of the thesis and future works.

1.2 Gravitational Waves

Gravitational Waves are defined as the perturbation/disruptions on the fabric of space-time occurring due to non-axisymmetric accelerating heavy mass objects like black holes, neutron stars, supermassive black holes, etc. Gravitational waves are represented as

$$h(t) = h_+(t) - ih_\times(t) \quad (1.1)$$

where h_+ and $h_\times$ represent the two independent plus and cross-polarization of the waveform.

1.2.1 Black Holes

Black holes are the regions of spacetime where gravitational fields are so strong that even light cannot escape through them. The last stable orbit after which no object can prevent infall inside the black hole is called the event horizon. We can classify black holes based on their masses into four types:

1. **Stellar-mass** black holes lie in the mass range of $2 - 150M_\odot$, generally originating from the core of the collapsing stars when all the energy sources of the star are exhausted [6].
2. **Intermediate-mass** black holes lie in the range of 10^2 to $10^5 M_\odot$ [7]. These black holes are too large to be formed by the core of a collapsing star. They are either formed due to the merging of two stellar-mass black holes or other objects by accretion. It is also proposed that such massive black holes can be formed inside the young star clusters due to multiple stellar collisions [8].
3. **Supermassive** black holes have masses in the range of 10^5 to 10^{10} and are often located at the centre of the galaxies [9]. NanoGrav detects GWs in the nanohertz frequency range. These waves are primarily emitted by inspiralling supermassive black hole binaries over cosmological timescales. These low-frequency signals are

measured through the precise timing of millisecond pulsars distributed across the sky [10].

4. **Primordial** black holes are hypothetical and are theorized to have existed just after the Big Bang and have no confined mass range [11].

Fig. 1.1 depicts the merging of two black holes or BBH systems and emitting GW upon the merger represented by the ripples in the spacetime. Black holes can also be classified

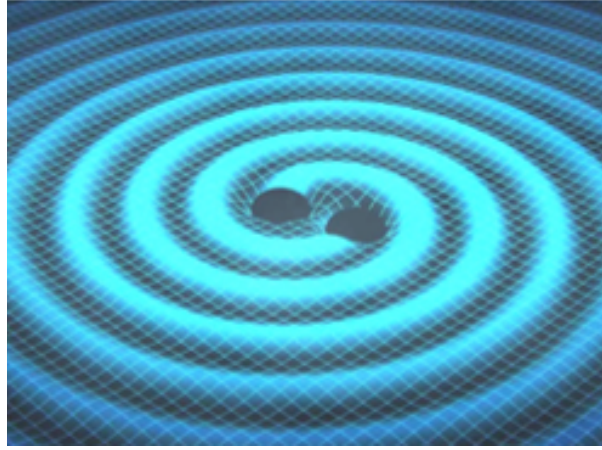


Fig. 1.1: Merging of BBH system and emitting GW (ripples in the spacetime)

based on their physical properties, which are inherent to the system, such as mass, spin, and charge. A non-rotating black hole with zero electric charge is called a Schwarzschild black hole [12]. A rotating black with zero charge is known as a Kerr black hole [13]. A black hole having both spin and electric charge is referred to as a Kerr–Newman black hole [14].

1.2.2 Neutron Stars

Neutron stars are fast-spinning, extremely dense objects mainly made of neutrons, with masses typically between $1 M_{\odot}$ and $2.5 M_{\odot}$. They are formed from the collapse of massive stars. Since both neutron stars and stellar-mass black holes result from similar stellar collapse processes, the type of compact object that forms depends on the mass of the remnant. If the remnant mass is greater than $3 M_{\odot}$, a black hole is expected to form; otherwise, a neutron star is formed [6]. The most massive neutron star observed so far is PSR J0952–0607, which has a mass around $2.5 M_{\odot}$ [15]. On the other hand, lightest black hole ever observed have $5 M_{\odot}$ which created an observational gap of $2.5 M_{\odot}$ to $5 M_{\odot}$. Interestingly, Ref. 16 reported the GW signal from the merger of a $23 M_{\odot}$ black hole with a compact object of $2.6 M_{\odot}$. Although it was unclear whether this object was a neutron star or a black hole, the observation confirmed that compact objects exist within this mass gap.

1.2.3 History of GW

Einstein showed the first ever theoretical existence of GWs in his 1916 paper [1] through the theory of GR. However, he predicted later that their detection would be highly infeasible due to their small strain amplitudes in his subsequent work [2]. Later, in 1969 [17], Weber reported the coincident excitation of two resonant bar detectors, suggesting the presence of GW radiation. However, his claims were not accepted well within the scientific community due to the lack of reproducibility and inconsistencies with theoretical predictions. Over time, research efforts shifted toward interferometric detection methods. These efforts turned into the first direct observation of GWs from the binary black hole merger on 14 September 2015 by LIGO Hanford (H1) and Livingston (L1) [3]. The GW signal extracted after applying traditional preprocessing techniques on the raw signal obtained from H1 and L1 is shown in Fig. 1.2 along with the summed wavelet transforms of data from two detectors. The Fig. 1.2 is regenerated from Ref. 3. Since this breakthrough, several GW detectors have been developed, including LIGO Hanford, LIGO Livingston, GEO 600, and the Kamioka Gravitational Wave Detector (KAGRA) [18], with LIGO India [19] currently under construction. On 17 August 2017, LIGO detected the first ever GW originating from the merger of BNS [20].

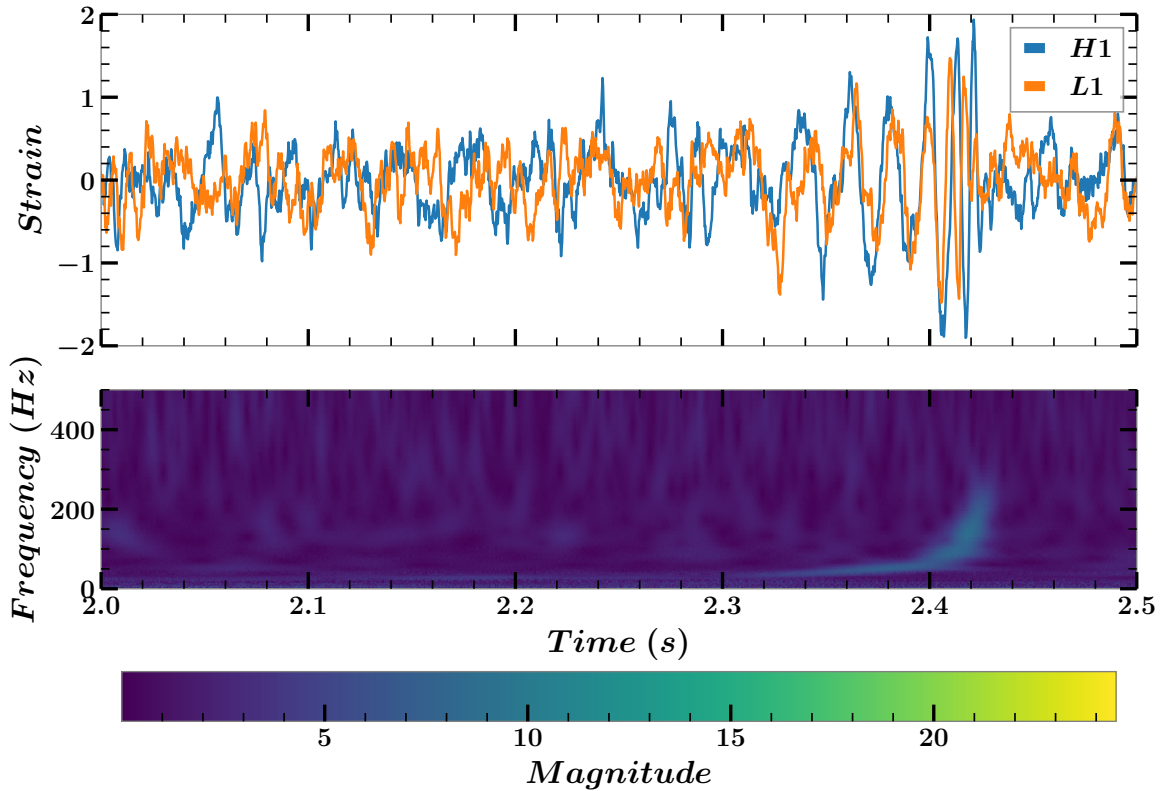


Fig. 1.2: GW signal for GW150914 event on the LIGO H1 and L1 and the summation of time-frequency evolution spectrum of the wavelet transform of the corresponding signals [3].

1.2.4 Detector

The LIGO is an advanced Michelson interferometer designed to detect GWs across a frequency range of 1 Hz to 1 kHz. A simplified design of the LIGO instrument is illustrated in Fig. 1.3, which demonstrates its operational principle. The Fig. 1.3 was obtained from Ref. 3. A laser beam emitted from a stable Nd:Yag laser source (wavelength of 1024 nm)

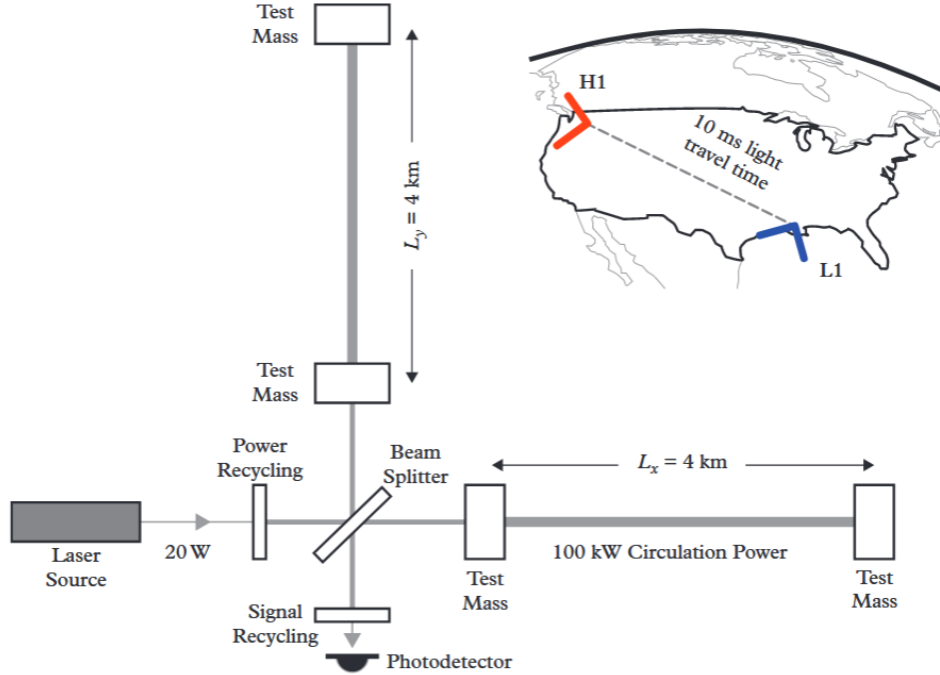


Fig. 1.3: Simple demonstration of LIGO architecture and map showing the locations of two LIGO detector H1 and Livingston L1 [3].

passes through a beam splitter, dividing it into two orthogonal beams that travel towards two test masses separated by a distance of 4 km. When the path lengths of the two arms are equal, the light beams interfere constructively and return through the beam splitter, resulting in no transmission and, thus, no detection at the photodiode, which is initially set at the dark fringes. However, when a GW passes through the detector, the differential change in the arm lengths (as depicted in Fig. 1.4) is detected by the photodiode, whose intensity is directly related to the amplitude of the GW. The GW strain is represented as,

$$\frac{\Delta L}{L} = h \quad (1.2)$$

where L is the length of detector and ΔL is the differential change in the arm length. $h \approx 10^{-21}$, and $L = 10^3$, therefore, $\Delta L = 10^{-18} \text{ m}$. Since, $\Delta\phi = \frac{2\pi\Delta L}{\lambda}$, which accounts to $\Delta\phi$ being of the order of $\frac{\Delta L}{\lambda} \approx 10^{-12}$ where λ represents the wavelength of laser. Detection of the amplitude of the order of 10^{-12} is still complicated, as most of the interferometric data is filled with noise.

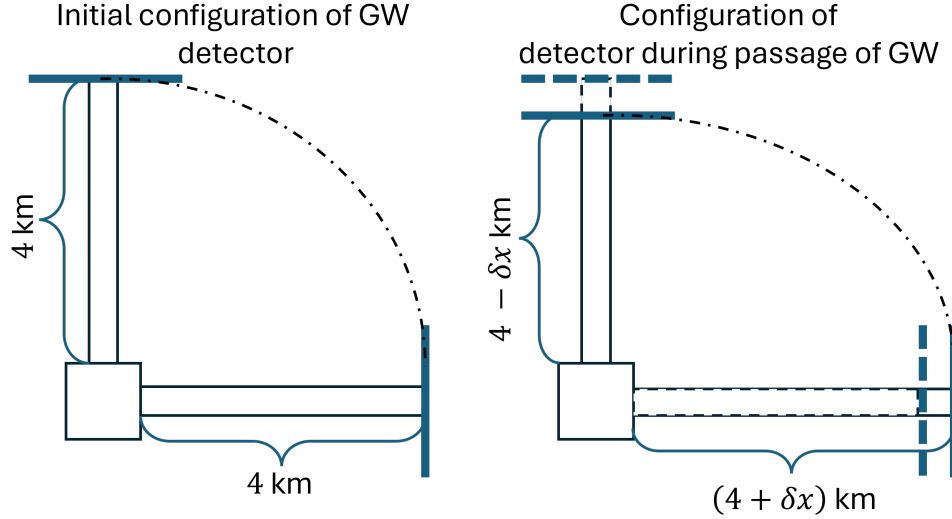


Fig. 1.4: Schematic of a Michelson interferometer used in GW detection. The left panel shows the setup when no GW is present, with both arms having equal length and no light reaching the photodetector due to destructive interference. The right panel shows the effect of a passing GW, which changes the arm lengths by stretching one arm and compressing the other. This causes a phase shift between the two beams, resulting in partial constructive interference and a measurable signal at the photodetector.

Noises in GW detector

GW detectors suffer from various kinds of noises, including quantum noise of the laser light, thermal noise of the mirror coatings, optic suspensions or gravity gradient noise [21]. Some other noises include noise from instrumental backgrounds, natural disasters, seismic activity, and any other activity causing the external vibration of the system.

Most of the noises are non-stationary in nature, or even if they are stationary, they mostly couple in a non-linear manner with GW detector data. This causes the detector data to be highly non-Gaussian and non-stationary, which ultimately leads to the scare of false alarm rates in GW detectors.

1.2.5 Types of GWs

Any non-axisymmetric accelerating object emits GW. They can be classified into several categories based on their origin type: Compact Binary Coalescence (CBCs) GWs, Continuous GW, Stochastic GW, and Burst GW [22]. GWs arise from the merging of heavy mass celestial objects revolving in a binary system around the common centre of mass, such as Binary Black Holes (BBH), Binary Neutron Stars (BNS), or Neutron Star Black Hole (NSBH) systems are called CBCs [22]. On the other hand, rapidly rotating neutron stars emit waves at a constant frequency and amplitude, which are referred to as Continuous GW [23]. We believe that more than one GW passes through Earth at

a given instant, but their amplitudes are so low that their precise detection is almost impossible. However, statistical studies may lead to their detection. These waves are called Stochastic GW [24]. Moreover, some waves that originate from unknown sources whose physics is currently unknown to us or are of very short duration are called Burst GW [25].

1.2.6 Phases of GWs from CBCs

GW waveforms arising from CBCs can be divided into three components, namely, the inspiral phase, merger phase and ringdown phase, as shown in Fig. 1.5.

1. **Inspiral:** This phase represents the revolution of two objects about their common centre of mass in a binary system. Objects far apart attracted towards each other experience force and gradually decrease their orbital radius, contributing to the release of GW. As they move, the orbital radius decreases, increasing the orbital velocity and hence decreasing the radius. This phase can be analytically modelled using Post-Newtonian (PN) formalism where two objects are separated far apart.
2. **Merger:** When the two objects come close enough such that the distance between them becomes less than the Innermost Stable Circular Orbit (ISCO) of the objects (a term commonly defined for black holes, which represents the last stable orbit up to which the orbits of the BBH system can be approximated as circular), their orbital velocity increases drastically, leading to the merging of their black holes, this is called as merger phase. During this phase, the orbital velocity of the system becomes comparable to the speed of light, and hence, pN formalism fails to provide an accurate estimation of the waveform. Numerical Relativity (NR) simulations are required to predict accurately the late-inspiral and merger phases of the system.
3. **Ringdown:** The final phase, called the ringdown, is characterized by the damped oscillations of the quasi-normal modes (QNMs) of the remnant black hole. If the remnant settles into a stationary state as predicted by general relativity, it is described by the Kerr solution, and the ringdown is governed by its spectrum of QNMs. This regime is well-modeled using black hole perturbation theory [26, 27].

Although there are no strict boundaries that define the transition between the two phases, this division is carried out to understand the system in a simpler way.

Binary parameters of signal from CBCs

Binary parameters of GW signal emitting from the quasi-circular BBH merger are described by **intrinsic** and **extrinsic** parameters [28]. Intrinsic parameters are inherent to the system and include the masses m_i of the black holes and dimensionless spin vector

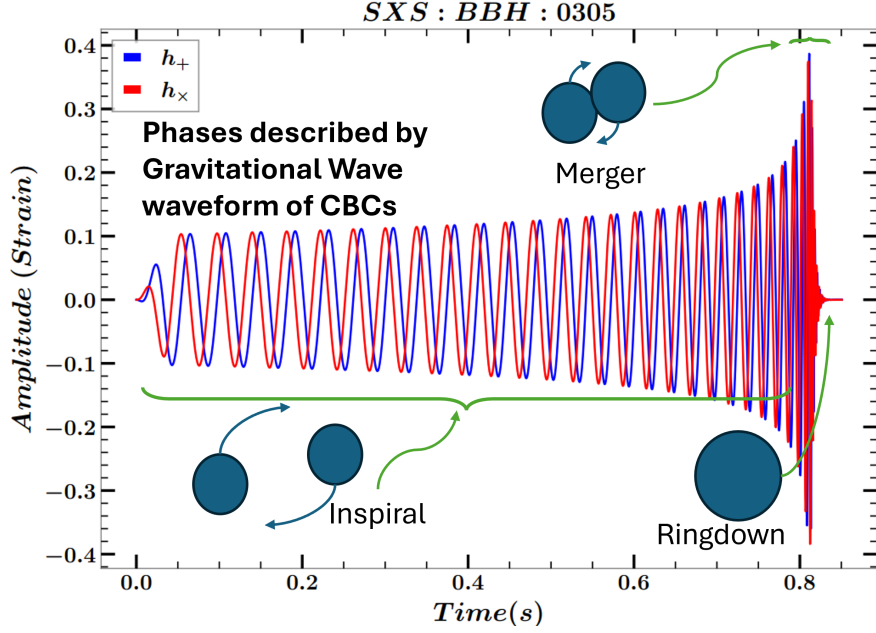


Fig. 1.5: Visual description of different phases of GW signal emitted from CBCs

$\left(\chi_i = \frac{c^2 S_i}{G m_i}\right)$, where S_i are dimensionful spin vectors. Extrinsic parameters, on the other hand, are observed by the observer and depend on the orientation and location of the system relative to the observer. They include Right Ascension (α), Declination (δ), Polarization angle (ψ), luminosity distance (d_L), orbital inclination angle (ι), time (t_c) and phase (ϕ_c) at coalescence [29].

As the merger deviates from quasi-circular orbits, two parameters, eccentricity and the argument of periapsis (the position of the minimum distance between the orbital path and the centre of mass of the system), are added to the parameter space [30]. Usually, the spin vectors of the black hole are projected along the orbital angular momentum of the system. But in some cases, spins of the BBH are not aligned (anti-aligned) with the orbital angular momentum (L) of the system, then these spins combine with the orbital angular momentum, to form total angular momentum $J = S_1 + S_2 + L$. The non-zero components of spins perpendicular to the orbital angular momentum give rise to spin precession of the orbits. For BNS mergers, tidal deformability becomes an important binary parameters which is included in the list.

Other parameters that can be used as substitutes for the above binary parameters are shown below:

- Symmetric mass ratio $\left(\eta = \frac{m_1 m_2}{(m_1 + m_2)^2}\right)$, Chirp mass $\left(M_c = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}\right)$ and mass ratios $\left(q = \frac{m_1}{m_2}\right)$ can be used as a substitute for the mass parameters [29].
- Effective Spin Parameters $\left(\chi_{\text{eff}} = \frac{m_1 \chi_1 + m_2 \chi_2}{m_1 + m_2}\right)$ can be used as an alternative parameter which condenses the information of both the spin parameters weighted with the corresponding masses of the black holes [29].

- Remnant properties of black holes like masses and spins are also the two parameters which are used as binary parameters of GW signal. Reference 31 investigated the remnant properties and GW waveforms of BBH merger for small mass ratio systems through a numerical approach motivated by Zeno's paradox.

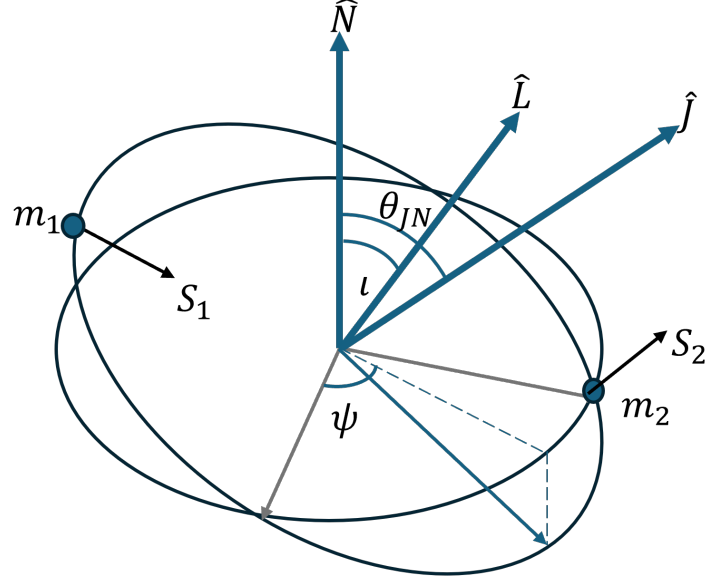


Fig. 1.6: Diagram depicting the binary parameters of a GW signal from a quasi-circular BBH merger system. The black holes with masses m_1 and m_2 have spin vectors $\vec{S}_1$ and $\vec{S}_2$. $\vec{L}$ is the orbital angular momentum, and $\vec{J}$ is the total angular momentum. $\vec{N}$ denotes the line of sight to the observer. The angles ι , ψ , and θ_{JN} describe the orientation of the system relative to the observer and affect the observed GW signal.

1.2.7 Binary Inspiral

Here, we derive the dynamics of the binary inspiral for the CBCs in the Newtonian approximation [28]. We assume two masses m_1 and m_2 , located at the position r_1 and r_2 , separated by a distance r can be written as $r = |r_1 - r_2|$. The two-body problem can be converted to a single-body problem when analyzing the motion along the centre of mass frame, where the reduced mass $\mu = \frac{m_1 m_2}{M}$ and $M = (m_1 + m_2)$.

$$\mu \ddot{r} = -\frac{Gm_1 m_2}{r^3} \hat{r} \implies \ddot{r} = -\frac{GM}{r^3} \hat{r} \quad (1.3)$$

Considering the case of circular orbits, R is the radius of the circular orbit, for which we can derive Kepler's law given below using equations of motion

$$\omega_s^2 = \frac{GM}{R^3} \quad (1.4)$$

, where ω_s is orbital angular frequency of the binary system. Since the amplitude of the GW increases along with an increase in the frequency of GW with time, it is often termed a chirp signal. Therefore, we define the chirp mass of this signal as,

$$M_c = \mu^{3/5} M^{2/5} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}. \quad (1.5)$$

Gravitational radiation is quadrupole in nature. The expression of the gravitational waveform from the binary system revolving in circular orbits is given by,

$$h_+(t; \theta, \phi) = \frac{1}{r} \frac{4G\mu\omega_s^2 R^2}{c^4} \left(\frac{1 + \cos^2(\theta)}{2} \right) \cos(2\omega_s t_{ret} + 2\phi). \quad (1.6)$$

$$h_\times(t; \theta, \phi) = \frac{1}{r} \frac{4G\mu\omega_s^2 R^2}{c^4} \cos(\theta) \sin(2\omega_s t_{ret} + 2\phi). \quad (1.7)$$

where r is the distance between the observer and the source and θ, ϕ represent the angle of inclination of the orbit and the orbital phase of binary stars, respectively (Note that we assume r, θ, ϕ remain constant for a fixed time). Hence, $\omega_s t_{ret} + \phi = \omega_s t + \alpha$, therefore, $\alpha = t - \omega_s r/c$, then by shifting the origin, we can write $\omega_s t + \alpha = \omega_s t$. We can rewrite the Equations (1.6) and (1.7) as

$$h_+(t; \theta, \phi) = \frac{1}{r} \frac{4G\mu\omega_s^2 R^2}{c^4} \left(\frac{1 + \cos^2(\iota)}{2} \right) \cos(2\omega_s t). \quad (1.8)$$

$$h_\times(t; \theta, \phi) = \frac{1}{r} \frac{4G\mu\omega_s^2 R^2}{c^4} \cos(\iota) \sin(2\omega_s t). \quad (1.9)$$

We can infer several important points from the above Equations (1.8) and (1.9). First, orbital angular frequency is half of the GW frequency ($\omega_{GW} = 2\omega_s$). Second, $h_+ \approx (1 + \cos^2(\theta))$ and $h_\times \approx \cos(\theta)$. Lastly, if we keep $\iota = \pi/2$, that is, we observe the binary system head-on, then only h_+ is present, while if we observe the binary system along its perpendicular axis that is at $\iota = 0$, $h_+ \propto \cos^2(2\omega_s t)$, and $h_\times \propto \sin^2(2\omega_s t)$. The power radiated per unit solid angle in quadrupole approximation is given by

$$\left(\frac{dP}{d\Omega} \right) = \frac{r^2 c^3}{32\pi G} \langle \dot{h}_{kl}^{TT} \dot{h}_{kl}^{TT} \rangle \quad (1.10)$$

$$= \frac{r^2 c^3}{16\pi G} \langle \dot{h}_+^2 + \dot{h}_\times^2 \rangle. \quad (1.11)$$

Upon substituting the Equations (1.8) and (1.9) in the Equation (1.11), and integrating them over the solid angle, we obtain the power radiated as,

$$P = \frac{32c^5}{5G} \left(\frac{GM_c \omega_{GW}}{2c^3} \right)^2. \quad (1.12)$$

The energy emitted in the spacetime during the merger process is derived from the energy present in the orbit of the system (E_{orb}), which is equal to the summation of kinetic and potential energies.

$$E_{orb} = E_{K.E.} + E_{P.E.} \quad (1.13)$$

$$= \frac{Gm_1m_2}{2R} - \frac{Gm_1m_2}{R} \quad (1.14)$$

$$= \frac{-Gm_1m_2}{2R} \quad (1.15)$$

As the radius decreases, the energy of the orbit increases (magnitude), and ω_s increases (according to Equation (1.4)). It then forces R to decrease more and vanish as $t \rightarrow \infty$ (Note that if R starts to decrease significantly, the initial approximation of circular orbits will start to fail). Differentiating 1.4, with respect to time, we get,

$$\dot{R} = -\frac{2}{3}R\frac{\dot{\omega}_s}{\omega_s} \quad (1.16)$$

$$= -\frac{2}{3}\omega_s R\frac{\dot{\omega}_s}{\omega_s^2} \quad (1.17)$$

Therefore, the above conclusion is true if and only if $\dot{\omega}_s \ll \omega_s^2$. From Equation (1.17) and the above argument, we can infer that $|\dot{R}| \ll \omega_s R$, which means that the tangential velocity of the masses should be much greater than the decay of the orbital radius.

Now, let us evaluate the back reaction of GW, as the orbital radius decreases, the energy of the orbit increases, simultaneously giving rise to orbital velocity, which in turn increases the rate of decay of the radius; this cyclic process is referred to as a back reaction. To calculate it, we substitute the Equations (1.4) and (1.5) into the Equation (1.15) to obtain,

$$E_{orb} = -\left(\frac{G^2 M_c^5 \omega_{GW}^2}{32}\right)^{1/3}. \quad (1.18)$$

It is interesting to note that at a constant ω_{GW} , the mass is dependent only on the chirp mass of the system. Since, $P = -dE/dt$, we can differentiate the Equation (1.18) w.r.t time and substitute the above calculated value of P in the Equation (1.12), into the Equation (1.18) and obtain

$$\dot{f}_{GW} = \frac{96}{5}\pi^{8/3}\left(\frac{GM_c}{c^3}\right)^{5/3}f_{GW}^{11/3} \implies M_c = \frac{c^3}{G}\left(\frac{5}{96}\pi^{-8/3}\dot{f}_{GW}f^{-11/3}\right)^{3/5}. \quad (1.19)$$

Equation (1.19) is an important equation used to estimate the chirp mass of the real GW by fitting the curve using the above equation of the frequency derivative.

1.2.8 Numerical Relativity

Numerical Relativity is a multidisciplinary field of physics encompassing relativity, magnetohydrodynamics, and astrophysics with the aim of solving numerically highly dynamical, strong-gravity scenarios from the Einstein field equations [32]. NR simulations provide precise solutions to the Einstein equations for the two-body problem in curved space-time.

Because of the low amplitude of GW signals, they are embedded deep inside the noisy detector data, which makes the process of detection nontrivial and predicts the binary parameters of the GW signal accurately. However, the advancements in the signal processing techniques provide us with the advantage to detect their presence provided we have the prior information about the template of the GW signal [33]. This is achieved by employing a matched filtering technique. The inspiral phase of GW waveforms can be predicted accurately by the Post-Newtonian (PN) expansion [34]. However, these analytical techniques fail to provide reliable solutions as strong gravity regimes and relativistic regimes kick into play.

NR simulations solve Einstein's equations by employing different state-of-the-art numerical techniques which include the 3+1 decomposition of the spacetime with 3 spatial and 1 temporal coordinate. Einstein equations consist of 10 coupled non-linear partial differential equations which describe the geometry of spacetime in the presence of mass and energy [35]. The simulations become necessary in the presence of strong gravity regime where theoretical predictions fail to provide reasonable solutions [36, 37, 38]. Several publicly accessible template waveform catalogs, generated via Numerical Relativity simulations, are available for research and data analysis purposes. Notable examples include the SXS waveform catalog [39], the Rochester Institute of Technology (RIT) gravitational wave catalog [40], and the Georgia Tech waveform catalog [41].

With the increasing development of computational techniques, several platforms including software packages and Python libraries have been developed, enabling researchers to perform NR simulations with comparatively less computational power and time. Some of which are illustrated below.

- EinsteinPy is a python package developed to perform symbolic and numerical calculations of GR. It includes built-in functions to work with spacetime metrics such as Schwarzschild, Kerr, and Kerr–Newman geometries. The package also provides symbolic tensor manipulation utilities, which can be effectively used for learning tensor algebra. Additionally, it offers visualization tools integrated with plotly and matplotlib, allowing users to analyze various spacetime geometries visually [42].
- NRpy+ is designed to bridge the gap between symbolic tensor algebra and efficient numerical code generation. It takes formulations such as the BSSN system and automatically generates highly-optimized C code that can be compiled and run directly.

Its emphasis is on pedagogical transparency and flexibility, which allows users to experiment with new formulations and understand the inner workings of NR solvers more easily [43].

- Einstein Toolkit is a large-scale, modular computational framework that is built for full-scale NR simulations. It is equipped to handle BBH mergers, BNS mergers, relativistic hydrodynamics, and GW generation. The toolkit integrates several components such as Cactus (core infrastructure), Carpet (AMR), and various thorns for evolution and initial data, offering a stable environment for long-term, production-level simulations [44].

Simulating eXtreme Spacetime (SXS)

The public catalog provides the dataset of the NR simulations for different configurations of mass ratios and spins, including the processing and non-precessing BBH simulations. It contains a total of 2476 non-deprecated normalized waveforms of BBH mergers, 10 NSBH mergers and 2 BNS mergers [45]. We use this dataset in our thesis and construct machine learning models and perform various analysis techniques stated below.

Approximate Solutions to NR

Due to the high computational cost of NR signals, LIGO and VIRGO employ approximant waveforms. These approximants are generated from Effective One Body (EOB) models [46], and calibrated with NR waveforms (for the inspiral phase) and combined with waveforms obtained from phenomenological fit with NR for later phases of system. Surrogate modelling of these approximate waveforms increases the rate of generation of these waveforms along with simultaneously increasing their accuracy [47]. Reference 48 demonstrated the greedy algorithm method for obtaining the reduced order basis element for set of template waveforms described uniquely by binary parameters of the signal and then used the empirical interpolation method to find the basis points of the waveform.

1.3 Signal Analysis

1.3.1 Fast Fourier Transform

Fast Fourier Transform (FFT) is an efficient algorithm for evaluating the Discrete Fourier Transform (DFT) of any time domain signal [49, 50]. DFT is used to calculate the Fourier transform of a function numerically [51]. We discretise the temporal space and then obtain the corresponding discretised frequency space using DFT [52]. Fourier transforms are helpful in analysing the signal in its frequency space and obtaining information about

the frequency spectrum of a time domain signal. The DFT of a signal is obtained as

$$X_k = \sum_{n=0}^{N-1} e^{-2\pi i k(n/N)} x_n. \quad (1.20)$$

1.3.2 Wavelets and Wavelet Transform

Wavelets are predefined functions of scale and shift parameters. They are used to decompose complex non-stationary signals. These predefined functions are called mother wavelets which are defined as,

$$\psi_{\tau,s}(t) = \frac{1}{\sqrt{s}} \psi\left(\frac{t-\tau}{s}\right) \quad (1.21)$$

where τ, s are the translation and scale parameters, respectively [53, 54]. Magnitude of scale greater than one dilates the mother wavelet and a positive value of shift translates the mother wavelet onto the horizontal axes.

Wavelet Transform is defined as the inner product of the given signal and a family of wavelets for a particular mother wavelet [55]. It can be mathematically represented as

$$W_\psi(f_{\tau,s}(t)) = \frac{1}{\sqrt{s}} \int \psi^*\left(\frac{t-\tau}{s}\right) f(t) dt \quad (1.22)$$

The wavelet transform is a powerful tool for performing time-frequency analysis [56]. It transforms a time-domain signal into its frequency components while retaining the temporal information at each frequency. There are two types of wavelet transform, Continuous Wavelet Transform (CWT) and Discrete Wavelet Transform (DWT) [57].

The Fourier transform provides information about the frequency content of a signal without retaining the information of the time at which those frequencies occur. This makes it more suitable for stationary signals (where the frequency content remains constant over time). In contrast, wavelet transform provides information about the dependence of frequency and time for a signal, which makes it particularly useful for analysing non-stationary signals [54].

Although the Short-Time Fourier Transform (STFT) provides information about both the time and frequency domains simultaneously by applying a window function to the signal and taking the Fourier transform over that window length. However, its disadvantage is that the window length remains constant throughout the entire transform. This limitation makes it difficult to gather information about non-stationary signals whose frequency content changes at a highly dynamic rate.

1.4 Machine Learning

Machine Learning (ML) is a branch of data science encompassed within the broader field of Artificial Intelligence (AI) focused on developing algorithms which enable computers to learn essential sequences and patterns present inside the data and predict outcomes based on the learned sequences, these sequences are learned using the deep neural networks which lie in the field of Deep Learning as shown in Fig 1.7. Algorithms related to ML

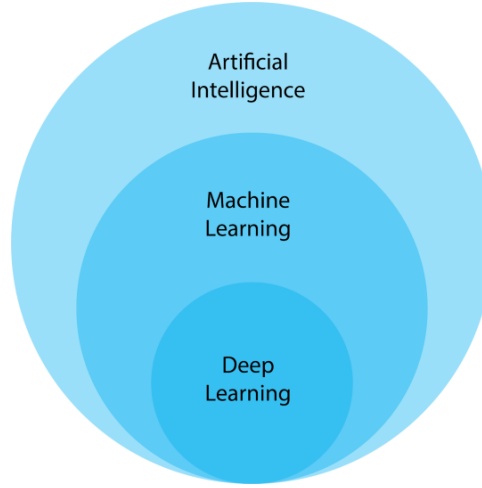


Fig. 1.7: Hierarchical relationship among AI, ML, and Deep Learning (DL). Deep Learning is a subset of Machine Learning, which itself is a subset of the broader field of Artificial Intelligence. Each inner circle represents a more specialized and data-intensive approach within its encompassing domain.

can be classified into three categories: Supervised learning, Unsupervised learning and Reinforcement learning [58].

In this thesis, we have made use of several ML models and techniques. We will be providing a brief introduction to each model employed in this thesis.

1.4.1 Neural Networks

Artificial Neural Networks (ANNs) are the state-of-the-art ML algorithm inspired by the working of a human brain. The basic unit of a neural network is called a neuron or a perceptron. A perceptron inputs a number and processes it by multiplying weights (w) and adding a bias (b) as a parameter. Finally, the number is output after applying an activation function to the intermediate output. Perceptrons can be mathematically represented as a function, $g_{\theta}(x) : x \rightarrow y$, which inputs x and outputs y for the given parameters θ .

$$g_{\theta}(x) = f(wx + b), \text{ where } \theta = (w, b) \in \mathbb{R}^2 \quad (1.23)$$

where $f(u)$ represents the activation function applied to the intermediate output $u = (wx + b)$. The perceptron with multiple inputs can be translated as

$$g_{\theta}(\vec{x}) = f\left(\sum_i w_i x_i + b_i\right) \quad (1.24)$$

where w, b are column vectors. The output of a neuron is the weighted sum of its inputs along with a bias term as shown in Fig 1.8. Activation function is a one-dimensional

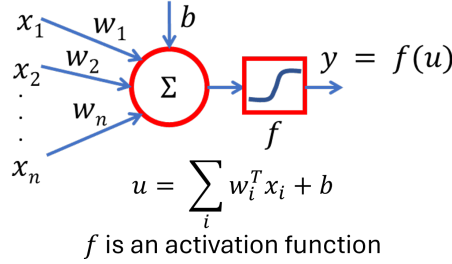


Fig. 1.8: Perceptron with multiple inputs

non-linear function, $f : \mathbb{R} \rightarrow \mathbb{R}$. Its use case is to induce the non-linearity in the model. Activation functions make the neuron capable of learning complex non-linear functions and modelling them effectively. Table 1.1 provides information about different activation function and their ranges for use in different cases. Perceptrons with multiple

Activation Function	Formula	Range
Sigmoid	$\sigma(x) = \frac{1}{1+e^{-x}}$	(0,1)
Tanh	$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$	(-1,1)
ReLU	$f(x) = \max(0, x)$	$[0, \infty)$
Leaky ReLU	$f(x) = \max(\alpha x, x)$	$(-\infty, \infty)$
Softmax	$\sigma(x)_i = \frac{e^{x_i}}{\sum_j e^{x_j}}$	(0,1)

Tab. 1.1: Different activation functions and their working ranges.

input parameters can be translated to a neural network that consists of input, output, and hidden layers. Each layer contains multiple perceptrons referred to as nodes. The number of nodes in the input layer and output layers is equal to the number of input and output features in the dataset. Each node in a given layer connects with all the nodes in the next layer. The simple neural network of 4 layers is shown here in Fig. 1.9.

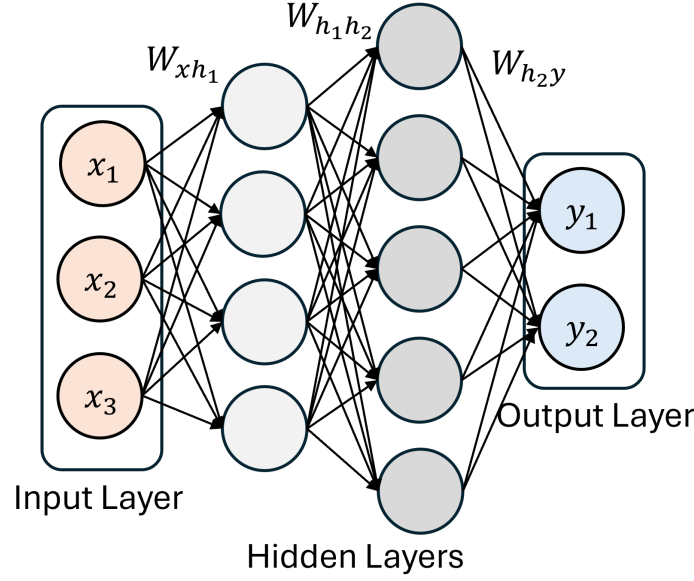


Fig. 1.9: Neural network with 3 nodes in the input layer, 4 nodes in the first hidden layer, 5 nodes in the second hidden layer and 2 nodes in the output layer.

1.4.2 Recurrent Neural Networks

Recurrent Neural Networks (RNNs) are class of ANNs that are specialized to learn sequential datasets. They learn the sequences present in a time series and predict the next sequences based on their previous learning. It works on the principle of employing the output of the previous cell as the input of the next cell. Fig. 1.10 demonstrates the transformation of NNs into RNNs where neural networks are capable of learning and predicting the stationary dataset by taking into account the contribution of x_t at a time while RNNs are capable of considering $x_{0:t-1}$ for an instant of time t . The information about the previous time steps is carried through previous hidden states given by the set: $h_{t-n}, h_{t-n+1}, \dots, h_{t-1}, h_t, \dots, h_{t+m}$ where m and n are any positive integers. Hidden states at time t is given as

$$h_t = \phi_t(x_t V_{in} + h_{t-1} W + b_h). \quad (1.25)$$

and output of RNNs at the same time step can be given similarly as

$$y_t = \phi_o(h_t U + b_o). \quad (1.26)$$

where b_o, b_h are the bias matrices. Hence, maintaining the order of the dataset becomes crucial for RNNs. One of the notable points about RNNs is the weight matrices remain the same throughout the network for a particular transformation.

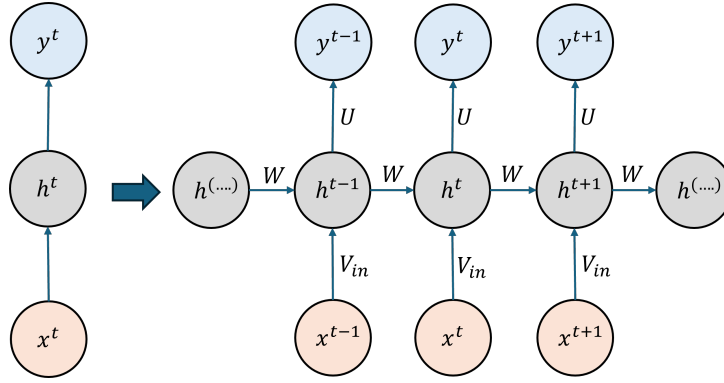


Fig. 1.10: Transformation of neural network into recurrent neural networks

Issue in RNNs

Similar weight matrices multiplying multiple times may lead to exploding gradient or vanishing gradient problems [59]. If the weight matrices contain small values < 1 , then the weights evolved up over a long time period and may lead to the vanishing of gradients $= 0$. Similarly, if the elements in the weight matrices are > 1 , then it may lead to large values for evolution for long time intervals, leading to exploding of gradients and hence causing the model to behave imperfectly. This problem act as the motivation for the introduction of the Long Short Term Memory (LSTM) Networks.

1.4.3 Long Short Term Memory Networks

As the name suggests, LSTMs are capable of learning short-term and long-term dependencies simultaneously [60]. This change accounts for the introduction of two different states inside the architecture controlling the flow of information separately. The hidden state is used to learn short-term dependencies, and the cell state is used to learn long-term dependencies. LSTM networks were designed to specifically deal with the two problems explained above: Vanishing Gradient and Exploding Gradient.

LSTMs work on the principle of gates and therefore, consist of three gates: input gate, forget gate and output gate. The input gate (i_t) controls the amount of information to be extracted from the current input data, forget gate (f_t) manages the information to be removed from the previously stored hidden states and sent to the cell state, and the output gate (o_t) decides the amount of information from the cell state to be sent out as output in the hidden state for the next layer. LSTMs are quite complicated to understand, they can be summarized mathematically as

$$o_t = \sigma(x_t W_{xo} + h_{t-1} W_{ho} + b_o) \quad (1.27)$$

$$i_t = \sigma(x_t W_{xi} + h_{t-1} W_{hi} + b_i) \quad (1.28)$$

$$f_t = \sigma(x_t W_{xf} + h_{t-1} W_{hf} + b_f). \quad (1.29)$$

where $W_{xo}, W_{xi}, W_{fo} \in \mathbb{R}^{d \times h}$ and $W_{ho}, W_{hi}, W_{hf} \in \mathbb{R}^{h \times h}$ are the weight matrices for the corresponding transformations and the b_o, b_i, b_f are the biases. A candidate memory cell state is defined as

$$\tilde{c}_t = \tanh(x_t W_{xc} + h_{t-1} W_{hc} + b_c). \quad (1.30)$$

The final cell state is given as

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t. \quad (1.31)$$

and finally, the current hidden state is given as

$$h_t = o_t \odot \tanh(c_t). \quad (1.32)$$

The detailed visual representation of neural networks is shown in Fig. 1.11 which denotes the gates present in the LSTM and different input and output states of LSTM for a particular timestep.

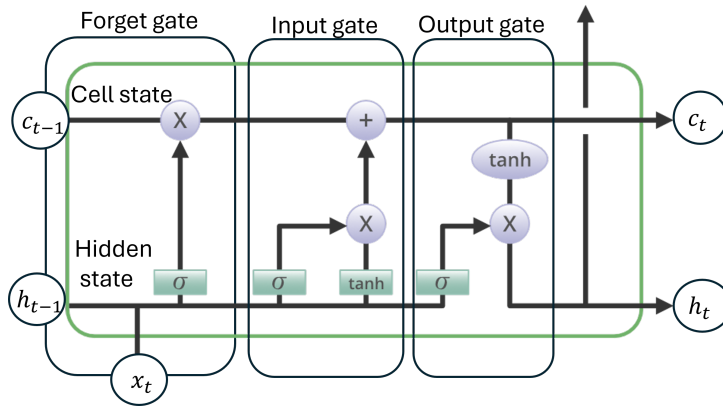


Fig. 1.11: Architecture and working of LSTM Networks.

1.4.4 Applications of Machine Learning in GW Astronomy

The need for ML algorithms in the field of GW astronomy stems from the fact that the current algorithms employed for detection are computationally inefficient as the number of detections is expected to grow from $O(100)$ to several orders of magnitude higher [5]. ML algorithms cover a long range of use cases in the field of GW astronomy.

ML has the ability to learn complex patterns and functions present in the dataset. They are expected to speed up the process of detection and parameter estimation in the coming years. Automating the data analysis methods, creating complex pipelines governed by the predictive nature of ML models and classifying the noise patterns present in the dataset and distinguishing them from GW are some of the tasks for which ML is widely used in this field [4].

Glitch Characterization and Classification

Glitches are non-astrophysical transient signals arising in detector data due to different environmental and instrumental noises. They can sometimes mimic the GW signal, which increases the chances of false detection. Hence, the characterization and classification of different noise elements becomes essential.

Classification of glitches is done via mapping the features of glitches with their glitch type, but due to the complex temporal and frequency evolution of glitches, their morphologies can become very difficult to analyze. Manual classification of glitches by the citizen scientist in the GravitySpy project of Zooniverse has prepared a dataset for supervised learning in classifying the glitches [61]. Classification through ML models and different algorithms have also been explored in this field. Three algorithms, namely Principal Component Analysis for Transients (PCAT), LALInfernece Burst (LIB), and wavelet detection filter combined with ML algorithm, were introduced in Ref. [62] and it was tested on real LIGO detector data [63]. Deep Convolutional Neural Networks (CNN) can be used for the analysis of the time-frequency representation of glitches [64]. Ref. 65 demonstrated the classification through the XG boost algorithm of a wavelet decomposed and denoised simulated data. They also proposed a pipeline working on the deep convolutional neural networks, which uses images of time-frequency analysis of signals to classify the noise transients present in the detector data [66]. A difference-boosting neural network algorithm that uses a combination of supervised and unsupervised classifiers to classify transient noise in data [67].

Auxillary channels contain signals obtained from photodetectors and seismometers installed throughout the two arms of the detector. These channels contain noise patterns and glitches purely from non-astrophysical origin [68].

Denoising of GW Signals

Interferometric data contains non-gaussian and non-stationary noise. These noises are either non-linear in nature by themselves, or they couple non linearly with the detector data to make its denoising complicated. Traditional denoising techniques of signal processing such as Weiner filtering are computationally expensive and do not remove the noise coupling from the detector data [69]. Principal Component Analysis (PCA) and dictionary learning are also computationally expensive and not viable for learning non-Gaussian noise. A deep learning algorithm for unsupervised learning has shown superior accuracy for denoising non-gaussian GW signals by training from simulated Gaussian noise [70]. A CNN was designed in Ref. 71 with an autoencoder configuration, which has shown more promising results than the conventional matched filtering approach. It showed the significance of deep learning-based search where the training of the dataset is required only once, and then the trained model can be employed for future purposes.

Modeling of GW Waveform

As discussed in earlier sections, the modeling of GW is essential for detection and parameter estimation purposes. One important point to emphasize is that modeling of GW must be accurate enough and computationally efficient to be used in the pipelines of GW analysis. ML in this domain is mainly used to construct the GW surrogate models using different approaches. Ref. 72 provides an in-depth study about different traditional techniques involving fitting and interpolation of existing waveforms, using Gaussian process regression for predicting waveforms and, for the first time, use of artificial neural networks in this domain. The advancements presented in Ref. 48 were further refined in Ref. 73 through the application of artificial neural networks for surrogate modeling. This enhancement significantly reduced the waveform generation time from 1794 ms for SEOBNRv4 (Spin-Aligned Effective One Body Numerical Relativity), 83.7 ms for SEOBNRv4opt, and 3.5 ms for SEOBNRv4ROM (Reduced Order Model) to 1.5 ms on a CPU and 0.5 ms on a GPU.

Other Use Cases of ML in GW astronomy

Deep learning models like sequence to sequence model was used in predicting the merger and ring down phases of the GW waveform templates by learning the inspiral phase of the waveforms [74]. LSTM networks have been used as the detection pipelines which could provide early warning detection upto 4 seconds earlier to the merger of system [75]. Injection of vacuum squeezed states of light in the detector, has increased their sensitivity since 2019. Ref. 76 introduces ML algorithms for effectively injecting vacuum-squeezed states of light into the detector.

1.5 Quantum Computing

Quantum computing is an interdisciplinary field lying at the intersection of multiple fields like quantum physics, computer science, electrical engineering, chemistry, biology and many more. Quantum computing is performed through the principles of quantum mechanics, providing us with the advantage of utilizing the properties of quantum mechanics like superposition, interference, and entanglement, which do not exhibit any classical analogue [77, 78].

The quantum-focused era began in the late 20th century when researchers focused on quantum algorithms to speed up classical computations. One of the first was developed by Deutsch, where he proposed a two-qubit algorithm for determining the type of a function [79]. Later, Deutsch and Jozsa generalized it for n qubits. One of the breakthroughs in this field was done by Grover and Shor, where Grover proposed an algorithm for searching an unsorted database with $O(N)$ complexity in classical computing to $O(\sqrt{N})$ in quantum

computing [80] while Shor provided an exponential speed-up over the classical algorithm of prime factorization to period finding algorithm in quantum computing [81].

Hadamard gate is a unitary operator, which converts the computational basis states $\{|0\rangle, |1\rangle\}$ into $\{|+\rangle, |-\rangle\}$. It is also known as the Quantum Fourier Transform (QFT) Operator of the single qubit states in the computational basis.

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad (1.33)$$

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle \quad (1.34)$$

CNOT gate is a 2-qubit gate which is used to create entanglement between two qubits. The matrix for the CNOT gate is defined as

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (1.35)$$

and its operation on the qubits in computational basis states for 2 qubits system can be given as

$$CNOT |00\rangle = |00\rangle, \quad (1.36)$$

$$CNOT |01\rangle = |01\rangle, \quad (1.37)$$

$$CNOT |10\rangle = |11\rangle, \quad (1.38)$$

$$CNOT |11\rangle = |10\rangle. \quad (1.39)$$

A simple demonstration of Bell state simulation is shown in Fig. 1.12 where a Hadamard gate was applied to the first qubit, and then a CNOT gate was applied on the second qubit with the first as control. The results of this experiment for the simulator and quantum computer are shown in Figs. 1.13a and 1.13b, respectively. The outcomes of quantum simulators are consistent with the theoretically calculated states. This is because the simulators operate in ideal, noise-free conditions. However, the same quantum circuits produce states that often deviate from the exact mathematical predictions. This discrepancy is due to different noise sources present in real quantum hardware. These noises include noise due to decoherence, gate implementation errors, and readout inaccuracies.

With the recent advancements in the field of quantum computing in the era of the second quantum revolution, various companies have invested in extensive research programs to develop scalable quantum hardware devices. The current comparison between quantum and classical computing can be understood using the graph shown in the Fig. 1.14

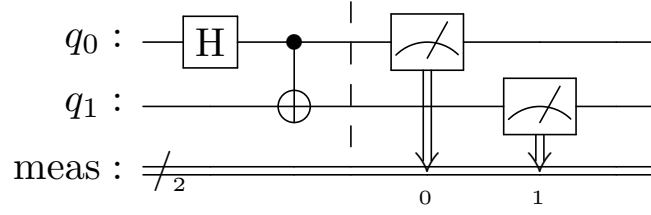


Fig. 1.12: Quantum Circuit to formulate the Bell state in the circuit

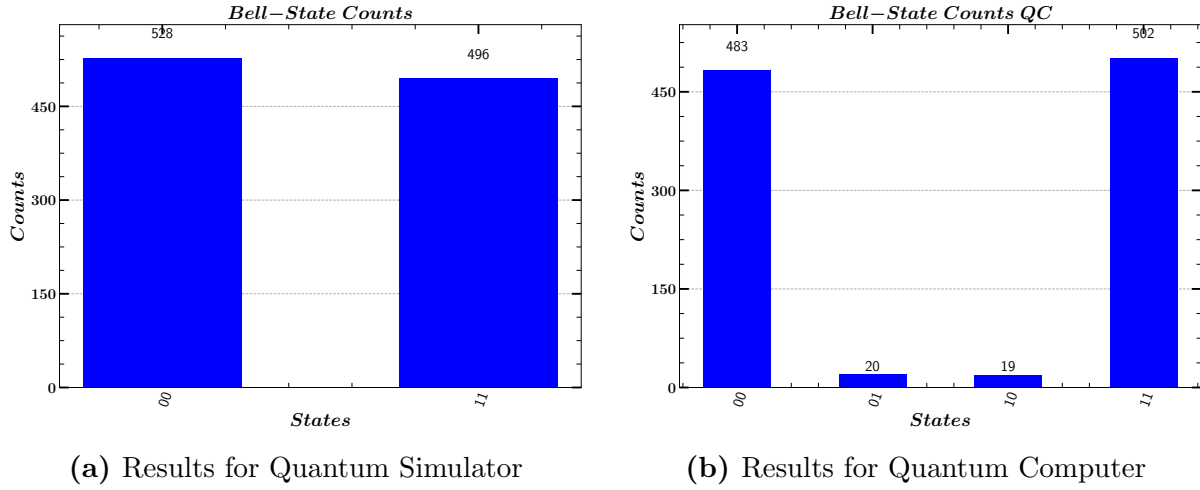


Fig. 1.13: Results of Bell State measurement on simulator and quantum computer: “IBM Kyiv”.

which was obtained from [82]. IBM has been actively working on developing supercon-

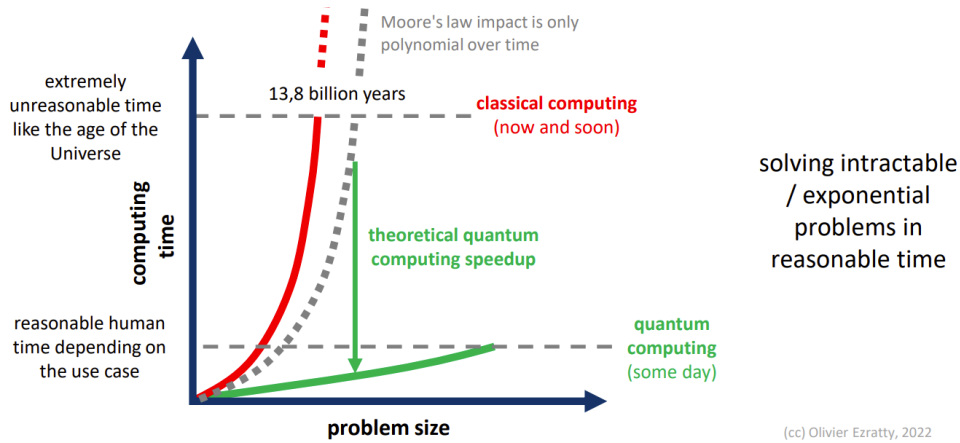


Fig. 1.14: Comparison of quantum and classical computing [82].

ducting qubit-based quantum processors of up to 1000 qubits, out of which they provide access to 128 qubits to free users for 10 minutes every month at IBM Quantum Experience [83]. Xanadu has photonic quantum computer, which utilises squeezed light and continuous-variable quantum states to perform computations with potential scalability advantages [84]. Google recently demonstrated quantum supremacy using a 53-qubit su-

perconducting processor, Sycamore [85]. Amazon Web Services uses Braket and provides an interface to multiple quantum hardware backends, including IonQ, Rigetti, etc. [86]. Similarly, Microsoft has Azure Quantum, which integrates simulation tools and hybrid solutions that include topological qubits as a future direction [87]. Recently, Microsoft has proposed a Majorana chip, the first ever topological qubit quantum processor, which has set up the revolution of topological quantum computing, showing significant progress towards fault-tolerant quantum computing [88].

1.6 Quantum Machine Learning

Quantum Machine Learning (QML) is the branch of data science that is formed at the convergent point of three specialized fields: quantum mechanics from physics and machine learning from data science [89]. It is one of the recent developments in the quantum industry that has shown considerable use of quantum technologies in determining solutions to industrial and academic problems [90, 91, 92, 93, 94].

QML combines the potential of quantum computing and machine learning to develop algorithms that could outperform their classical counterparts. A significant breakthrough in this area was made by Carleo and Troyer [90], who showed that neural networks can represent quantum many-body wavefunctions using a variational approach. This idea opened up new directions for studying entanglement in neural-network-based quantum states, including using Restricted Boltzmann machines (RBMs) to encode both area-law and volume-law entangled systems [91]. Quantum Algorithms (QA) have also found immense applications in deep learning [95], convolutional networks, and quantum-inspired natural language processing [96]. For instance, quantum training of deep RBMs showed better efficiency than classical methods [95]. Their expressiveness and structure can also be studied using effective dimension [97]. One of the issues that arise naturally in deep QNN models is the barren plateau, which affects the trainability of a variational quantum circuit. These barren plateaus can be detected using fisher information metric [98, 92]. Some QNNs have also been shown to train faster and generalize better than classical models [92].

Along with discriminative models, generative quantum models have also gained considerable importance. For example, the Quantum Boltzmann Machine (QBM) can handle the issue of non-commuting observables using bounds on quantum probabilities [99]. Other generative models have shown exponential improvements in learning and inference over classical methods [100]. Some basic machine learning tasks also have quantum versions like data fitting with exponential speedups under certain conditions [101], and classification methods such as quantum support vector machines (QSVM) [102] and quantum k-nearest neighbors [94], which benefit from faster distance or kernel computations. Quantum Convolutional Neural Networks (QCNNs) follow a layered structure similar

to classical CNNs and have found application in phase recognition and quantum error correction [93]. QML methods have been developed for supervised, unsupervised, and reinforcement learning tasks [103], which shows their wide range of applications. In data analysis, Quantum Principal Component Analysis (QPCA) can extract features from low-rank quantum states more efficiently [102]. There are also quantum-inspired classical methods like swarm optimization based on quantum ideas, which have improved performance in many classical problems [104]. Altogether, these developments show the growing potential of QML in both theory and practice.

1.6.1 Quantum Neural Networks

Quantum Neural Networks (QNNs) are the quantum analogue of classical neural networks that aim to exploit the quantum advantage while performing ML tasks. The architecture of QNNs can be divided into three subparts, namely, feature maps, ansatz and the final measurement of the circuit as shown in Fig. 1.15.

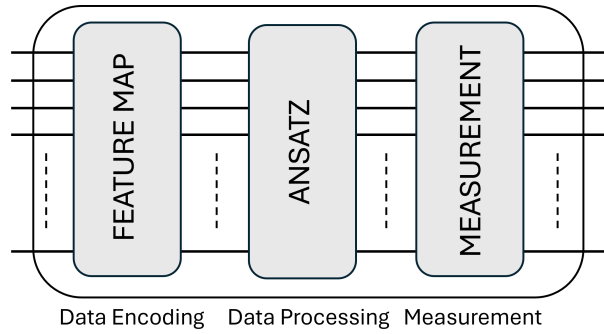


Fig. 1.15: The architecture of QNN consisting of feature map, ansatz, and final measurement of quantum circuit.

Feature Maps

Feature maps are defined as the higher dimensional functions that map our classical dataset into the quantum dataset. It basically transforms the classical inputs into the quantum states in a Hilbert space, where the information is embedded into the quantum circuits. These functions can be defined as

$$\mathcal{T} : x \rightarrow |\psi(x)\rangle. \quad (1.40)$$

This mapping is done through different encoding strategies described below in detail [105].

1. **Basis Encoding:** It refers to the encoding where the classical inputs are stored in the form of the basis states. This is usually helpful when mapping real numbers into

quantum states, where the binary notation of that real number can be written as the quantum states. For an example, number 15 can be represented as 1111 in binary notations, hence 4 qubit system can be used to represent this number.

$$\mathcal{T}(15) = |1111\rangle \quad (1.41)$$

2. **Amplitude Encoding:** It refers to the encoding scheme where information is encoded in the form of amplitude of the states. The probability of the outcome of each state is the output of the model. For example, encoding of the array of real values $X = [2.3, 1.5, 3.2, 0.9]$ into a quantum state using amplitude encoding can be written as we first compute the normalization factor

$$\mathcal{N} = \sqrt{2.3^2 + 1.5^2 + 3.2^2 + 0.9^2} = \sqrt{18.39}. \quad (1.42)$$

Thus, the quantum state representation is

$$|\psi\rangle = \frac{2.3}{\sqrt{18.39}}|00\rangle + \frac{1.5}{\sqrt{18.39}}|01\rangle + \frac{3.2}{\sqrt{18.39}}|10\rangle + \frac{0.9}{\sqrt{18.39}}|11\rangle. \quad (1.43)$$

3. **Angle Encoding:** It refers to the encoding strategy when the normalized dataset is stored as the angles of the $\mathcal{R}_{x,y,z}(\phi)$ (rotation) gates. The number of input parameters to encode in the quantum circuit is equal to a number of qubits in the quantum circuit.

$$\mathcal{R}_y(\phi) = \begin{bmatrix} \cos\left(\frac{\phi}{2}\right) & -\sin\left(\frac{\phi}{2}\right) \\ \sin\left(\frac{\phi}{2}\right) & \cos\left(\frac{\phi}{2}\right) \end{bmatrix} \quad (1.44)$$

For a single qubit state, ψ with parameter ϕ , use of $\mathcal{R}_y(\phi)$ gate can be represented mathematically as,

$$\mathcal{R}_y(\phi)|\psi\rangle = \mathcal{R}_y(\phi)|0\rangle + e^{i\theta}\mathcal{R}_y(\phi)|1\rangle. \quad (1.45)$$

Qiskit SDK offers three predefined feature maps that can be used in QML models. ZZFeatureMap (shown in Fig. 1.16) works for quantum circuits having a number of qubits > 1 , where $\mathcal{R}_z$ gates are employed along with CNOT gates, which allows the model to use the quantum entanglement property in quantum circuits [106].

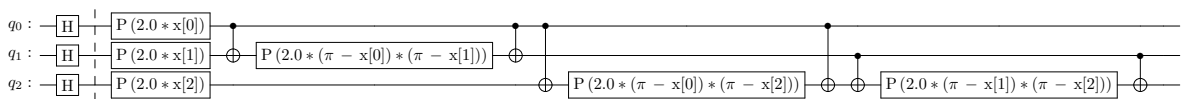


Fig. 1.16: ZZFeatureMap applied for 3 qubits encoding 3 input parameters and no repetition of feature maps (reps = 1). The text inside $P()$ represents the argument of the $\mathcal{R}_z$ gates.

Ansatz

Ansatz are Parametrized Quantum Circuits (PQCs) that apply unitary transformation ($U(\Theta)$) on the states prepared by the feature maps. They are parameterized by angle $\Theta = (\theta_1, \theta_2, \dots, \theta_n)$, where θ_i correspond to single weight/parameter in the QNN. These parameters are tuned using classical optimization algorithms like COBYLA, L_BFGS_B. This part of QNN is analogous to NN in classical ML, where weights are modified during the backpropagation algorithm which is driven with an objective to minimize the loss function. Qiskit provides some predefined ansatz for direct use in the ML models. RealAmplitudes is one of the ansatz provided by Qiskit and it is shown in Fig. 1.17 and Efficient SU2 is another circuit provided by Qiskit and shown in Fig. 1.18.

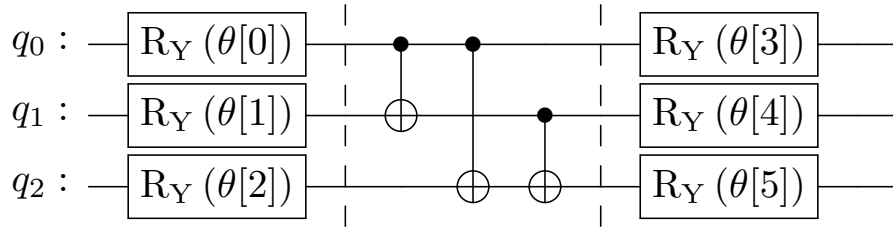


Fig. 1.17: 3 qubits Real Amplitudes parametrized quantum circuit.

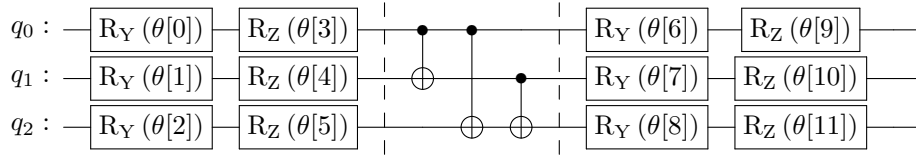


Fig. 1.18: 3 qubits Efficient SU(2) parametrized quantum circuit.

1.6.2 QML for Simple Deterministic Problems

Single Feature Problem

Here, we define a simple deterministic problem for single feature and draw comparisons between accuracies of ML and QML for simplistic problems to establish the use case of QML. We consider a binary classification problem for a single feature as input vector. The problem was defined as follows.

$$\mathbf{x} = x \quad (1.46)$$

If $\mathbf{x} > 0.5$, the result is 1, otherwise 0 where, $x \in (0, 1)$. We run Multi-Layer Perception (MLP) classifiers which gave an score of 1 for both the train and the test dataset. The model consists of three hidden layers as (1, 4, 1) and uses Adam optimizers with tanh activation functions to predict the results of a classification problem.

Now, we compare the accuracy of this ML model with the QML model. QML model was defined using the Sampler and the Neural Network Regressor obtained from Qiskit Machine Learning library. L-BFGS-B optimizer was used and maximum iterations were given as 50. However, the model converged at only 8 iterations and gave the score as 0.96 for the training and 1.0 for the test dataset. Since the sampler was chosen in the model, the accuracy is not always constant whenever the model is run, but it remains nearly same for all the runs. The objective function for the QML model is shown in Fig. 1.19.

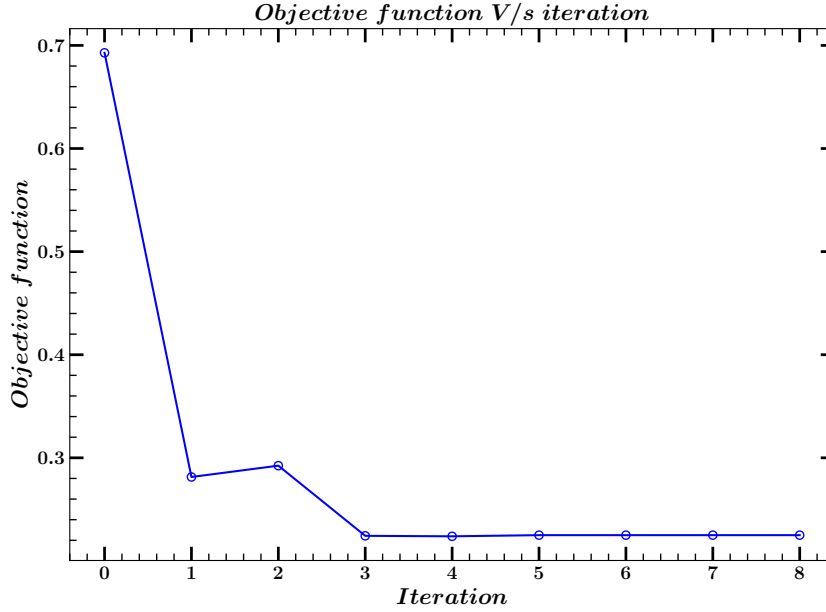


Fig. 1.19: Objective function value evaluated at each iteration where the model stops at 8th iteration only due to the convergence of the objective function value.

This implies the comparable accuracy of the ML and QML models hence signifying the role of QML models.

Two Feature Problem

Here, we define a simple deterministic problem for carrying out the comparison between the Classical and Quantum ML models and draw out basic inferences based on a prediction of that model. We consider a binary classification problem with a feature vector

$$\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2, \quad (1.47)$$

where each component $x_1, x_2 \in (-1, 1)$. The target variable y is defined as a binary label based on the sign of the product of the two features. Specifically, the label assignment function is given by

$$y = \begin{cases} 0, & \text{if } x_1 \cdot x_2 > 0, \\ 1, & \text{if } x_1 \cdot x_2 \leq 0. \end{cases} \quad (1.48)$$

The dataset is shown in Fig. 1.20, as expected the data points located in the first quadrant are assigned. We apply two classical ML techniques, MLP Classifiers from scikit learn

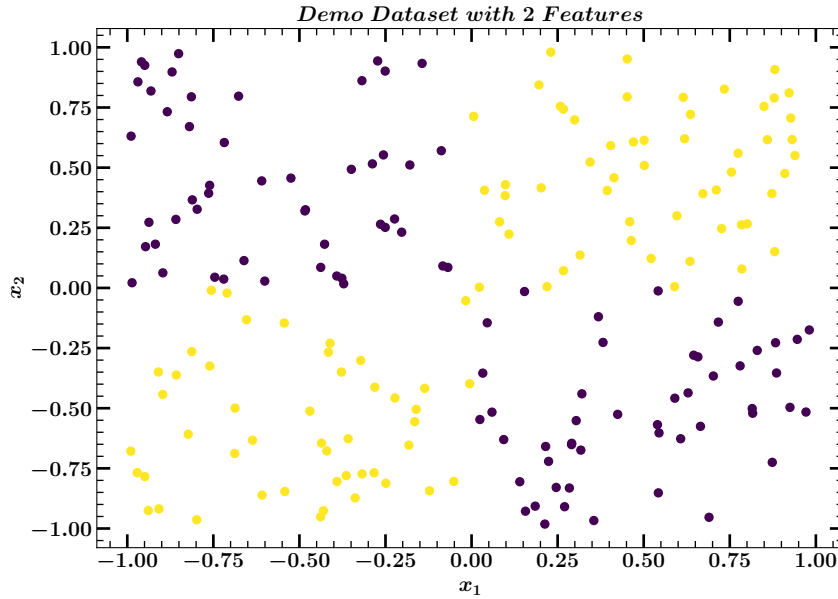


Fig. 1.20: Demo dataset for the simplistic and deterministic two-feature problem.

and NN from Tensorflow, to classify the above dataset into the given labels. Although both models performed simple feed-forward neural networks, NN from the TensorFlow package provides more freedom to initialize different activation functions to different layers, and also provides functions to set the varying learning rate, which is not available in the MLP Classifiers. We achieved an accuracy of 100% and 92% on the train and test sets, respectively, for MLP Classifiers. On the other hand, neural networks achieved an accuracy of 91% and 90% on training and test sets, respectively. We compare the accuracy of these simplistic models with QML models of Variational Quantum Circuits (VQC) and Neural Network Classifiers. We obtained a training accuracy of 96.7% and a test accuracy of 75% on the same dataset. The loss function of the Variational Quantum classifiers is shown in Fig. 1.21. The evolution of the weights for the COBYLA optimizers is shown in Fig. 1.22. The weights stabilize as the model achieves the maximum iterations.

1.6.3 Quantum Long Short Term Memory Networks

Quantum Long Short-Term Memory (QLSTM) networks are specialized networks that integrate the outer structure of classical LSTMs with QNNs as their inner components. The neural networks in classical LSTMs are replaced with QNNs or variational quantum circuits in QLSTMs, as shown in Fig. 1.23 [107]. QLSTMs can be used to learn time series sequences similar to classical LSTMs, but QLSTMs have the advantage of utilizing

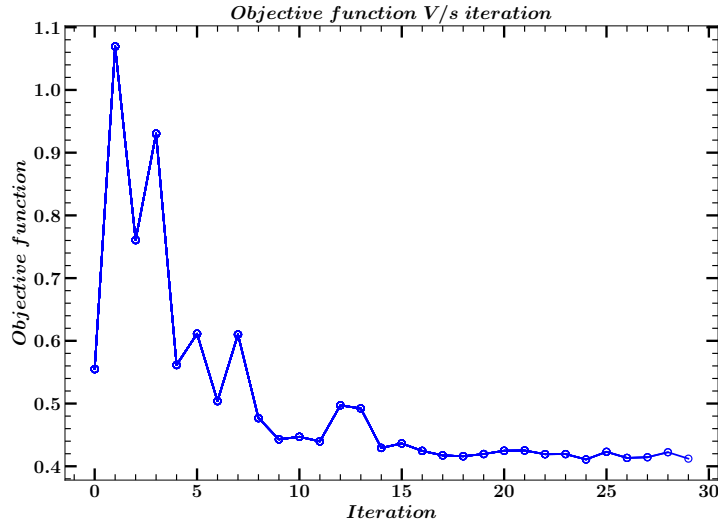


Fig. 1.21: Loss function evolution for each iteration.

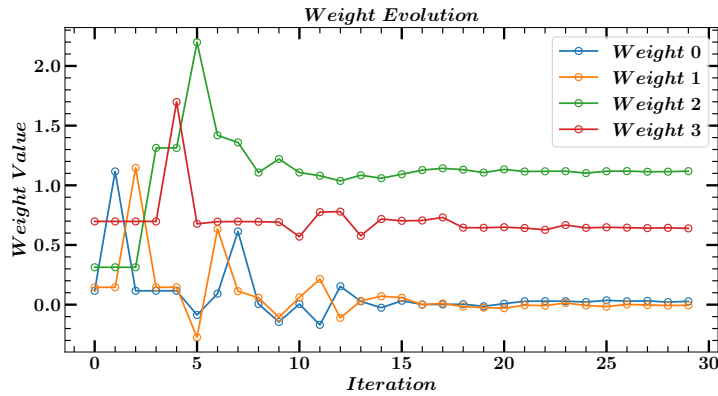


Fig. 1.22: Weight function evaluation for COBYLA optimizer.

the unique properties of quantum mechanics, which can be used to model systems in more complex ways compared to classical LSTMs. QLSTM networks extend classical LSTM architectures by replacing traditional classical NN with QNN, such as variational quantum circuits, to potentially exploit quantum parallelism for sequential learning tasks [107]. These networks maintain the same gating mechanisms as classical LSTMs while embedding quantum operations in their recurrent units, allowing them to process temporal data with reduced dimensionality and enhanced feature encoding. QLSTMs have been proposed for general-purpose time series learning tasks, with experimental studies demonstrating their competitive performance in low-resource classical settings and highlighting their potential advantage in representational capacity and generalization [108]. In the domain of gravitational wave astronomy, QLSTM models have recently been incorporated into federated learning frameworks to enable collaborative, privacy-preserving training across distributed datasets collected by GW observatories. This approach not

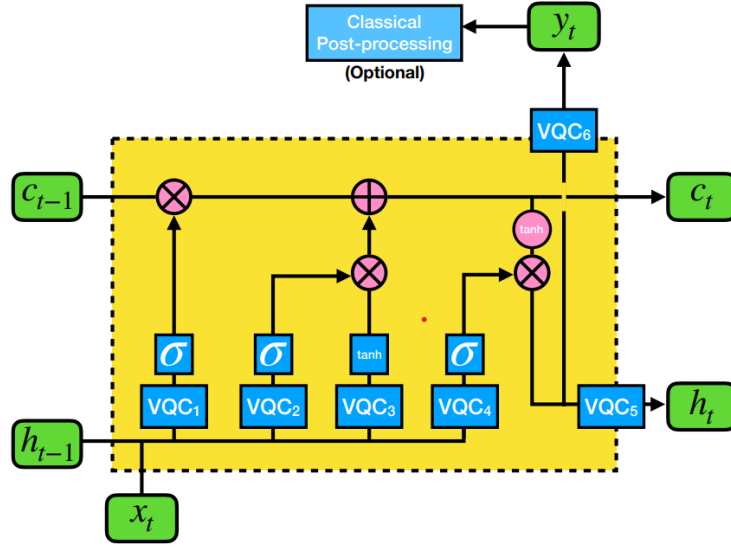


Fig. 1.23: Architecture of QLSTM showing the difference between the classical and quantum LSTMs where variational quantum circuits replaces deep neural networks [107].

only supports scalable training but also leverages quantum-enhanced models to improve signal detection and classification accuracy in noisy environments [109]. These developments collectively indicate that QLSTM networks are a promising direction for quantum-enhanced time series modeling, especially in data-intensive scientific applications such as gravitational wave signal analysis.

2. OUR CLASSICAL MACHINE LEARNING MODELS

2.1 Outline

Classical ML models have been used in the past for various purposes in the field of GW astronomy, as described above in Chapter 1. Here we propose two ML models for enhancing the detection of GW in the LIGO detector data. First, we train feed-forward neural networks to determine the information about the source of the GW signal. Second, we train the LSTM autoencoders to detect the presence of the GW signal inside the LIGO detector data and obtain the GW waveform for the corresponding detection.

2.2 Model 1: Feed Forward Neural Networks

We deal with two different types of information about a GW signal, one numerical and the other physical. Numerical information is the information which we obtain after analysing the signal through different signal processing techniques in different domain spaces and after carrying out various analysis. On the other hand, physical information is the information inherent to the system. This information describes the parameters of the system from which the GW was emitted, which is also known as the binary parameters of the GW signal. Our aim is to connect these information through a higher dimensional function which can map each input to its respective output. This mapping requires dealing with huge number of parameters and learn complex patterns embedded inside the dataset, motivating us to apply machine learning techniques to determine the mapping function. Hence, we employ neural networks to carry out this mapping.

2.2.1 Preprocessing of Data

We obtain the dataset of NR waveforms from the SXS waveform catalog [39], which provides simulated GW signals for CBCs. It includes a dataset for the mergers of BBH, BNS, and NSBH.

Each waveform is described by a unique identifier defined as SXS : (BBH/BNS/NSBH) : XXXX, where XXXX denotes a randomly assigned number that is not correlated with

the physical parameters of the system, but is based on order in which simulations were performed. SXS catalog covers a broad range of simulations, from non-eccentric to eccentric binaries (usually for high mass ratio systems), precessing binaries (for systems whose spin angular momentum is not aligned with the orbital angular momentum), or spinless or spin-aligned (systems whose spin angular momentum lies along the orbital angular momentum of the system) binaries.

Each waveform is uniquely described by the physical parameters of the waveform. Physical parameters include the binary parameters of the waveform as described in the introduction chapter, as well as some other parameters that were essential to run the simulation.

The data for each parameter is provided for two instances: the initial parameters, which were initialized at the start of the simulation and the reference parameters, which were recorded when the binary system achieved quasi-static equilibrium. After the simulation starts, the system is initially not in a quasi-static equilibrium state. To achieve the equilibrium, the masses of the black holes are reduced iteratively by a factor of 10^{-5} until the reference point. Until then, the system emits a spurious signal that mimics GW but originates purely from numerical errors, which is referred to as Junk radiation. Therefore, the waveforms need to be preprocessed before being employed for any learning tasks in the ML models. The steps of preprocessing are described below.

1. **Extracting the dominant mode:** The catalog provides waveforms for multiple modes of GW signal, $l = 2$, $m = \pm 2$ (dominant mode) and several other subdominant modes of spin-weighted spherical harmonic decomposition of strain. Since we are concerned with the dominant mode, which carries most of the information about the signal, we extract out only $l = 2$, $m = \pm 2$ mode of the waveform.
2. **Cropping and Scaling:** Cropping the waveform up to reference time to remove the spurious signal present inside the waveform. The time axis of the waveform is given in dimensionless quantity. To convert it into units of seconds, we scale the time axis by multiplying it with the total mass of the BBH system. We obtain waveforms ranging from $M = 40$ to $M = 95$ with unit spacing, creating the dataset for almost 1,00,000 waveforms.
3. **Equi-spacing of intervals:** Waveforms are not recorded at equal intervals of time. Hence, we interpolate our waveform on an equally spaced interval of time.
4. **Tapering:** We taper the final waveform from both ends by creating a turkey window (also known as tapered cosine window) with shape parameter $\alpha = 0.1$ to taper it from both sides. Tapering smoothly reduces the amplitude of the curve to zero, which is essential to minimize the spectral leakage while evaluating the Fourier transform of the signal.

5. **Padding:** Different NR waveforms have different lengths, and the duration of the waveforms is highly dependent on the physical parameters of the system. Therefore, to use the waveforms in evaluating time-frequency analysis, FFT and in several machine learning algorithms it is a good approach to make all the length of data of equal size.

Each waveform is preprocessed before being used further into the analysis for the next steps. Fig. 2.1 provides the visual description of the preprocessing of the waveforms.

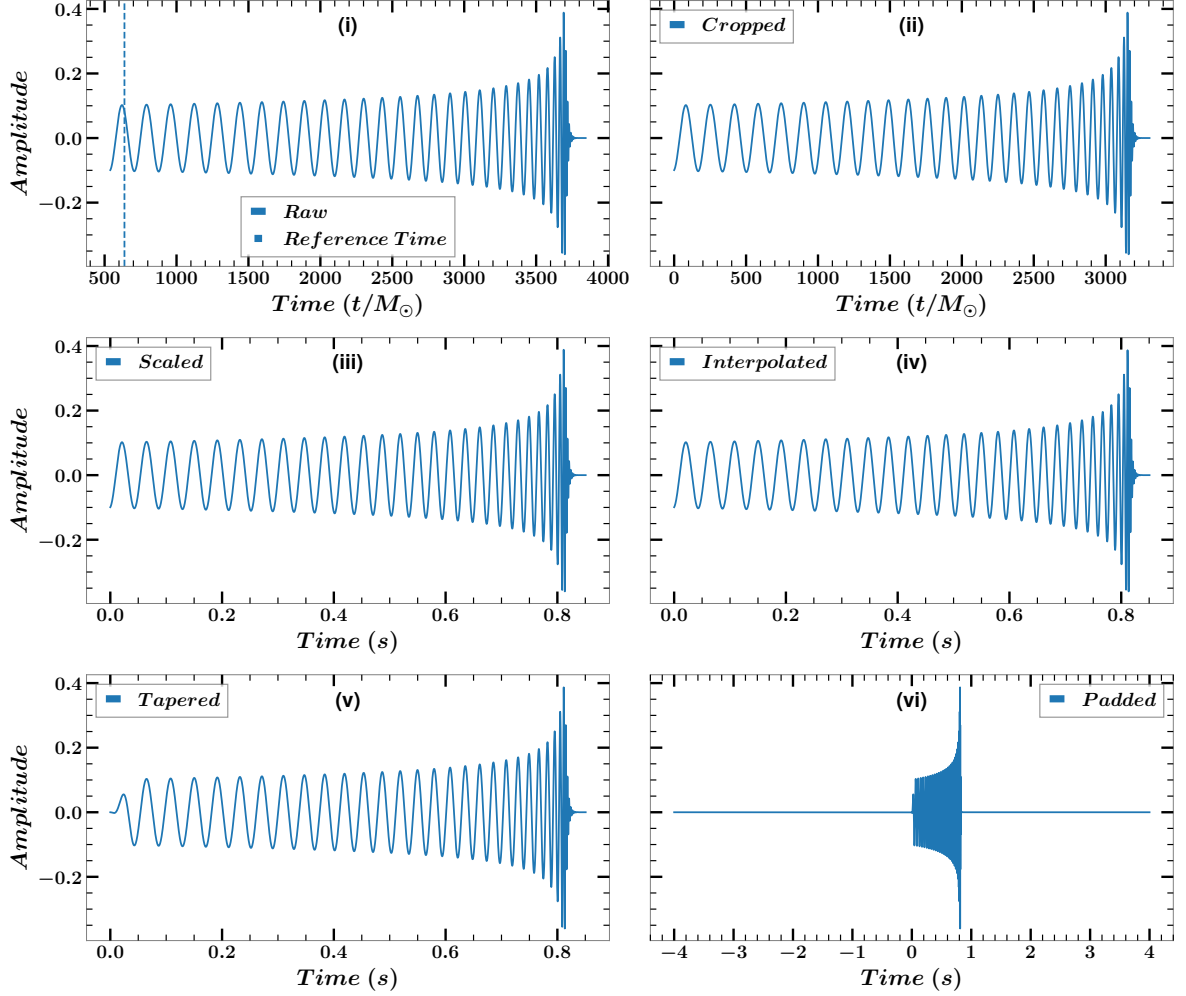


Fig. 2.1: Preprocessed steps for SXS:BBH:0305. (i) Reference point in the signal, (ii) Cropping of the signal, (iii) Scaling of the signal, (iv) Interpolating the signal to equispaced intervals of time, (v) Tapering both the ends, (vi) Padding of the signal

2.2.2 Waveform Analysis

As introduced in the first chapter, we perform different signal processing methods to analyze the signal in different parameter spaces. First, we analyze the preprocessed

waveforms in the frequency domain. We evaluate the FFT of the time series signal and represent the signal's amplitude in frequency space. Since the GW signal is a chirp signal (whose amplitude and frequency simultaneously increase with time), we plot the time-frequency analysis using the wavelet transform and analyze the curve in detail. For example, Fig. 2.2 depicts the waveform analysis for SXS:BBH:0305 signal which is the NR signal corresponding to GW150914 event. The signal spans 0.8 seconds in the temporal domain and achieves a maximum strain of approximately 0.4, denoting the time of the merger. The CWT for the corresponding signal is displayed in Fig 2.2, highlighting the rise in the frequency and amplitude of the signal, depicting its chirp nature. Besides CWT, the rotated FFT of the signal shows a peak at around 25 Hz. This peak corresponds to the flat line in the CWT plot and depicts the range of the inspiral region. This coincides with the fact that the GW signal remains in the inspiral phase for most of its duration during the merger process. The peak frequency in FFT corresponds to the inspiral frequency of the GW signal, and half of it is equal to the orbital frequency of the system in the inspiral region. Apart from the BBH waveforms, we also analyze

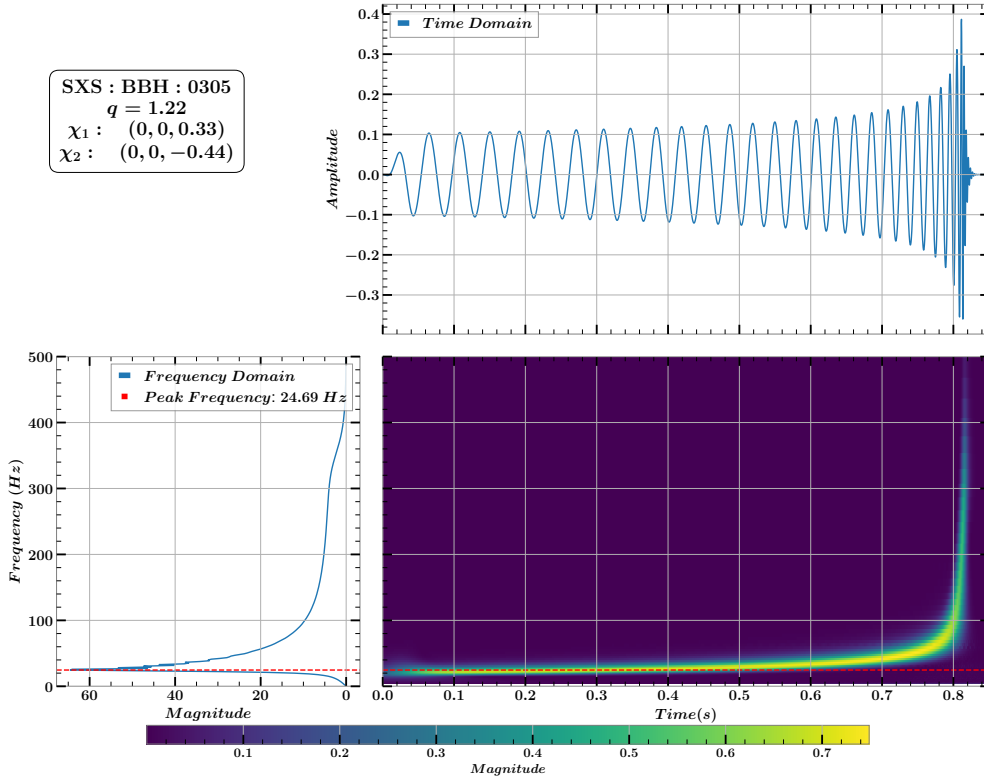


Fig. 2.2: Waveform analysis of NR waveform SXS:BBH:0305 depicting the Time Frequency (Wavelet Transform) evolution and Fourier transform of time domain signal.

the waveforms emerging from BNS merger and NSBH mergers. We perform the analysis of SXS:NSBH:0009 which is a NSBH merger shown in Fig. 2.3. The time domain signal of NSBH mergers is also explained through GWs emitting from CBCs. Here, we can observe the inspiral frequency of the GW signal is equal to 40.09 Hz. The time-frequency evolution shows a smooth transition from merger to ringdown phase and is less abrupt as compared to BBH mergers in Fig. 2.2. This can be explained due to the fact that NSBH mergers often leads to the formation of black holes. This simulation did not produce a kerr black hole or the kerr black hole produced did not have enough spin magnitude to emit GWs in the spacetime. We also performed an analysis of GW emissions from

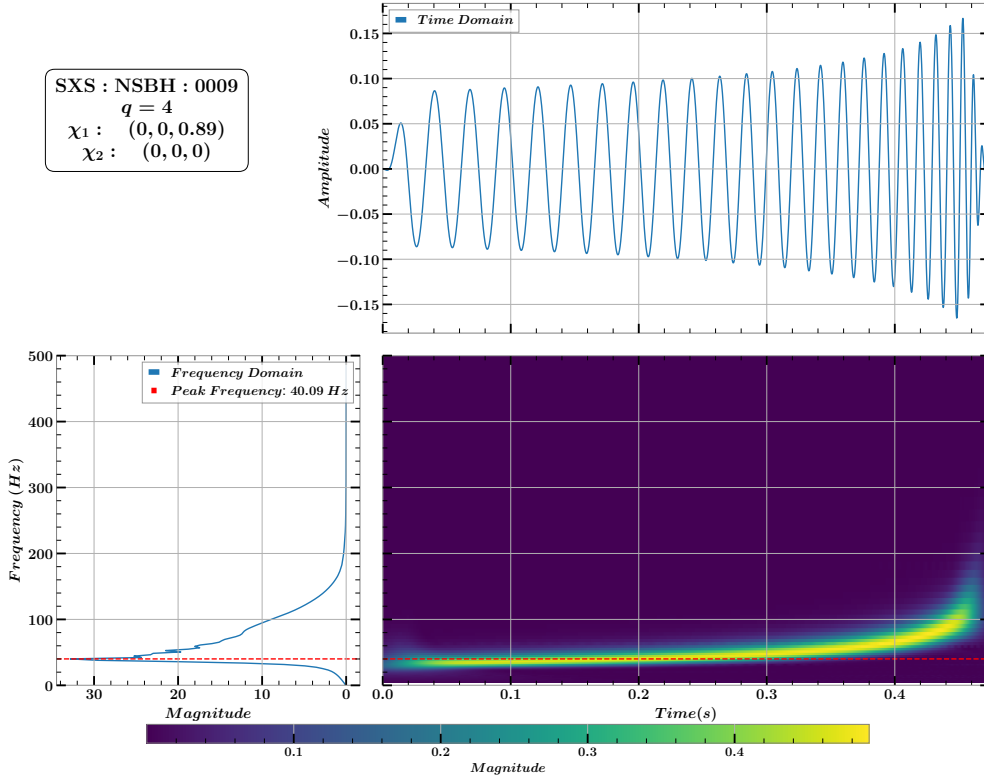


Fig. 2.3: Waveform analysis of NR waveform SXS:NSBH:0009 depicting the Time Frequency (Wavelet Transform) evolution and Fourier transform of time domain signal.

BNS mergers. This merger may lead to the formation of either an unstable neutron star, which eventually collapses to form a black hole, or directly convert it into a black hole. We analysed NR simulation SXS:BNS:0001 which is shown in Fig. 2.4. We can observe that the inspiral frequency of the GW signal was 24.17 Hz. We can also observe a clear pattern in the ringdown phase of GW signal. This implies that the merging of these two neutron stars leads to the formation of an unstable neutron star, which emits GW for a

small period of time and then collapses to form a stable Kerr black hole.

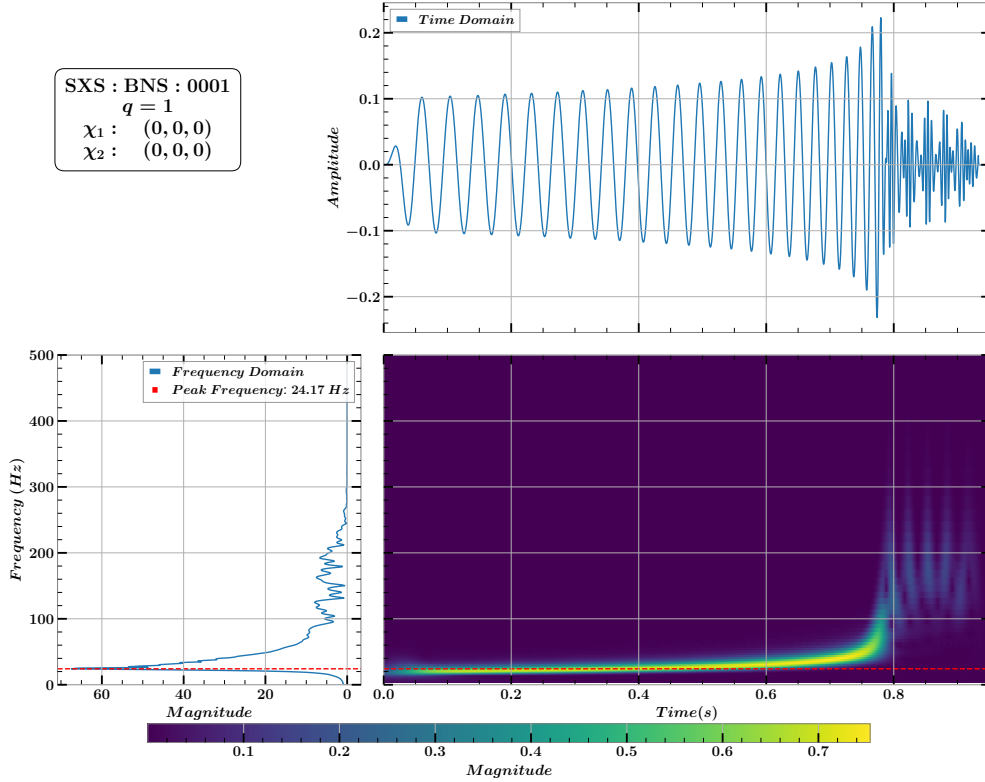


Fig. 2.4: Waveform analysis of NR waveform SXS:BNS:0001 depicting the Time Frequency (Wavelet Transform) evolution and Fourier transform of time domain signal.

2.2.3 Methodology

Our aim is to define a simplistic model of the FFT arising purely from the numerical techniques and observational skills that follow the principles of physics established in the theory of GW.

We model the amplitude of the FFT of the NR signal by fitting it using the fitting functions. These fitting functions arise from the observational and theoretical results of the GW theory. We observed the exponential increase in the amplitude until the inspiral frequency of the merger is reached. This motivates us to fit it with an exponential increase function. During the inspiral phase, the amplitude of the GW signal in the frequency domain decreases as a function of $f^{-7/6}$ [28], which is a power law decay. This inspired us to fit the second phase with the power law decay. The third phase, where theoretical predictions fail, we fit it using a different parameter but with a power law decay only.

Finally, at the ringdown phase, the amplitude in the time domain decreases exponentially, which in the frequency domain is translated as the Lorentzian decay of the amplitude [28]. The fitting function used for modelling the amplitude of the FFT of the signal is given as

$$\mathcal{A}(f) = \begin{cases} Ae^{f/\alpha}, & 0 < f \leq f_0 \\ Bf^{-\beta}, & f_0 \leq f \leq f_1 \\ Cf^{-\gamma} & f_1 \leq f \leq f_2 \\ \frac{D}{1 + \left(\frac{f-f_2}{\delta}\right)^2} & f_2 \leq f \end{cases} \quad (2.1)$$

where the parameters A, B, C, D are constants evaluated using continuity equations at the boundaries. f_0, f_1, f_2, f_3 are the different frequencies where the model shifts its fitting function. From the continuity equations, one can write as

$$A = \frac{A_0}{e^{\frac{f_0}{\alpha}}} \quad (2.2)$$

$$B = Ae^{\frac{f_0}{\alpha}} f_0^\beta \quad (2.3)$$

$$C = Bf_1^{\gamma-\beta} \quad (2.4)$$

$$D = Cf_2^{-\gamma} \quad (2.5)$$

where A_0 represents the maximum amplitude of the FFT curve, f_0 represents the frequency of maximum magnitude in the FFT curve, f_1 is the point of maximum curvature of the CWT curve and f_2 is the frequency at maximum amplitude in the CWT curve.

Procedure of Fitting FFT Using the Above-Defined Model

Here, we will explain the procedure of fitting the model using the above equations. As described in the Chapter 1, the merging of the BBH system encompasses three major phases. Physicists are often interested in studying the phase transition of a system to obtain different physical properties of the system. Inspired by this thought, we evaluate the transition points of the merger process using numerical and observational analysis. The FFT of the NR signal shows a peak at the inspiral frequency, which concludes that the system remains in the inspiral region for most of the duration of the merger process, which also provides the orbital frequency of the system at the inspiral stage of the merger ($\omega_{GW} = 2\omega_s$). Therefore, we consider this as one of the fitting parameters of the GW signal, which is shown by the red line in Fig. 2.5 (ii),(iii).

Next, we aim to determine how the system transitions from the inspiral phase to the merger phase. Theoretically, we know that the ISCO frequency is considered the last stage of the inspiral region. However, we consider that we are unaware of the system's parameters and calculate the transition point arising purely from numerical intuitions.

We trace the 2-dimensional CWT by taking the maximum value at each instant of time, as shown by the black line in Fig. 2.5 (ii). Since the tracing of the curve is not entirely continuous, we apply mean averaging at a window length of size 15. Due to heavy discontinuities in some of the FFT, parameters might not smooth the curve with such a tiny size window length. Therefore, we iteratively modify the window size and stop the loop at a window size that results in the frequency between $f = 50$ and $f = 150$ to achieve the best transition point (The frequencies 50 and 150 were obtained from the observational analysis of the waveforms, because after 150 Hz most of the waveforms reached in the merger phase and before 50 Hz most of the waveforms had the inspiral frequencies). After obtaining the smoothed time-frequency evolution of the signal, we calculate the curvature of the curve numerically and assign the transition frequency as the frequency at maximum curvature. Now, we describe how we calculate the curvature of the point. For a one-dimensional (1D) curve defined as $y = f(x)$, the curvature κ at a point x can be written as

$$\kappa(x) = \frac{y''(x)}{[1 + (y'(x))^2]^{3/2}}. \quad (2.6)$$

While performing the computations on the computer we discretize the system as follows. The data is a set of sampled points (x_i, y_i) with uniform spacing h , the first and second derivatives are numerically approximated using central difference methods as

$$y'_i \approx \frac{y_{i+1} - y_{i-1}}{2h}, \quad y''_i \approx \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2}. \quad (2.7)$$

Substituting the Equation (2.7) into the Equation (2.6) yields the curvature at each point as

$$\kappa_i = \frac{\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2}}{\left[1 + \left(\frac{y_{i+1} - y_{i-1}}{2h}\right)^2\right]^{3/2}} \quad (2.8)$$

We determine the point of maximum curvature by evaluating $i_{\max} = \arg \max_i |\kappa_i|$. This is the point of maximum curvature, which can be written as $(x_{i_{\max}}, y_{i_{\max}})$.

Finally, the final frequency is obtained by evaluating the maximum frequency in the CWT trace curve as shown by the blue line in Fig. 2.5 (ii), (iii) and Fig. 2.6. This frequency regards the frequency at which the final decay in the FFT occurs.

By using the above schemes and techniques, we fit our model using the equations defined above in Fig. 2.1. We obtain the fit as shown in Fig. 2.7. The curve is basically divided into four segments which fit with an exponential increase in the first segment, power decay in the second and third segments with different decay rates and finally, the Lorentzian decay. We fit all the waveforms present in our dataset in a similar fashion and express the time-frequency evolution, and create a dataset that contains all the parameters obtained from the numerical fit of the amplitude. The fitting of the FFT for different waveforms is shown in Fig. 2.8 It describes the fitting for waveforms having the

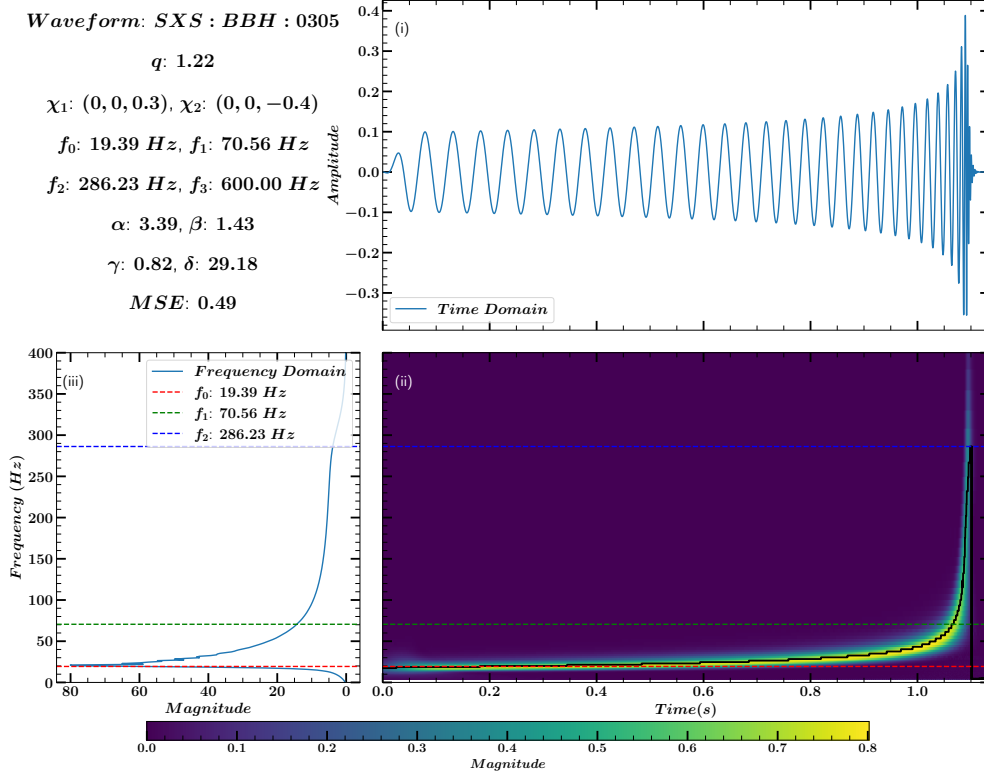


Fig. 2.5: Waveform analysis of SXS:BBH:0305 NR signal where (i) denotes the time series signal, (ii) denotes the time-frequency evolution of the signal, and (iii) denotes the frequency spectrum of the same time series signal.

same total masses but different different mass-ratios.

2.2.4 Forming the Neural Network Table

Now, we form a neural network table that contains the fitting parameters of the FFT fitted parameters and binary parameters of the GW signal. These binary parameters include m_1 , m_2 , χ_1 and χ_2 , which in total becomes 8 parameters. We represent spins in spherical co-ordinates, which refers to (r_1, θ_1, ϕ_1) and (r_2, θ_2, ϕ_2) . Hence, we form a neural network table that consists of 8 input parameters and 8 output parameters.

2.2.5 Data Analysis

We analyze the dataset to gather more information about the parameters and to understand the nature of variation and dependencies on other parameters of the neural network table. We have the dataset for the 1,36,180 waveforms, which is extrapolated from the normalized dataset of 2476 waveforms by multiplying with total masses ranging from M

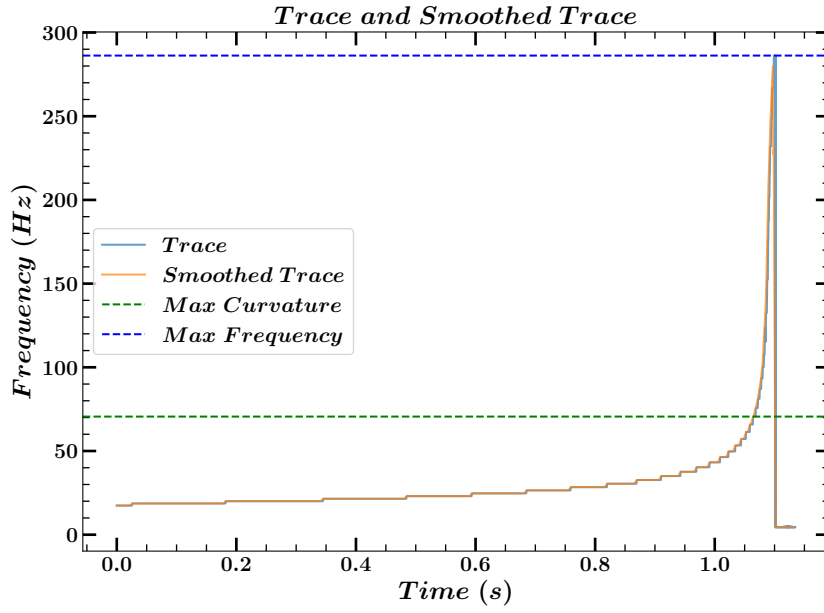


Fig. 2.6: Trace and smoothed trace of time-frequency evolution of SXS:BBH:0305 using mean averaging window of length 15 and depicting the point of maximum curvature and point of maximum frequency.

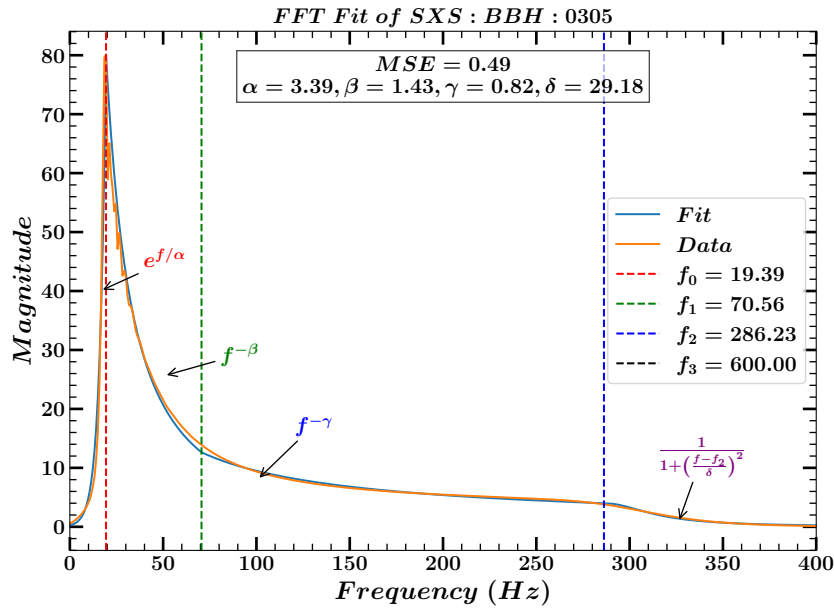


Fig. 2.7: FFT fit of SXS:BBH:0305 signal whose time domain signal is shown in the Fig. 2.5 (i).

$= 40$ to $M = 95$. We split the dataset into two parts: BBH systems with zero spin magnitude and BBH systems with non-zero spin magnitude. This gives us 10,285 waveforms for the spinless systems, and the remaining 1,25,895 waveforms belong to the non-zero spin BBH systems. Before we begin analyzing the dataset, we calculate the Z-score for each

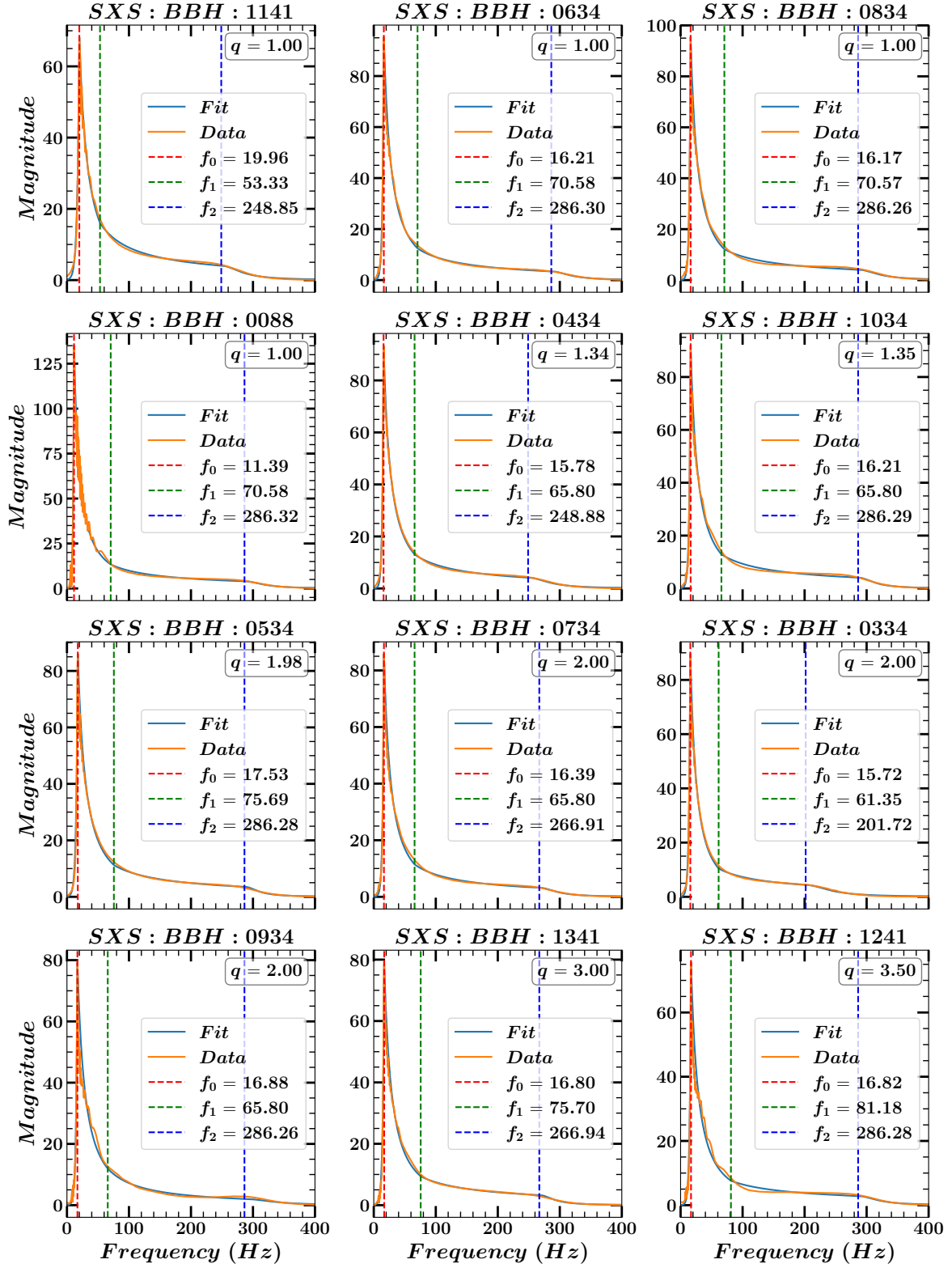


Fig. 2.8: FFT fit of different waveforms with same total masses and different mass ratios.

parameter and select only those data points that fall within the 2σ confidence interval, which roughly covers 95.45% of the total data. Z-score is defined as a statistical measure that provides the distance of the given point from the mean of the normalised dataset

with respect to the standard deviation of the dataset.

$$z = \frac{x - \mu}{\sigma}, \quad (2.9)$$

where x is the points of the dataset, μ is the mean and σ is the standard deviation of dataset.

Spinless BBH systems

We begin by analyzing the data for the spinless BBH systems and draw inferences that can be used while applying ML models to our dataset. Our dataset consists of fitting parameters and binary parameters of GW signal. We begin with analyzing the distribution of the dataset on the sample space.

The distribution of fitting parameters is shown in Fig. 2.9. Parameter α exhibits highly skewed distribution peaking around 2 and the majority of distribution lying in the region 0 and 5. This suggests most waveform morphologies experience steep peaks before reaching the inspiral frequency. Parameter β shows a Gaussian distribution with a peak around 1.35; this gives a close match with the theoretical result, which suggests the decline of amplitude as $f^{-7/6}$ in the inspiral region till the beginning of the merger phase. Similar inferences can be carried out for the parameter γ , which also shows a Gaussian distribution peaking around 0.8. Parameter δ shows a positively skewed distribution where most of the values lie in the range 10 and 50, but the long tail extends up to 60, indicating some waveform morphologies have a low decay rate δ . Parameter A_0 shows a Gaussian distribution with a peak at around 100. Parameter f_0 shows a steep peak with a tail at 40 Hz, meaning most of the systems achieve inspiral frequency from 0 to 40 Hz. f_1 peaks at around 70 Hz, demonstrating most of the system's transition with respect to FFT. f_2 shows a multimodal distribution ranging from 100 to 350 Hz, indicating its dependence on remnant masses on black holes and spins.

The distribution of output parameters m_1 and m_2 is shown in Fig. 2.10. m_1 shows a shifted gaussian distribution and m_2 is uniformly divided into the dataset. Parameters α and β correlate positively, meaning the more the steep increase, the more the steep decay of the curve. The correlation of α is also positive with f_0 because as α decreases, the steepness increases, shifting the curve towards the left in the frequency domain. Hence, decreasing the f_0 simultaneously, which in turn makes the correlation with β positive too. Now, we explain the correlation matrix as shown in Fig. 2.11. Masses m_1, m_2 are inversely correlated with δ . m_2 shows an inverse relation with the parameters $\beta, \gamma, \delta, f_0$. We observe a strong positive correlation between δ and f_0, f_1 . Similarly, f_0, f_1 are strongly correlated, meaning the shift of f_0 transforms smoothly to shift the f_1 as well. Similarly, the inference can be drawn for f_1 and f_2 . Now, we analyze the variation of the fitting parameters with the binary parameter m_1, m_2 of GW signal and take out the inferences

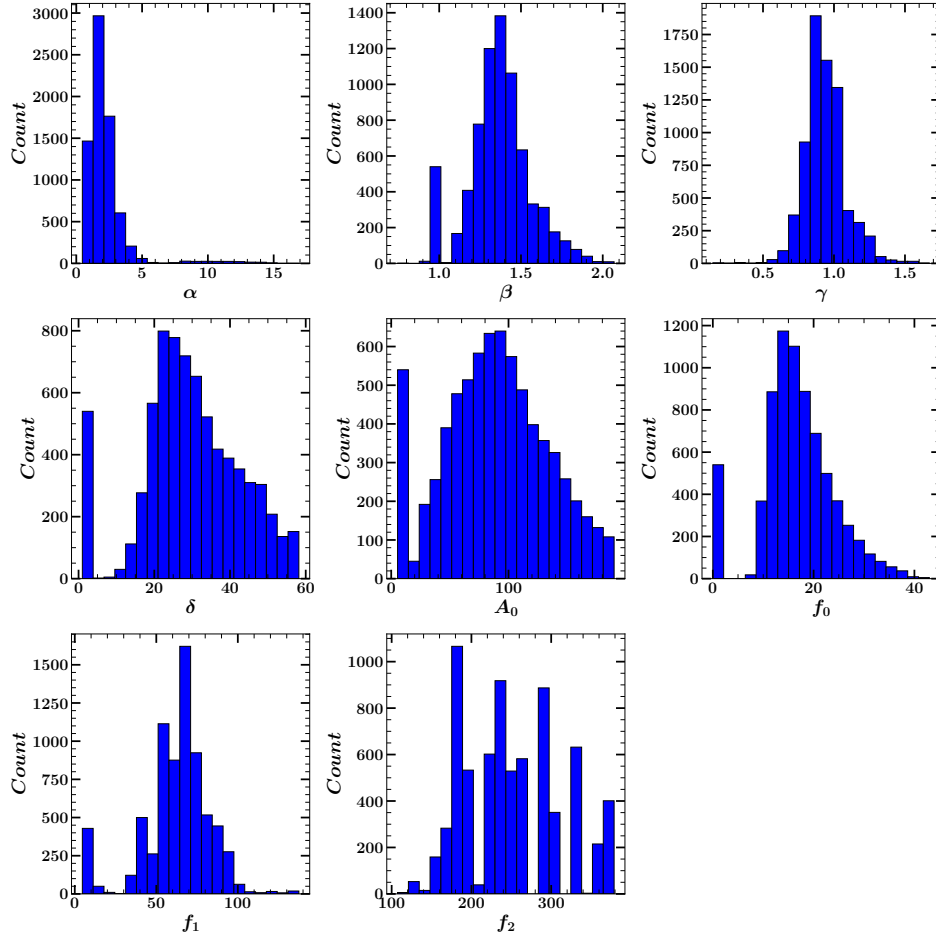


Fig. 2.9: Distribution of the fitting parameters of the GW signal for the systems having zero spin magnitude (spinless BBH systems).

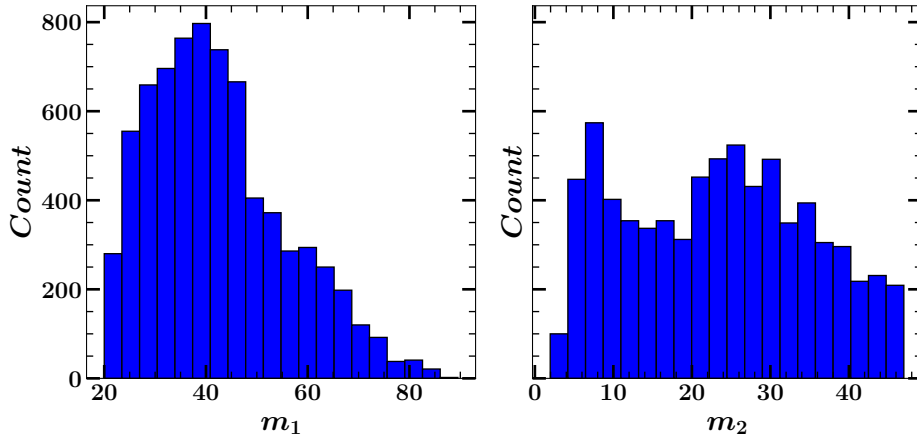


Fig. 2.10: Histogram plots showing the distribution of the binary parameters of the GW signal for the systems having zero spin magnitude (spinless BBH systems).

from curves shown in Fig. 2.12. To understand Fig. 2.12, we should note one important thing that points lying on the positive slope line indicate the waveforms with the same

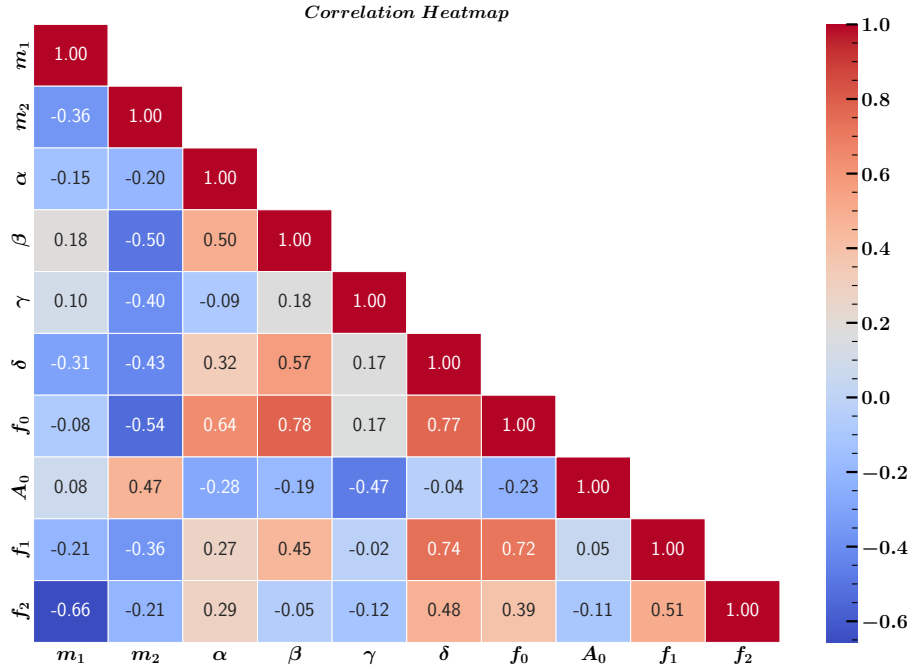


Fig. 2.11: Pearson Correlation matrix for the parameters in neural network table for spinless BBH systems

mass ratios but different total masses and points lying on the same negative slope line indicate waveforms with the same total masses of the system but different mass ratios. f_0 (the inspiral frequency) of the system is inversely proportional to the total mass of the system, which is clearly visible in Fig. 2.12 (i). We also notice that f_0 is higher for high mass ratio binaries and decreases gradually to equal mass ratio systems. This implies higher mass ratio binaries inspiral at higher frequency. This is because the merger of a higher mass ratio BBH system spans more time in the inspiral region, hence leading to a gradual increase in the inspiral frequency of the system. This pattern is gradually transferred to the f_1 as it is higher for a higher mass ratio, meaning the system reaches to merger at a higher frequency than equal mass ratio systems. The trend, however, in f_1 is more complicated as it is simultaneously dependent on the total mass of the system. The maximum amplitude of the GW signal in the frequency domain increases with increasing total mass of the system, indicating that high-mass systems produce GW signals with higher amplitudes in the frequency domain. The parameter α seems independent of the masses, mass ratio or total masses of the system. Parameter δ is inversely related to the total mass of the system. As the total mass increases, the parameter δ reduces. This effect is considerably higher at equal mass ratios and reduces as we move towards higher mass ratios. β and γ follow a similar pattern where they are higher for higher mass ratio binaries and decrease as the data approaches equal mass ratio binaries. From the above

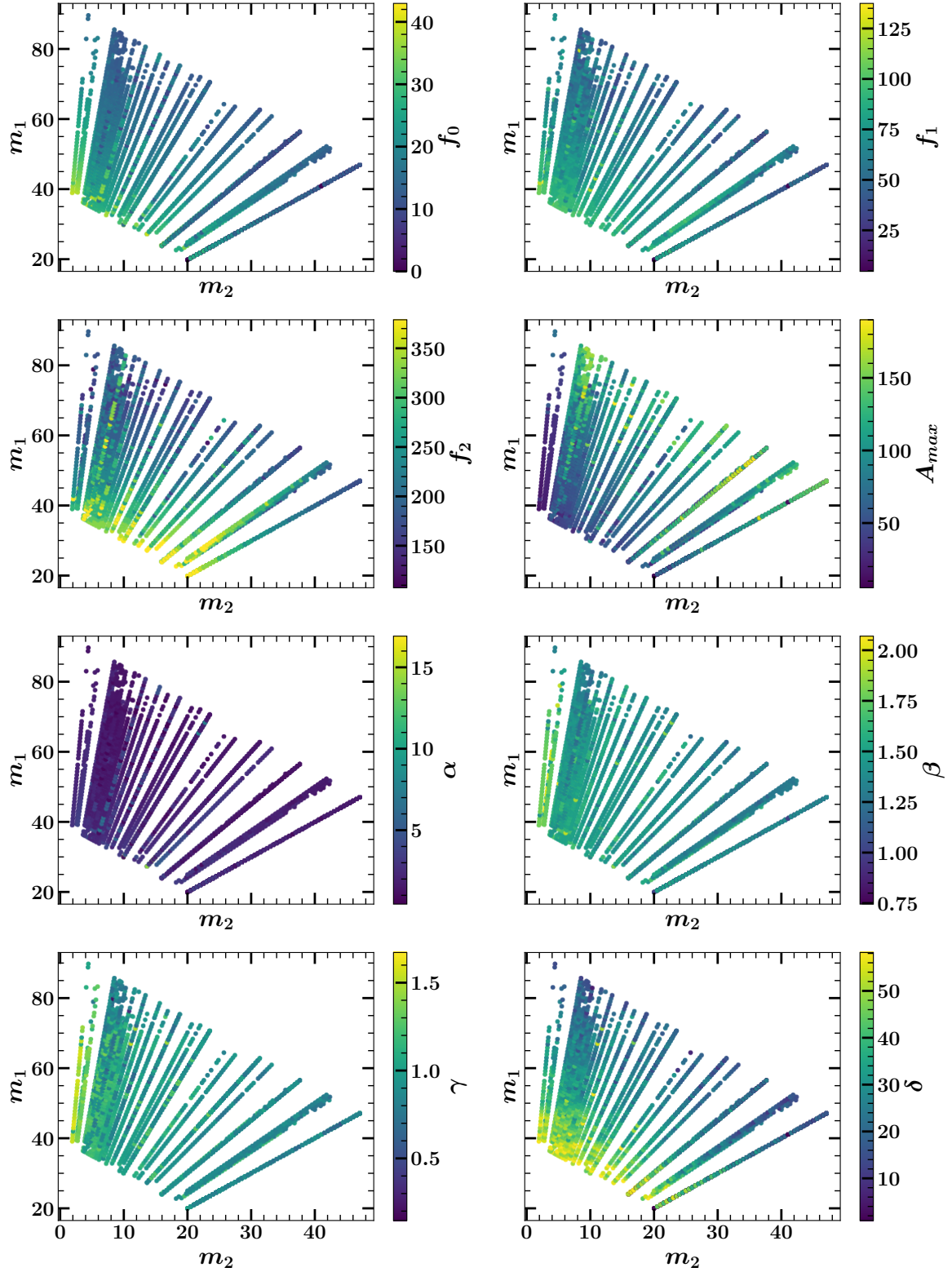


Fig. 2.12: Variation of the fitting parameters with respect to the m_1 and m_2 (binary parameters of GW) for spinless BBH systems.

analysis, we can conclude the following notable points. These points will be useful while defining ML models and their input and output features.

- The fitting parameter α exhibits negligible correlation with the binary masses. We need to investigate it further, which could be done in future work.
- The inspiral frequency parameter f_0 is strongly and inversely related to the total mass $M = m_1 + m_2$ of the system. Higher total mass systems tend to be inspiral at lower frequencies, consistent with post-Newtonian theory. Additionally, f_0 increases with increasing mass ratio, implying higher mass-ratio systems inspiral at relatively higher frequencies.
- The merger frequency parameter f_1 shows a positive correlation with both f_0 and the masses of the black holes.
- Parameters β and γ , associated with the slope and curvature in the frequency domain amplitude, show similar correlation trends with m_1 , m_2 , and mass ratio. Both parameters are higher for systems with larger mass ratios for constant total mass and lower total masses for constant mass ratios of the system.
- A strong correlation exists among frequency parameters f_0 , f_1 , and f_2 , as well as between δ and both f_0 and f_1 .
- We can also conclude the nonlinear nature of correlations and dependence between physical and fitting parameters of the waveform morphology which can not be completely understood directly. Hence, we employ NN models to understand the complex behaviours in the dataset.

Spin BBH Systems

Here, we analyze the dataset of the neural network table for spin BBH systems. First, we analyze the distribution of the fitting parameters, which are the output parameters here in the neural network table. The distribution curves are shown in Fig. 2.13. The distribution of the parameter α appears to be roughly symmetric and centred around a value slightly above 2, suggesting a consistent influence of α across the dataset. In contrast, β possesses a positively skewed distribution, peaking near 1 with a long tail. This implies that for most samples, the value of β remains small, but there are some cases with significantly larger values. The parameter γ possesses a value around zero only. The parameter δ has a unimodal distribution skewed slightly to the right, centred around 25. This indicates that the majority of waveforms exhibit a moderate rate of decay, with a few cases showing slower falloff. The amplitude parameter A_0 , f_0 are broadly distributed, peaking near 100, with a slight asymmetry in Gaussian nature. The distribution of f_1 , is broader and exhibits some signs of multimodality.

Now, we analyze the distribution of binary parameters of GW signal as shown in Fig. 2.14. m_1 , m_2 show a normal distribution where m_1 ranges from 20 to 80 and m_2

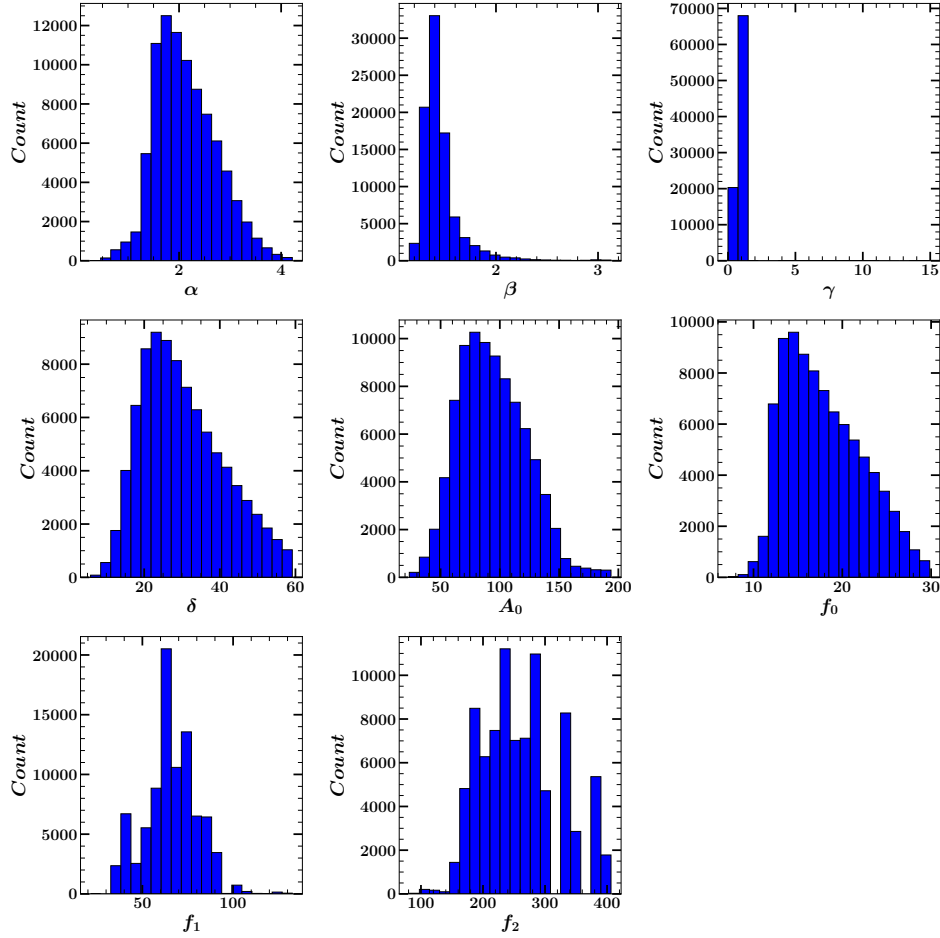


Fig. 2.13: Distribution of the fitting parameters of GW signal for the spin BBH systems.

ranges from 0 to 45. The spin parameters are demonstrated in the spherical coordinates system. As these parameters we have extracted from the NR dataset, these are the configuration of the spin parameters for which they have run the simulations. ϕ shows a uniform distribution for the black holes. The number of simulations ran for aligned and anti-aligned configurations of black holes is high. Spin magnitudes of the black holes are distributed such that majority of simulations were performed for certain discrete values of spin magnitude like 0.8, 0.1.

The Pearson correlation matrix for all the parameters is shown in Fig. 2.15. We observe that the spin parameters are almost uncorrelated with the other parameters in the dataset. Only masses and fitting parameters show correlation. Out of which m_1 shows negative correlation with f_1, f_2, f_0, δ , on the other hand m_2 is positively correlated with A_0 as shown previously.

Now, we analyze the variation of m_1, m_2 with the fitting parameters. The relational plot for the above configuration is shown in Fig. 2.16. f_0 follows a similar pattern as in the Fig. 2.12, where high mass ratio binaries accompanied by low total mass have considerably high inspiral frequency, which gradually decreases as the total mass increases and mass

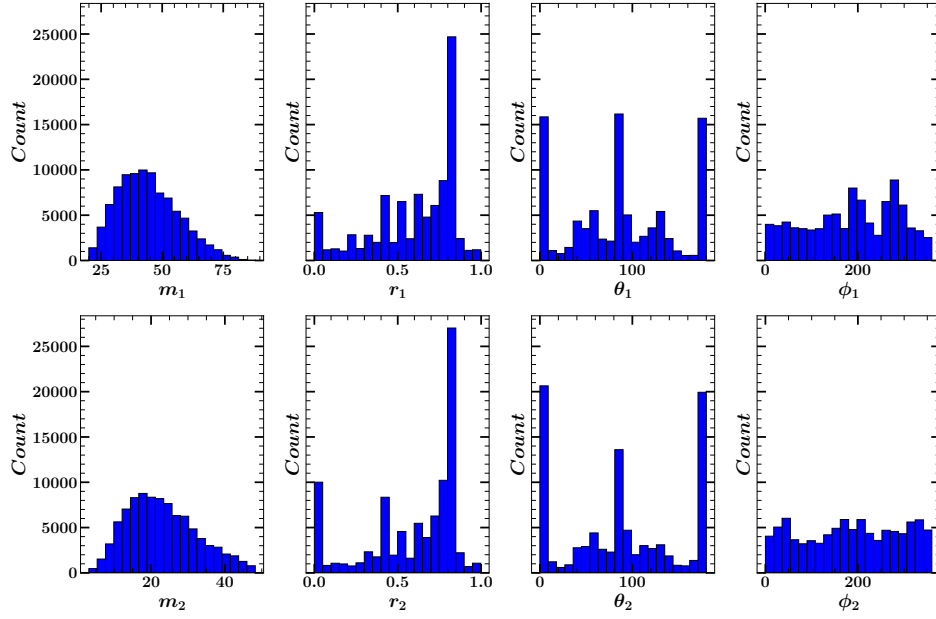


Fig. 2.14: Distribution of the binary parameters of GW signal including the masses and spin components of the black holes for the spin BBH systems.

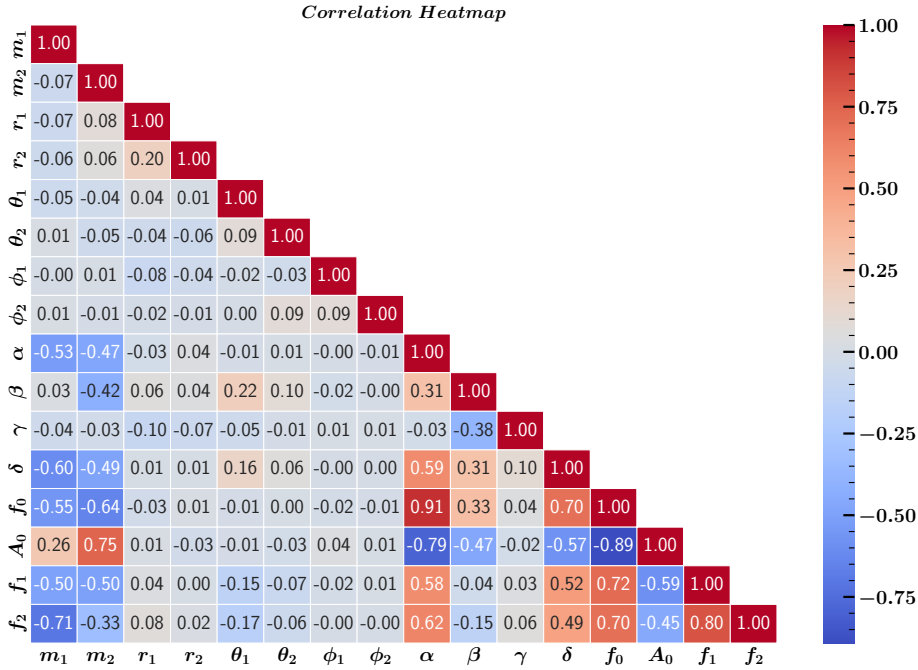


Fig. 2.15: Pearson correlation matrix for the spin BBH systems.

ratio decreases. Fitting parameters β and γ do not show any significant dependence with respect to the masses of the binary system. Nor do they show any dependence on the spins of the system as evident from the correlation plot shown in Fig. 2.15. Hence, they

can be removed from the parameters of the neural network table. Other conclusions of the system are similar to the conclusions of spinless BBH system model. The following

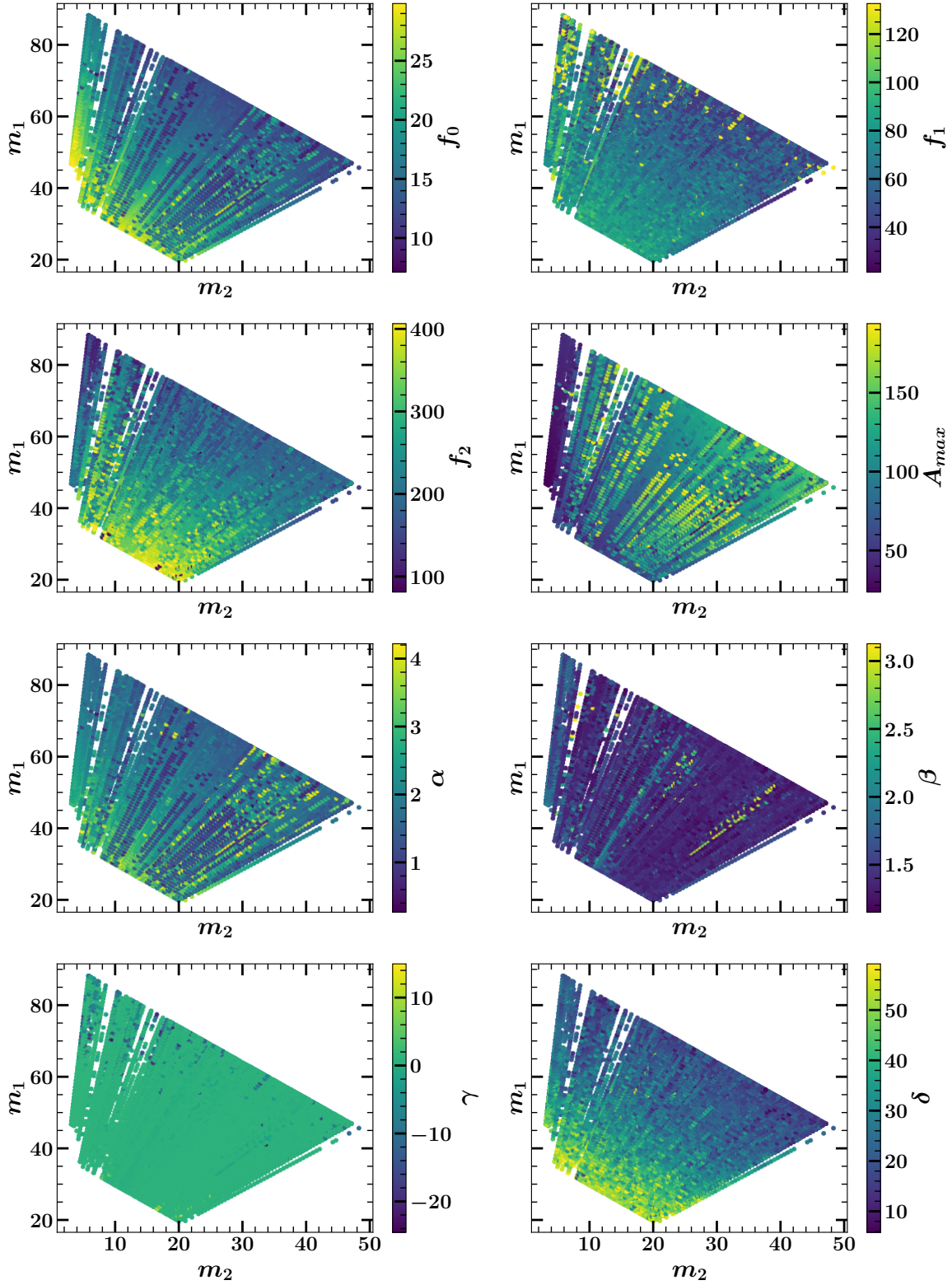


Fig. 2.16: Variation of fitting parameters of GW signal with respect to the m_1 and m_2 for spin BBH systems.

points can be concluded from the above analysis, which will be useful in defining the ML model and their input and output features.

- The parameter γ is narrowly distributed around zero. It shows no significant correlation with any of the binary parameters, including spin parameters.
- Frequency-related parameters f_0 and f_1 exhibit broader distributions. f_0 displays a pattern where high mass-ratio and low total mass systems exhibit higher inspiral frequencies. f_1 shows signs of multimodality, hinting at complex merger dynamics across spin configurations.
- Spin parameters represented in spherical coordinates show the uniform distribution in azimuthal angle ϕ and discrete peaks in spin magnitudes (e.g., 0.1, 0.8). This indicates the sampling strategy of the NR simulations. Most simulations were conducted for aligned or anti-aligned spin configurations.
- The Pearson correlation matrix reveals that spin parameters are largely uncorrelated with both the binary masses and fitting parameters. Correlations are primarily present between mass parameters and fitting parameters.
- β and γ do not show significant variation with m_1 or m_2 , nor with spin degrees of freedom.
- Overall, the trends in spin BBH configurations are similar to the case of spinless BBH configurations. However, the spin parameters are additional for this case.

2.2.6 Model

Our model aims to map the information encoded in the fitting parameters to the binary parameters of GW signal using a higher dimensional function. It can be regarded as fitting a higher dimensional curve using loads of parameters. We define two neural network models, one model for the spinless BBH systems and another model for BBH systems with non-zero spin parameters.

Spinless BBH Systems

Here, we propose two neural networks that input the fitting parameters and output the binary parameters of the signal.

Model 1: We define a simplistic neural network model from TensorFlow library that contains 2 hidden layers and one output layer containing 32, 16 and 2 neurons, respectively. The model is trained with the dataset at a learning rate = 0.00001 for 200 epochs with a batch size of 64. The activation layer for each layer of the neural network was kept as ReLU (Rectified Linear Unit, ranging from (0, 1)). The dataset was

normalized with the maximum of each column in the data frame and given as the input to the model.

Model 2: We also define a model that outputs M (total mass) and q (mass-ratio) instead of m_1 and m_2 . This motivation came from the above analysis of the dataset, which shows clear patterns with respect to mass ratios and total masses of BBH systems rather than individual masses of the BBH system. The model was defined using the TensorFlow library. The model consisted of three layers: 2 hidden layers of size (32, 16) and one output layer of neurons 2. The pictorial demonstration of the model is shown in the Fig. 2.17. It was trained at a learning rate of 0.00005 for 100 epochs with a batch size of 64. The activation function of each layer was kept as ReLU and a fraction of the test dataset was used as a validation set for each epoch.

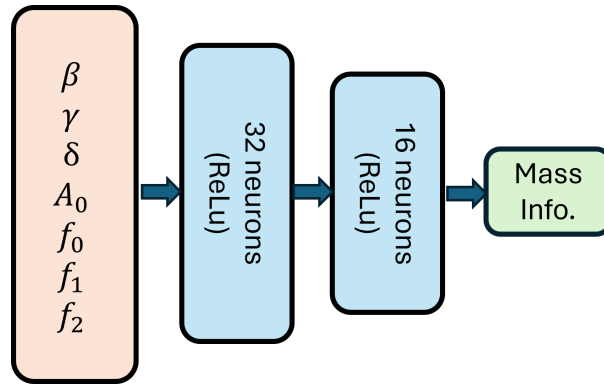


Fig. 2.17: The pictorial demonstration of the architecture of the NN model defined for spinless BBH systems.

Spin BBH Systems

Here, we train our model in the reverse approach because from the above analysis in Fig. 2.16, we can conclude that the β and γ parameters are independent of the binary parameters of GW signal. Hence, we remove these parameters from our neural network table. The input parameters of the neural networks are the binary parameters of the signal and the output parameters are the fitting parameters of GW signal.

We implement a deep learning feed-forward neural network with six fully connected layers where each layer is followed by a Batch Normalization layer and a Dropout layer at a rate of 0.1. These two layers were introduced so that the model could converge with the stability and prevent overfitting of the dataset. The 6 fully connected layers have (512, 512, 256, 128, 64) number of neurons, respectively. Each hidden layer unit had the activation function as ReLU. The repetition of two layers of 512 neurons was implemented for the model to learn non-linear effects in the datasets. The final layer is an output layer containing 6 neurons with a linear activation function. The model was implemented using the TensorFlow library with learning rate = 0.0005 and batch size =

64 for 50 epochs. The learning rate was reduced using the ReduceLR function, which reduces the learning rate of the model as the loss starts to stabilize. The patience of function was kept as 5 epochs with a reducing factor of 0.5 up to a learning rate of 10^{-6} . Corresponding pictorial demonstration of the model is shown in the Fig. 2.18

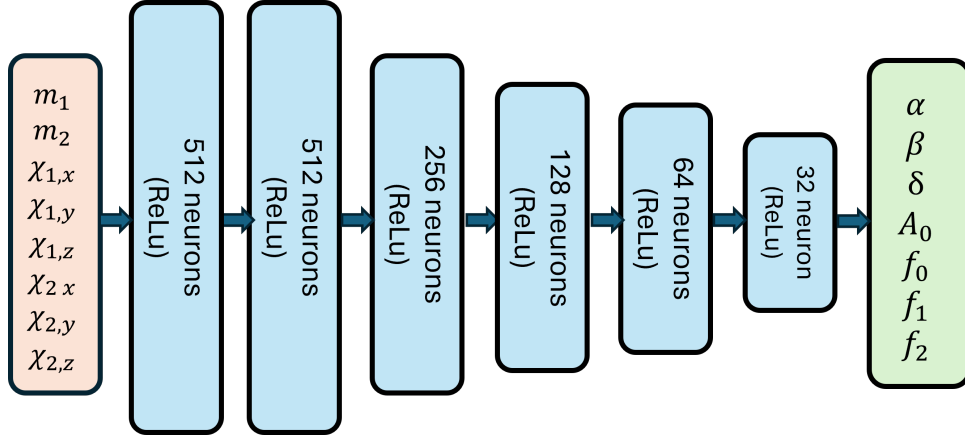


Fig. 2.18: The pictorial demonstration of the architecture of the NN model defined for spin BBH systems.

2.3 Model 2: LSTM Autoencoders for Denoising GW signal

We use the LSTM autoencoders to denoise the GW signal and detect the presence of the GW signal in the noisy interferometric data. LSTM autoencoders have the ability to denoise noisy data and detect the presence of an anomaly in the data and various other similar use cases. Inspiring by the use case of LSTM autoencoders and following a similar approach taken by Ref. 75, we built a model which is not only capable of detecting the presence of GW but is also capable to denoise the data and predict the embedded waveform.

2.3.1 Preprocessing of Data

We create a simulated Gaussian noise dataset with mean = 0 and standard deviation = 1 for 64 seconds, sampled at a sampling frequency of 4096 Hz shown in Fig. 2.19. We call this dataset as noise. We create another dataset, which we call as waveforms, which is built by appending the NR waveforms together up to the length equal to noise. Preprocessed waveforms are padded up to the length of the nearest power of 2 and combined together as shown in Fig. 2.20. The total number of waveforms that were used to inject the signal was 50. Padding was done to maintain uniformity and make the model learn the absence of GW signal in the noise. Waveforms are chosen such that

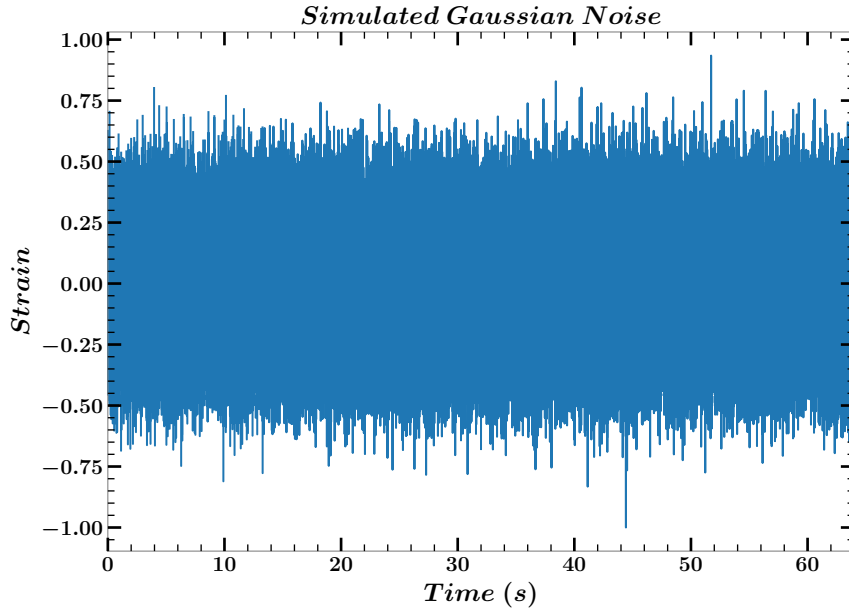


Fig. 2.19: Normalized simulated Gaussian noise extending up to 64 seconds in the temporal domain and sampled at a sampling frequency of 4096 Hz.

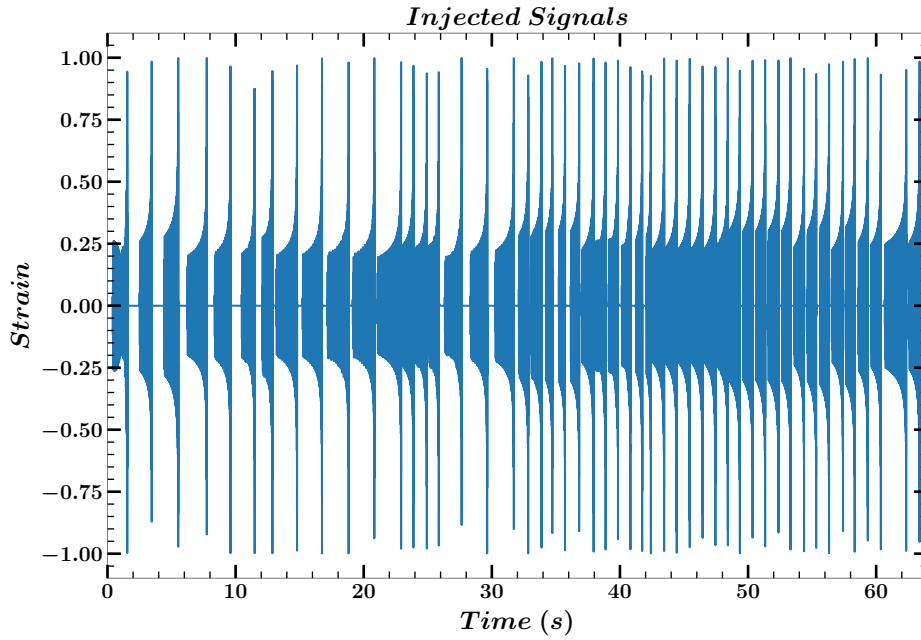


Fig. 2.20: Normalized waveforms obtained from NR database and padded up to the nearest power of 2 and appended until the length of noise becomes equal to the length of waveforms of the signal.

the length of waveforms equals the length of the noise. Then, we add both the noise and waveforms to create the final signal and normalize it to the maximum amplitude of the signal. The final signal dataset is shown in Fig. 2.21. We divide our normalized signal with injected waveforms into overlapped epochs shifted to 1 data point, where each

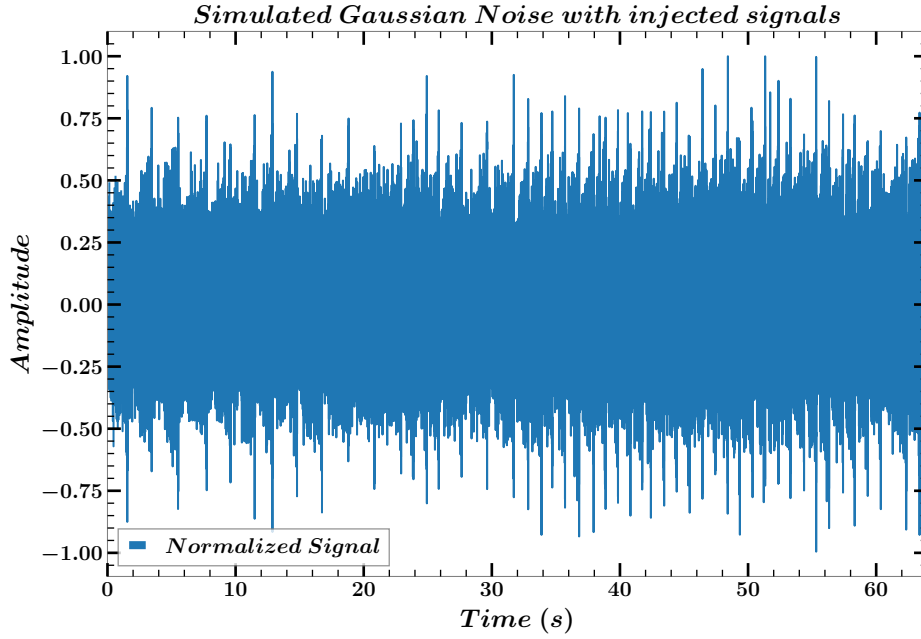


Fig. 2.21: Normalized signal where injected waveforms are added to the noise.

epoch size equals 128 data points. We divided the dataset into train and test datasets, where shuffling of the dataset was not allowed to maintain order in the time series signal. The train and test datasets were divided in the ratio 80:20 ratio and the model used the test dataset as the validation set for each epoch. Apart from the simulated dataset, we tested our model on the real LIGO dataset. We downloaded the official data uploaded at the Gravitational Wave Open Science Collaboration (GWOSC) platform [110] to get the event dataset recorded from LIGO Hanford and Livingston. We extract signals sampled at 4096 Hz for a duration of 4 seconds around the event. The raw dataset as shown in Fig. 2.22(i), was preprocessed by applying a bandpass filter on the signal from 2 Hz to 1024 Hz as shown in Fig. 2.22(ii). The bandpassed signal was then whitened to normalize the amplitude of the frequencies in the signal as shown in Fig. 2.22(iii). Whitening a signal means adjusting it so that all frequencies have similar strength. This is done to remove any strong patterns or trends in the noise, making the signal easier to analyze. In the frequency domain, it helps to flatten the spectrum so that no single frequency dominates over others.

2.3.2 Data Analysis

We perform a detailed analysis of the LIGO noise dataset, which includes the FFT and wavelet transforms. Figure 2.23, illustrates the raw time domain signal of the 4 sec signal sampled at 4096 Hz of LIGO detector data obtained from H1. Figure 2.23 also illustrates the corresponding wavelet transform of the signal, we can directly observe the patterns in the noise of LIGO detector data which is repeating after a varying interval of time.

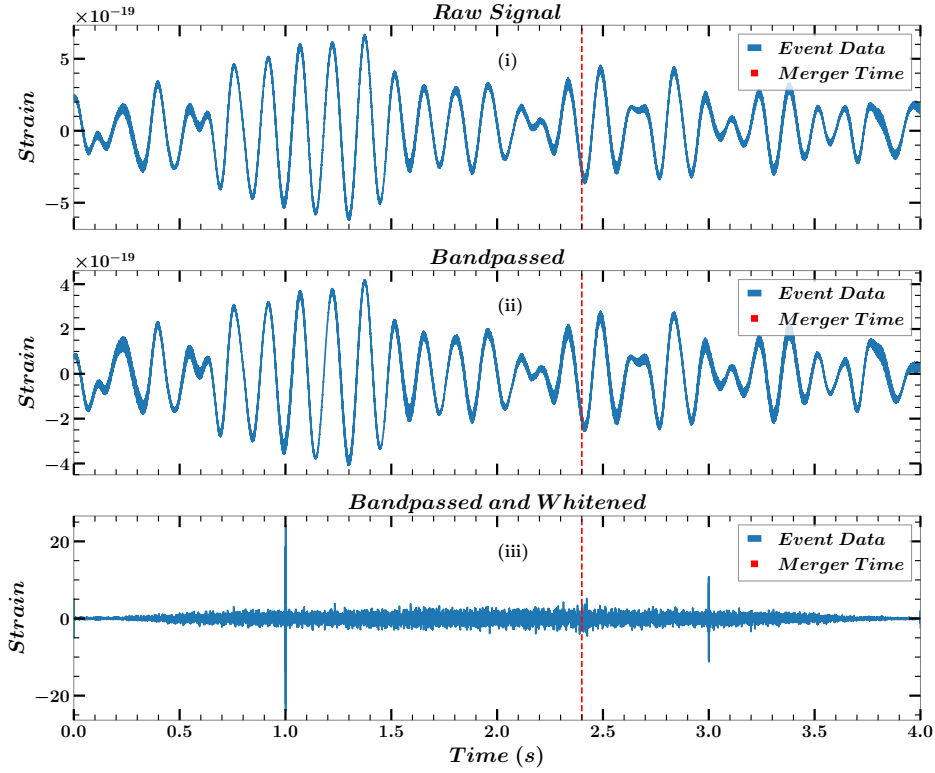


Fig. 2.22: Preprocessing steps demonstrated for the segment of a dataset of 4 seconds in the time frame in which GW150914 was recorded, where the event merger time was 1126259462.4 in GPS seconds (at 2.4 seconds in our picture).

These patterns could be learnt through ML models.

On the contrary, we also analyzed the signal in which the GW event GW150914 occurred. We perform the FFT of the each signal shown in the Fig. 2.22, and deduce the effect of the preprocessing and understand the motivation behind the preprocessing of the signal. The FFT of the time domain signals represented in Fig. 2.22 is shown in Fig. 2.24, which depicts the FFT of raw signal, bandpassed signal (between 2 Hz to 1024 Hz) and finally the whitened signal. It is evident from the frequency spectrum that the signal becomes significantly flattened beyond 1024 Hz, which can be attributed to the application of a bandpass filter. This observation also provides a rationale for employing a whitening filter, which is designed to equalize the power spectral density of the signal by suppressing the frequency-dependent noise and ensuring that each frequency contributes equally in amplitude. In contrast, the remaining two spectral curves demonstrate an exponential attenuation in amplitude at higher frequencies, highlighting the necessity of whitening for effective signal preprocessing.

Figure 2.25 shows the analysis of Fig. 2.22 using the CWT plot (time frequency analysis) which describes the evolution of signal in frequency domain with time. As clearly shown in Fig. 2.25, the GW signal becomes visible only after the whitening process is applied, highlighting the importance of whitening in signal analysis. Furthermore, we

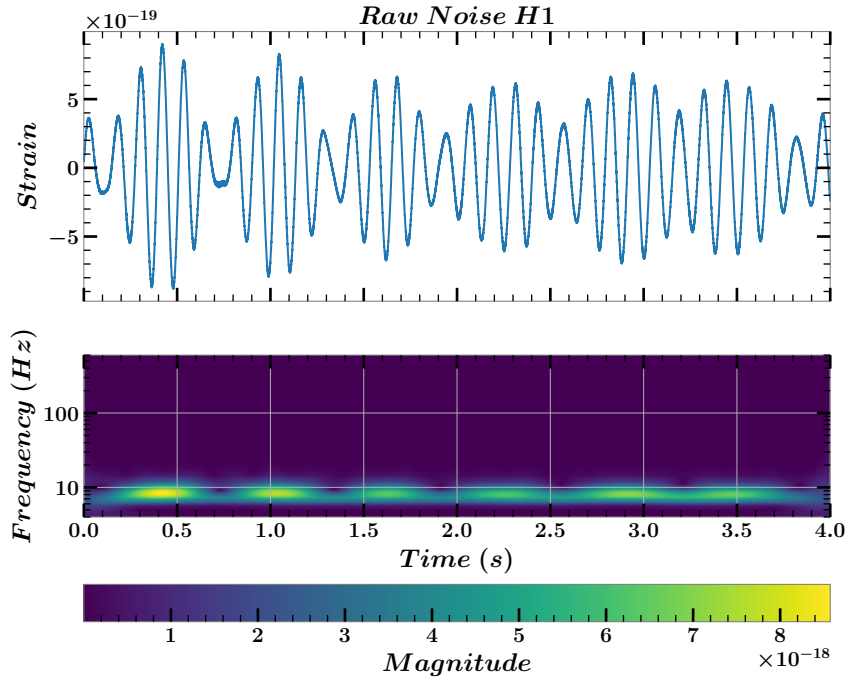


Fig. 2.23: Raw time domain signal of LIGO H1 for a duration of 4 sec sampled at 4096 Hz and its corresponding CWT plot showing the time frequency evolution of the signal.

can clearly observe the time-frequency evolution of the GW signal, which matches well with the waveform modeling in the previous section.

2.3.3 Model

We construct an LSTM autoencoder to denoise waveforms by removing noise from a dataset containing injected NR waveforms. The model comprises seven layers, where the first three layers form the encoder, which maps the input data from a high-dimensional space to a lower-dimensional compressed representation. The middle layer, a RepeatVector, serves as a bottleneck, storing the encoded representation and preparing it for reconstruction. The final three layers form the decoder, which maps the compressed representation back to the original high-dimensional space.

The encoder consists of three stacked LSTM layers with output dimensions (32, 16, 8). The first two layers return sequences, while the last layer outputs only the final hidden state, reducing the dimensionality. The Repeat Vector layer, with 8 units, replicates this encoded representation across time steps to serve as input to the decoder. The decoder is a mirror image of the encoder, comprising LSTM layers with output dimensions (8, 16, 32), all of which return sequences to reconstruct the original input.

The final output layer is a Time Distributed layer with a single unit per time step,

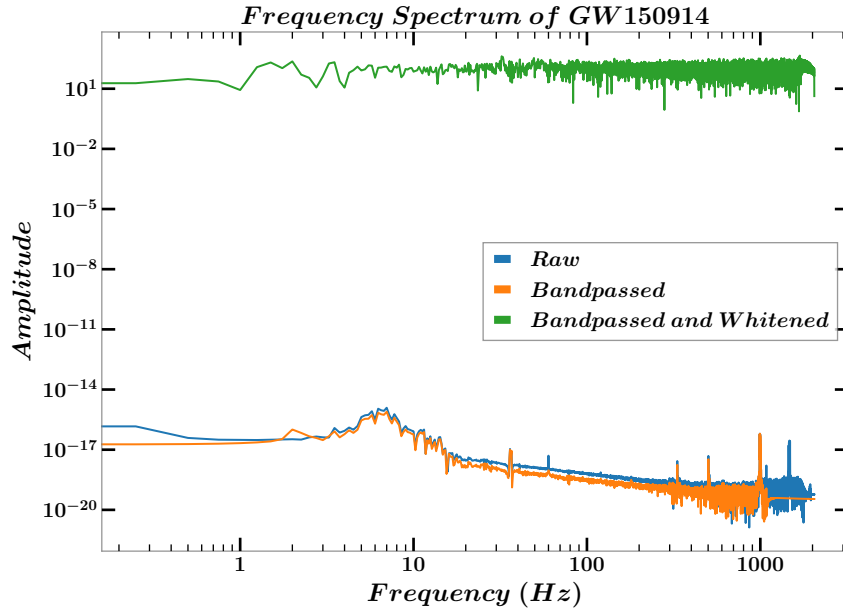


Fig. 2.24: Frequency domain signal of the raw, bandpassed and whitened time series signal which clearly shows the need of the preprocessing to increase the amplitude of the higher frequency signal upto the amplitude comparable to the lower frequency signal (in other words, flattening the signal in the frequency domain)

producing the denoised waveform. The model is trained for 40 epochs with a learning rate of 0.0001 and a batch size of 128. The pictorial demonstration of the model is shown in the Fig. 2.26.

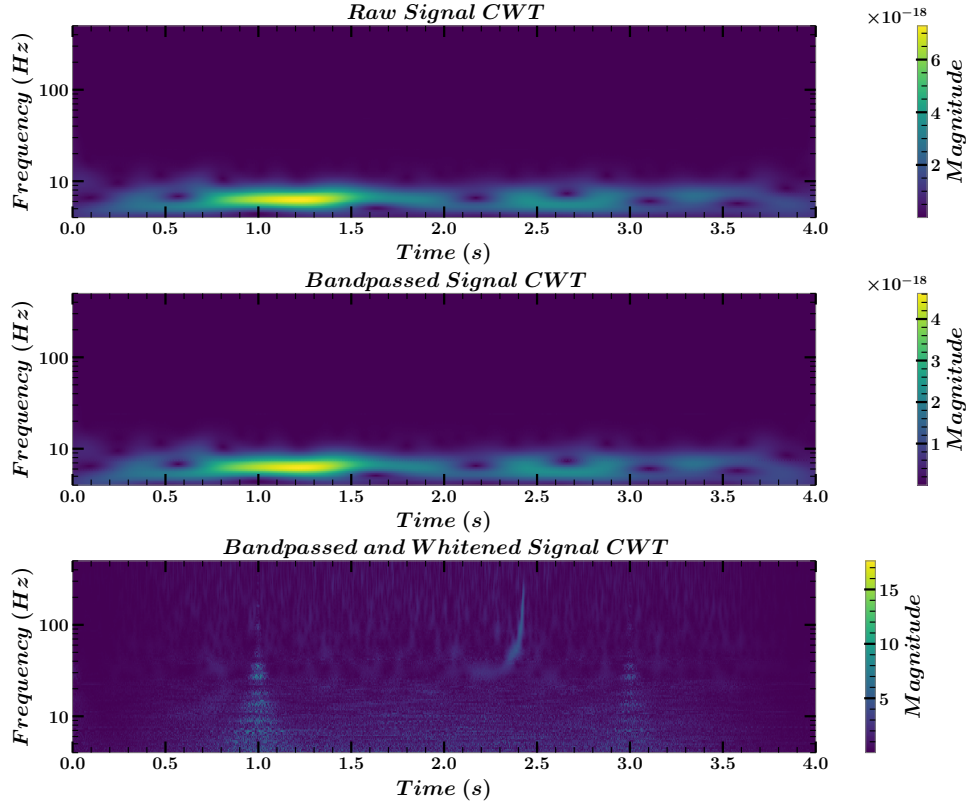


Fig. 2.25: CWT plots or time frequency analysis of GW signal for raw, bandpassed and whitened GW signal. The last plot shows the time-frequency evolution as observed in NR signals denoting the detection of GW.

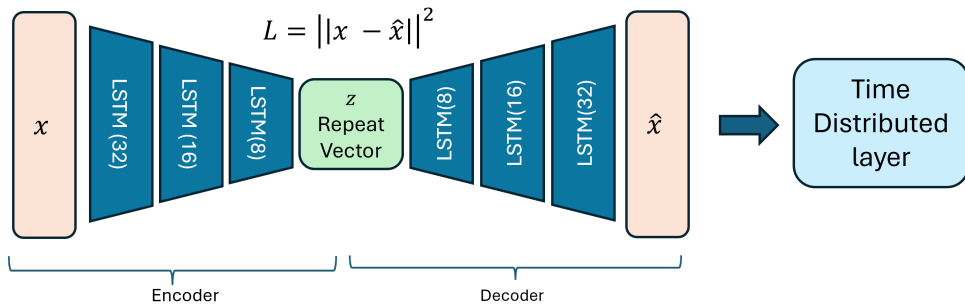


Fig. 2.26: LSTM autoencoder model for detecting GW signal and predicting waveforms.

3. OUR QUANTUM MACHINE LEARNING MODEL

3.1 Outline

Quantum Neural Network (QNN) aims to utilize the quantum advantage of superposition and entanglement in ML models and techniques. Here, we illustrate our QML model to show the use case of quantum computing in GW astronomy. We propose a QNN model that works on the neural network table generated in the Chapter 2 and predicts the physical parameters of the system. It is the quantum analogue of the NN model.

3.2 Model 3: Quantum Neural Network Model

As discussed in Chapter 1, working of QNN requires defining the feature map to transform our classical data into quantum states, and then initializing the ansatz which behaves as the hidden layers in the QNN. These ansatz contain rotation gates where rotation angles are treated as the weight parameters. Here, we propose two models for spinless BBH systems. The first model is the demo model, which establishes the importance of QNN and its ability to detect patterns in the dataset. The second model performs on complete data of the spinless BBH system and results in the physical parameters of the GW signals.

3.2.1 Restricted Spinless BBH Systems Model

We call this model "restricted" as we restrict the number of input and output parameters while working on the dataset of spinless BBH systems. We establish a quantum map using a QNN and predict the parameters of the GW signal for those particular features. This simplistic model takes two input features and outputs a single feature. The input features of the model were m_1 and m_2 , and the model was trained to output f_0 . We chose this particular combination of features because we observed a clear pattern in the variation of f_0 with respect to m_1 and m_2 . Hence, we wished to test whether the model is able to capture the patterns visible to us or not.

The dataset was normalized between $(0, \pi/4)$ to utilize the one-to-one nature of encoding in the feature maps using the angle encoding approach. The ZZFeatureMap was used as the feature map with 3 repetitions to encode the dataset into the quantum state. RealAmplitudes was used as the ansatz with 8 repetitions, and the output was obtained using the EstimatorQNN, which evaluated the expectation value of the Z operator on the first qubit. The quantum circuit is shown in Fig. 3.1, where a barrier is used to separate the feature map and the ansatz defined in the quantum circuit. The Neural Network Regressor was used with the COBYLA optimizer, with a maximum of 60 iterations. The loss function was the squared error, which computes the mean squared error between the true and predicted values. The codes were run on the quantum simulator only.

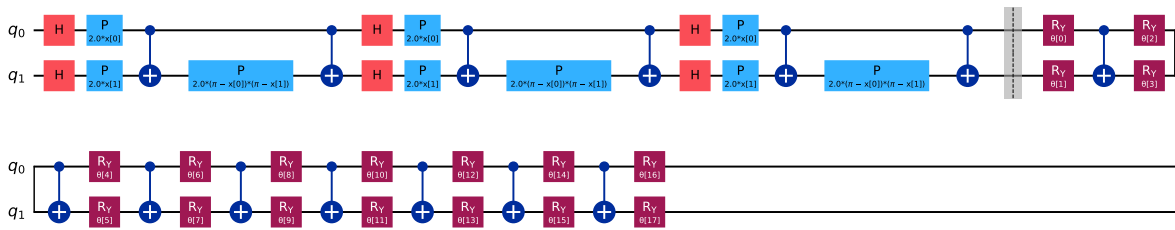


Fig. 3.1: Quantum circuit showing the feature Map of the quantum circuit (ZZFeatureMap with 3 repetitions to encode the dataset into quantum states) and ansatz (RealAmplitudes with 8 repetitions) separated by barrier.

3.2.2 Complete Spinless BBH Systems Model

We work on the complete features of spinless BBH systems, which include 7 input features and 2 output features. For this, we define the Multi Output Regressor function using the Neural Network Regressor class from the Qiskit Machine Learning library with the help of scikit-learn. We then train our model for spinless BBH systems with 7 input features and 2 output features. The architecture of the model includes 7 qubits, where the quantum circuit comprises a ZZFeatureMap with 2 repetitions as the feature map and RealAmplitudes with 8 repetitions as the ansatz of the quantum circuit. The quantum circuit is shown in Fig. 3.2. We evaluate the output using EstimatorQNN with the COBYLA optimizer and a maximum of 30 iterations.

We also ran the complete spinless BBH system model with the integration of TorchConnector, where we employed the same quantum circuit as shown in Fig. 3.2, and evaluated the output in a similar fashion as defined above (using the EstimatorQNN) for the complete spinless BBH system model. TorchConnector integrates Qiskit's QNN models with PyTorch's deep learning framework to enable treating quantum circuits as layers in a classical neural network. This allows backpropagation through the quantum layer using PyTorch's autograd mechanism. The EstimatorQNN was used in the model to estimate the expected value of the observables. We defined the observables as $Z \otimes I_{26}$

and $I \otimes Z \otimes I_{2^5}$. Adam optimizers were used for optimizing the MSE loss function with a learning rate of 0.01, and the model was trained for 14 epochs with a batch size of 32.

Similarly, we can define the QNN models for spin BBH systems and create a quantum map which can effectively transform fitting parameters of GW signal into the binary parameters of GW signal or vice versa.

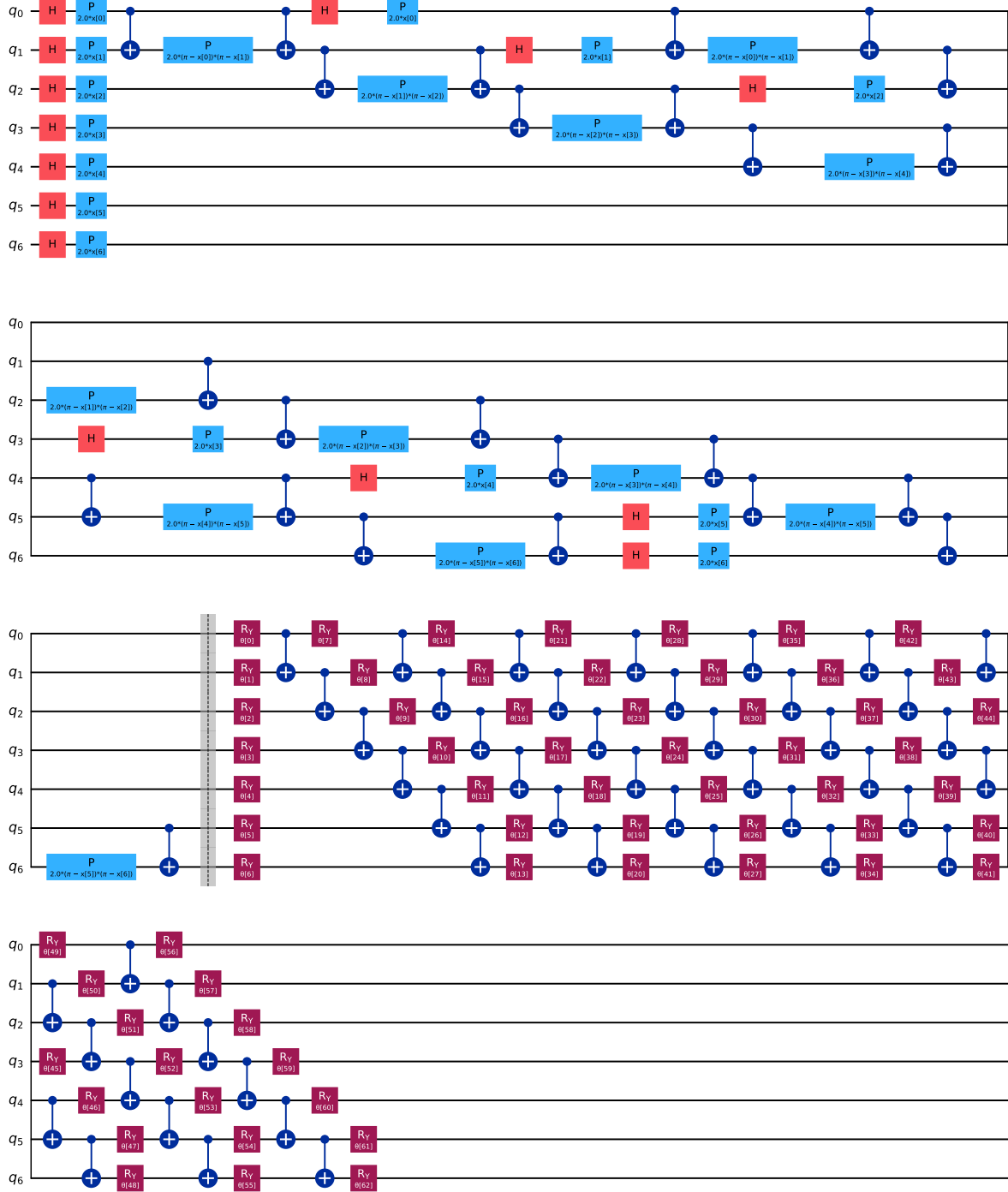


Fig. 3.2: Quantum circuit for 7 qubits denoting the feature Map of the quantum circuit (ZZFeatureMap with 2 repetitions to encode the dataset into quantum states) and ansatz (RealAmplitudes with 8 repetitions) separated by barrier.

4. RESULTS

4.1 Model 1: Neural Networks Model

As defined in Chapter 2, we proposed two different kinds of models for two different datasets, spinless BBH systems and spin BBH systems. Here, we show the results of both the models and highlight the key findings from our model. We evaluate the NN models based on two metric to validate the accuracy of the model. First, Mean Square Error (MSE) of the models which determines the mean of the squared difference of the predicted and actual values of features. Second, we also evaluate the model based on R^2 score of the model which indicate the measure of the percentage of variances existing in the dataset have been accurately covered by the model.

4.1.1 For Spinless BBH Systems

After preprocessing of the dataset of fitted waveforms, the training dataset contained 5819 waveforms while the test dataset contained 1518 waveforms.

Model 1: This model was trained using the fitting parameters of GW signal and outputs m_1 and m_2 as the output parameters. The prediction of the model gave the MSE = 0.02 for the training dataset and 0.019 for the test dataset with a R^2 score of 0.54 and 0.57 on the training and test dataset, respectively. The loss function of the training and validation set has been shown in Fig. 4.1, which shows the monotonic convergence of the loss function. This indicates that the model has the capability to generalize the predictions for unknown dataset because validation loss of the model is similar to training loss, indicating the model did not overfit and underfit at all. However, the low R^2 score may lead to some discrepancies about the model's performance.

Model 2: However, when we trained the model to predict M and q instead of m_1 and m_2 , the performance improved noticeably. The model gave a lower MSE of 0.0083 on the training set and 0.0069 on the test set. Along with that, the R^2 score also improved. it was 0.65 for the training set and 0.68 for the test set. The training and validation loss curves are shown in Fig. 4.2. Table 4.1 compares predicted and actual values of M and q with the combined loss of the particular entry.

Actual Parameters		Predicted Parameters		Total Error
M	q	M_{pred}	q_{pred}	
0.691489	0.075014	0.691167	0.077671	7.10×10^{-6}
0.457447	0.375068	0.454468	0.372280	1.66×10^{-5}
0.500000	0.050009	0.500545	0.054272	1.85×10^{-5}
0.893617	0.175023	0.894329	0.170680	1.94×10^{-5}
0.787234	0.100017	0.790779	0.095869	2.98×10^{-5}

Tab. 4.1: Comparison of normalized actual and predicted component masses (M, q) for spinless BBH systems for Model 2.

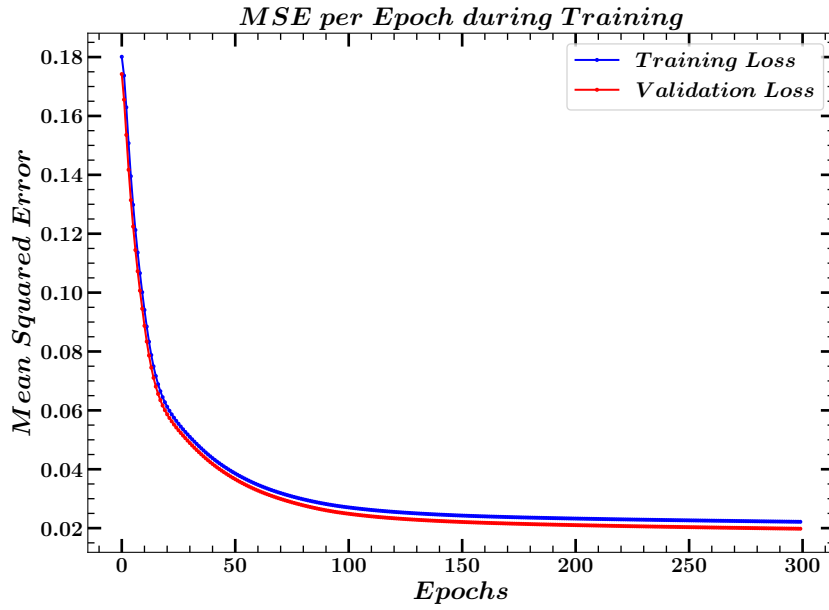


Fig. 4.1: Model 1 for spinless BBH system: Trained to output m_1 and m_2 of the binary black hole system.

4.1.2 For Spin BBH Systems

The model was trained on a dataset containing 70,662 waveforms for the training dataset and kept 17,666 waveforms as the test set. The model achieved an accuracy of $\text{MSE} = 0.00392$ on the training set and an MSE of 0.0041 on test set. The R^2 score of the model on the training set was 0.82, and on the test set was 0.80. Therefore, my model could trace the 80% of the variances in the dataset. The training and validation loss of the model is shown in Fig. 4.3, where both the training and validation loss decreased monotonically. Both the functions were close enough to each other at the time when the loss function began to stabilize signifying that model performed pretty well and did neither overfit nor underfit. The comparison of the predicted and actual values of the output parameters are shown in Table 4.2, where it shows the best 5 entries of the predictions with the minimum error.

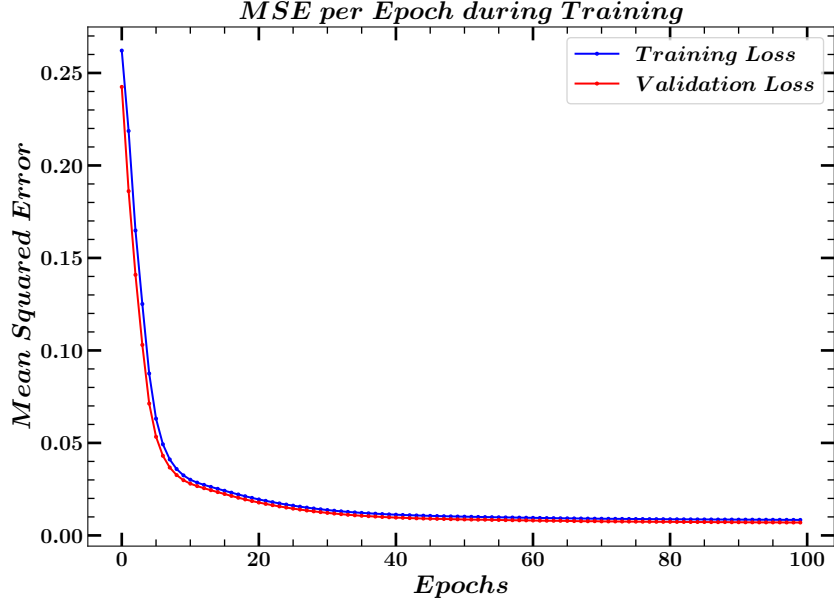


Fig. 4.2: Model 2 for spinless BBH system: Trained to output the total mass M and mass ratio q of the binary black hole system.

Actual Parameters							Predicted Parameters							Total Error
α	β	δ	A_0	f_0	f_1	f_2	α_{pred}	β_{pred}	δ_{pred}	$A_{0,\text{pred}}$	$f_{0,\text{pred}}$	$f_{1,\text{pred}}$	$f_{2,\text{pred}}$	MAE
0.598	0.464	0.763	0.329	0.796	0.656	0.810	0.616	0.467	0.761	0.330	0.796	0.650	0.819	0.039
0.406	0.443	0.369	0.549	0.478	0.402	0.496	0.399	0.445	0.360	0.552	0.479	0.422	0.495	0.041
0.534	0.440	0.567	0.455	0.652	0.571	0.756	0.538	0.442	0.575	0.445	0.646	0.572	0.766	0.041
0.576	0.447	0.476	0.466	0.621	0.496	0.657	0.563	0.450	0.477	0.460	0.620	0.512	0.653	0.044
0.404	0.438	0.387	0.599	0.488	0.431	0.532	0.417	0.440	0.397	0.600	0.481	0.425	0.536	0.044

Tab. 4.2: Comparison of Normalized Actual and Predicted Parameters with Total mean absolute error.

4.2 Model 2: LSTM Autoencoders

Here, we also evaluate the model's accuracy based on two metrics. First, MSE which calculates the mean square error of the dataset with respect to the predicted values of the model's output. Second, we evaluate the model using the normalized cross correlation function which evaluates the measure of the overlap of the predicted waveform with the actual waveforms.

The model's training took around 900 core minutes in the high-performance cluster. The training loss and validation loss for the model is shown in Fig. 4.4, which depicts the model's training accuracy and the capability to generalize the prediction for the unknown dataset. The loss function decreases monotonically for both the training and test datasets. The model was able to predict the presence and trace out the waveforms with an MSE of 0.0019 for the training dataset and an MSE of 0.002 for test dataset of simulated gaussian noise. This depicts that the model neither overfits nor underfits the dataset and can generalize the prediction pretty well for the unknown dataset. The

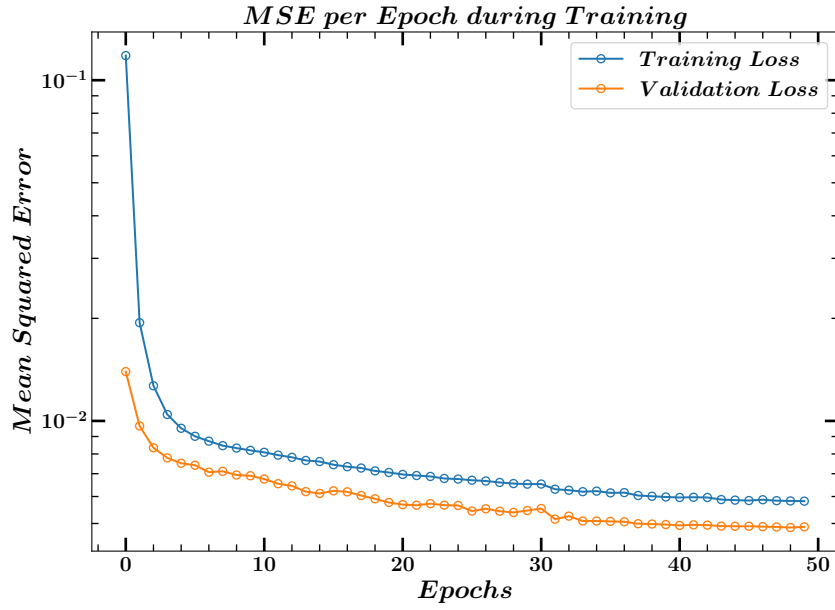


Fig. 4.3: Training and Validation loss for spin BBH System

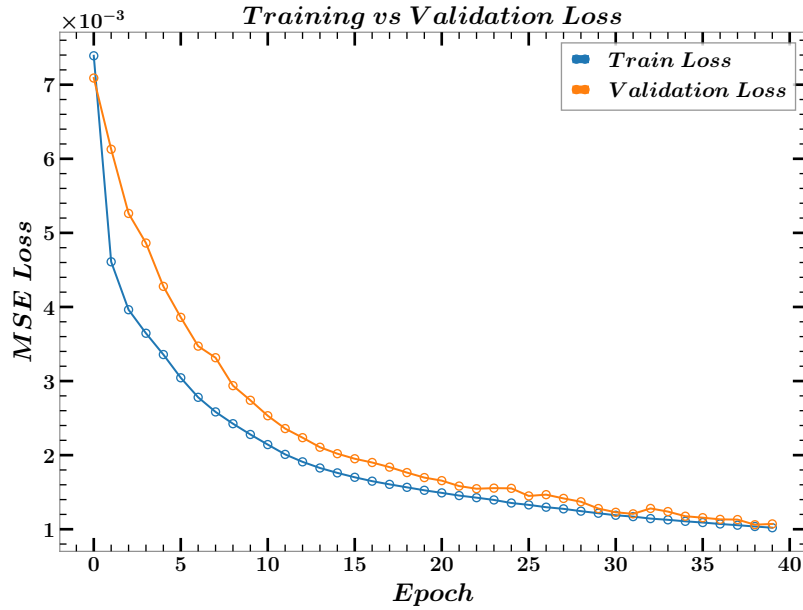


Fig. 4.4: Monotonically decreasing training and validation loss for the LSTM autoencoders.

training predictions are shown in Fig. 4.5, where the model seems to capture the inspiral region of the waveform but fails to capture the merger phase accurately because of the sudden change in the amplitude from the inspiral region. Similar to the training dataset, the model was able to predict the test set up to the inspiral region and was unable to predict the merger phase accurately, as shown in Fig. 4.6. However, our model was able

to deal with certain complications in the dataset.

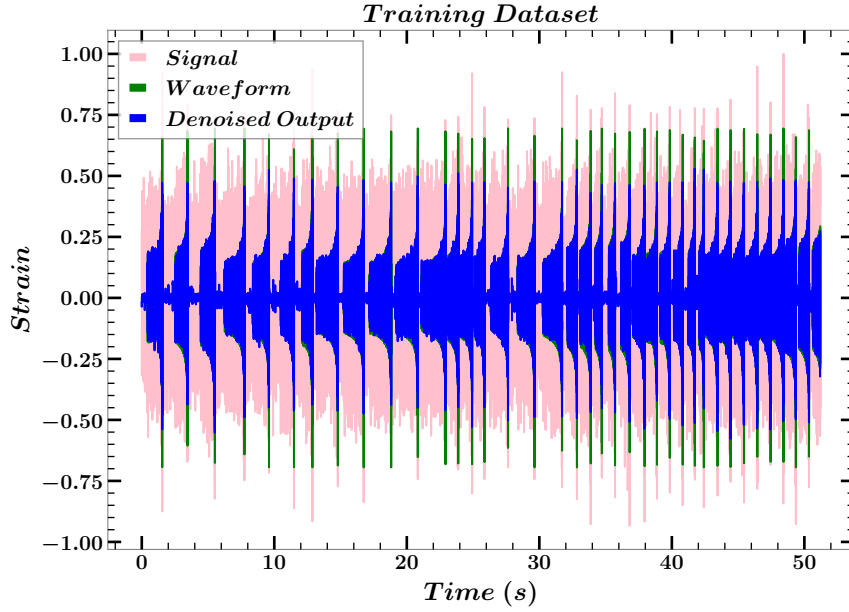


Fig. 4.5: Prediction of the model on the training dataset with an MSE: 0.0019.

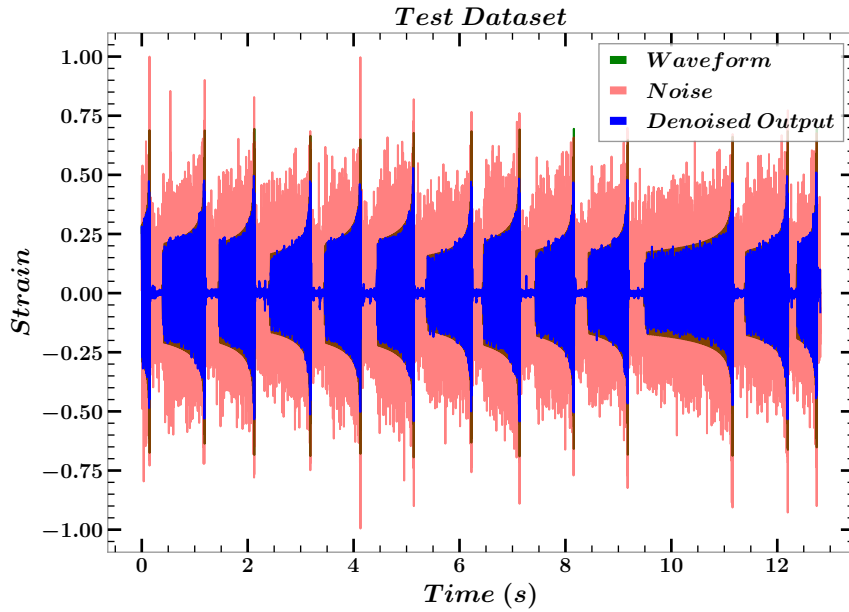


Fig. 4.6: Prediction of the model on the test dataset with an MSE: 0.002.

Fig. 4.7 shows the zoomed-in image of the test dataset prediction for the simulated dataset. The normalized cross-correlation of the waveforms was equal to 95.5% (meaning the model was able to trace out the actual waveform from the noisy simulated dataset with an accuracy of 95.5%). This was because the model was trained only on the NR

signals having the amplitude as half of the amplitude of simulated noise. We propose that similar approach could be used for training the dataset on the LIGO noise rather. Now, we take the LIGO dataset from the Hanford observatory, in which the GW event

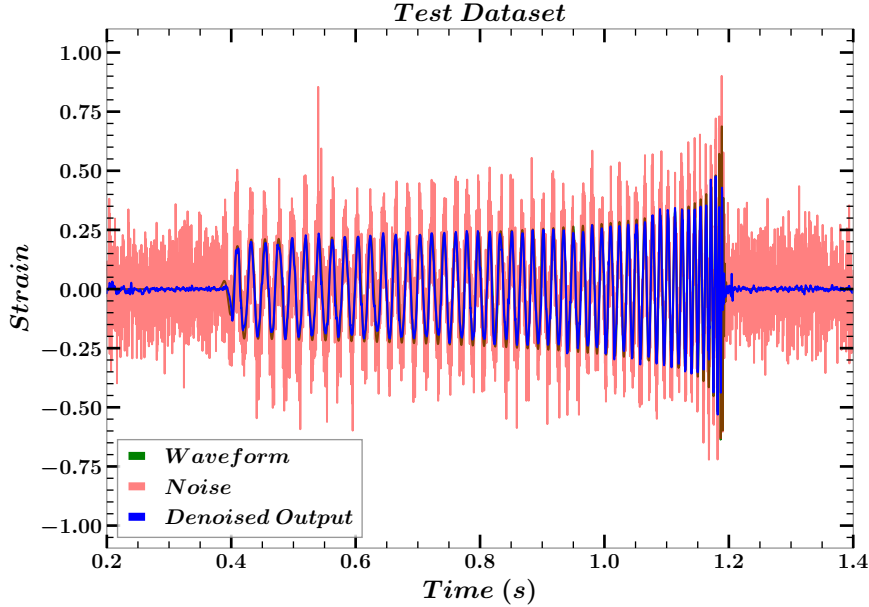


Fig. 4.7: Test dataset for prediction on a particular waveform.

was previously recorded and give it as input to our model. Our model was not only able to predict the presence of the GW signal inside the detector data but was also able to capture the waveform present inside it. The results for the event GW150914 are shown in Fig. 4.8, where the GW event is recorded at 2.4 seconds (actual time at GPS = 1126259462.4). The first picture denotes the prediction of the model in the provided time frame, while the second picture is cropped to show the time frame where only the denoised GW signal was present. The original preprocessed signal was shown in Fig. 2.22. The denoise GW signal, along with the noise floor of the LIGO detector data, is shown in Fig. 4.9. Similarly, the model was also able to predict the event GW170729 accurately at the time of the merger and could trace out the waveform embedded inside the LIGO noise.

4.3 Model 3: QNN Model

4.3.1 Restricted Spinless BBH Systems

The prototype model of QNN for the spinless BBH systems where 2 input features were mapped to a single output feature performed well on the quantum simulator. The objective function value was plotted against each iteration for 60 iterations as shown in

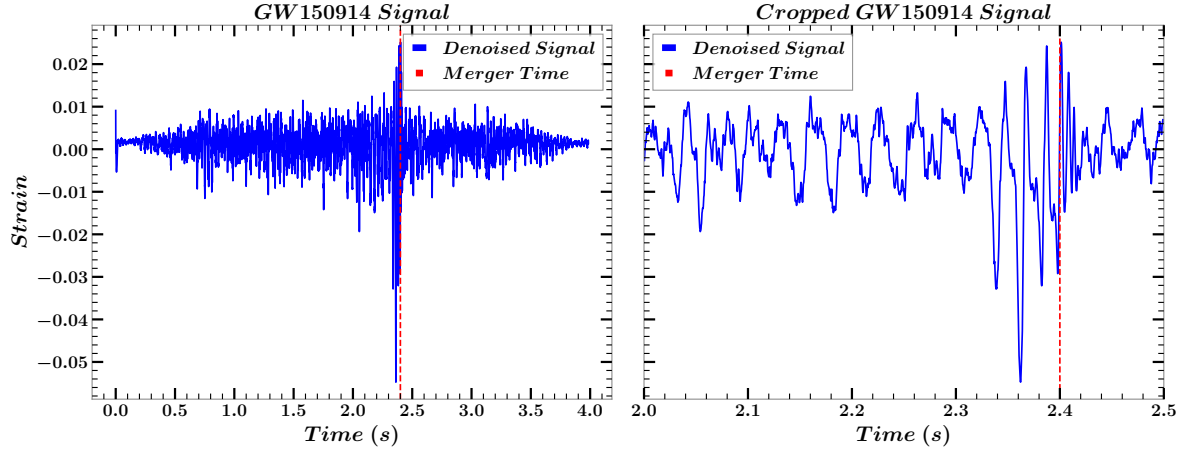


Fig. 4.8: GW150914 signal detected, which matches accurately with the time of merger given in the GWOSC database.

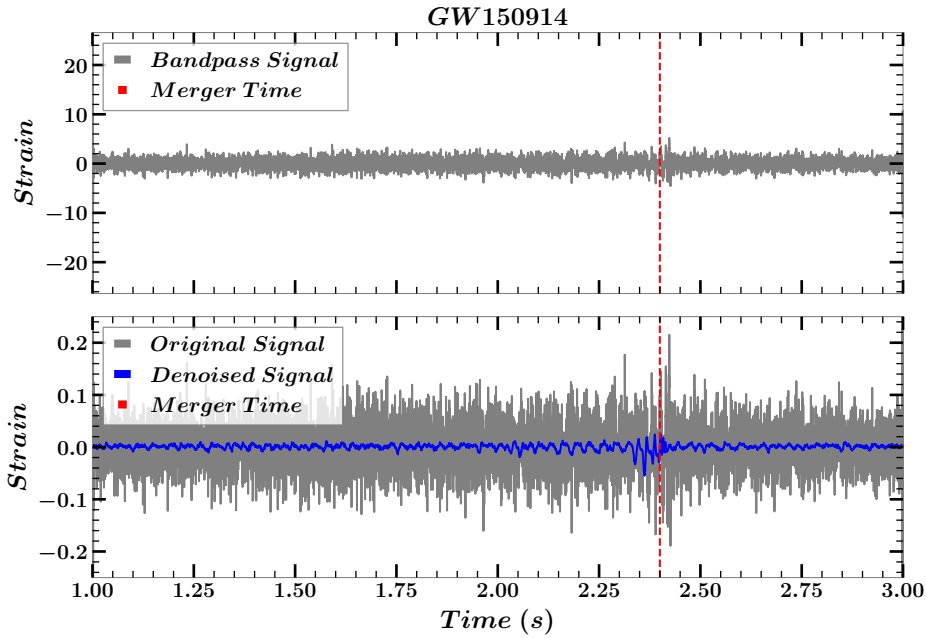


Fig. 4.9: GW150914 signal represented along with the noisy LIGO data showing the comparison of the model.

Fig. 4.10. The oscillating nature of the objective function is due to the working of optimizers in QML. We also capture the evolution of the weights with each iteration and observe that all the weights converge to a stable value as the model approaches the maximum number of iterations. The plot of the evolution of weights with iterations is shown in Fig. 4.11. Note that the objective function value does not always imply the value of the loss function defined in the problem. Rather, it may include some regularization terms which the model sets automatically according to the problem. We achieved an MSE of 0.03167 on the training dataset and an MSE of 0.03483 on the test dataset.

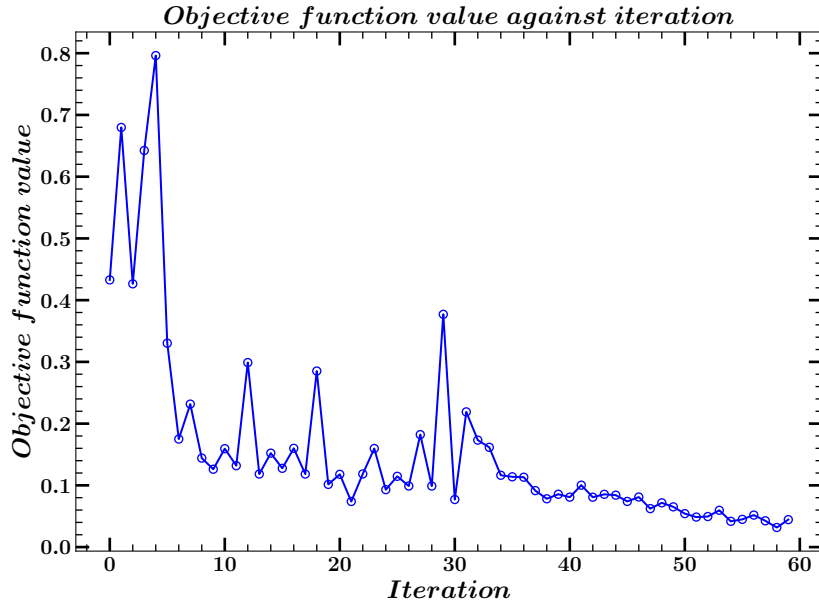


Fig. 4.10: Objective function value varying with each iteration for 60 iterations for spinless BBH systems.

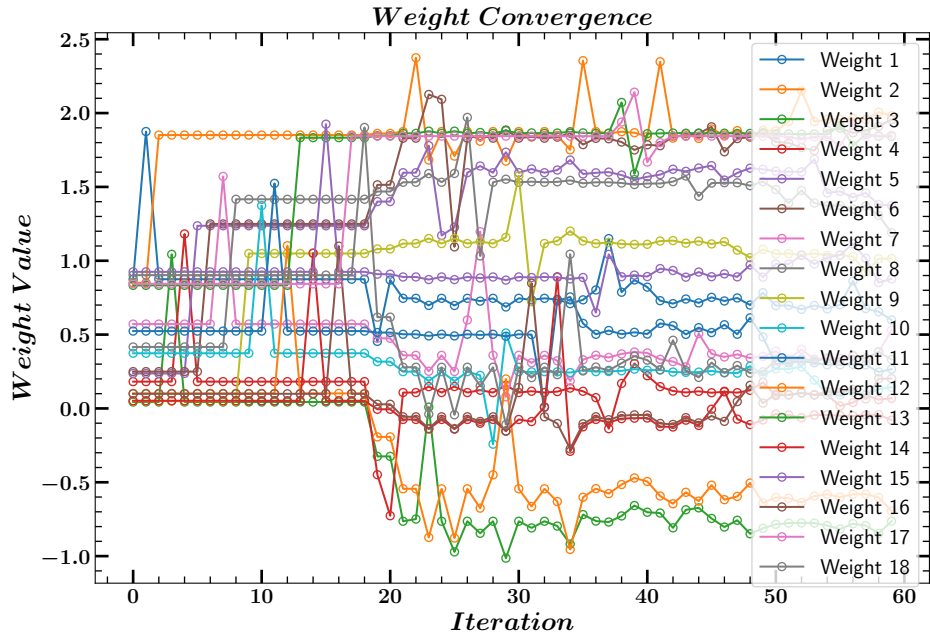


Fig. 4.11: Convergence of the weight factors in the ansatz with respect to each iteration.

4.3.2 Complete Spinless BBH Systems

We performed the Multi-Output Regressor of the Neural Network Regressor class as the QNN. This outputs two features, m_1, m_2 of the complete spinless BBH systems model. This model resulted in an accuracy of 0.083 on the training dataset and 0.080 on the test dataset. The objective function value of the above model for each iteration, up to 60

iterations, was evaluated and is shown in Fig. 4.12. The accuracy of these models can be improved by using different optimization techniques and defining different observables for the system. The loss function for the torchconnector QNN model is shown in Fig. 4.13.

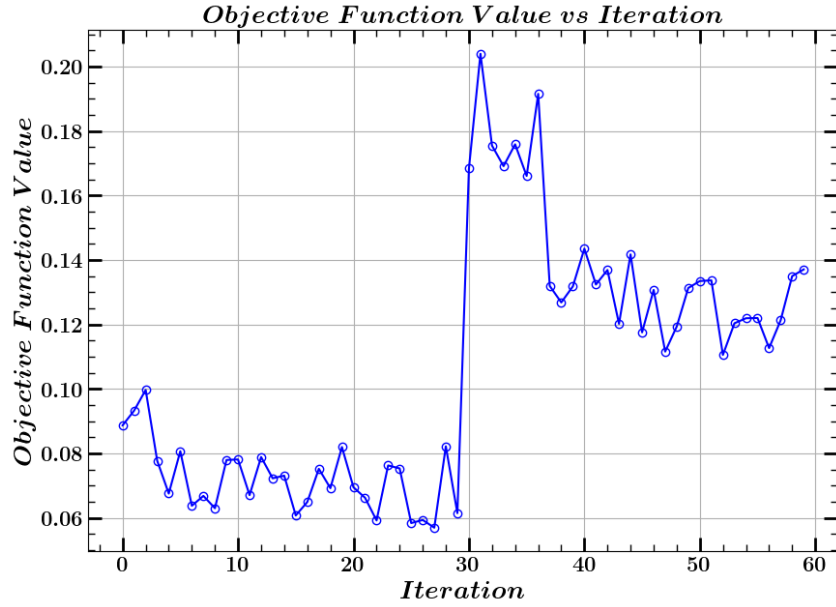


Fig. 4.12: Objective value function evaluated at each iterations up to 60 iterations.

The loss function decreases smoothly and monotonically and converges within 14 epochs giving an MSE of 0.03 and 0.04 on the train and test dataset, respectively. The accuracy of this model can be improved by basic techniques like increasing the depth of the quantum circuit but also by trying different optimization techniques.

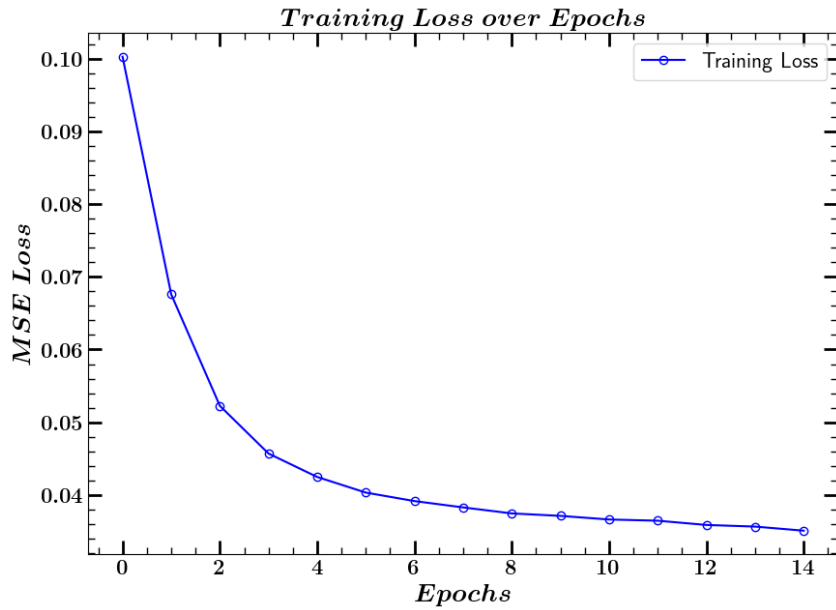


Fig. 4.13: Loss function of the torch connector model for complete spinless BBH Systems.

5. SUMMARY AND FUTURE WORKS

This thesis is focused on exploring ML techniques for enhancing GW astronomy. We defined several models aimed at tasks specific to the detection and characterization of GW signals. A comprehensive analysis of waveforms obtained through NR simulations was performed using different signal processing techniques, including time-domain analysis, frequency-domain analysis, and time-frequency analysis. These explorations revealed important patterns in the GW signal dataset, which motivated us to fit the amplitude of the FFT of NR signals using a physics-inspired fitting approach. The obtained fitting parameters were mapped to the binary parameters of the GW signal using higher-dimensional mappings defined through NN, or vice versa. We divided our dataset into two disjoint sets, defined as spinless BBH systems and spin BBH systems. Separate models were defined for each dataset.

For the spinless BBH system, we set the input features as the fitting parameters of the GW signal and the output features as binary parameters. The model was mapped in the fashion $7 \rightarrow 2$, where number of input features were 7 and number of output features were 2.

For spin BBH systems, the reverse approach was employed. NN were defined such that the binary parameters of the GW signal were treated as input features and mapped to the fitting parameters of the GW signal. The model defined for this case was complex due to the high number of parameters and the intricate relationships between them, whose inferences could not be deduced through visualization.

During data preprocessing and model refinement, certain spectral features were excluded from the mapping task based on their limited observed variance with respect to the binary parameters. However, it is acknowledged that some of these omitted features may possess nontrivial correlations or contribute to latent clustering structures that were not evident through our initial analysis. Inclusion of these parameters in newly defined models could lead to improved accuracy in the regression task.

A significant limitation of the above model is the exclusion of phase information from the FFT signal. Phase information is an important criterion which, combined with amplitude, encodes the complete information about the GW signal. Therefore, its inclusion could result in better predictions by the model. However, it may also introduce challenges, particularly in the complex modeling of the phase at high frequencies due to the merger and ringdown phases of the GW signal, which involve drastic changes in phase

characteristics.

We also explored techniques to extract GW waveforms from noisy datasets obtained from LIGO detector data using LSTM autoencoders trained on simulated Gaussian noise embedded with NR signals for different mass ratios. The model demonstrated the capability to extract GW signals from LIGO noise for high SNR cases. However, it exhibited limitations in detecting GW signals with low SNR from LIGO detector data. This can be attributed to the fact that the model was trained on embedded signals whose maximum amplitude was equal to that of the simulated noise. Nevertheless, we believe this limitation can be addressed by training the model to include NR signals with lower amplitudes.

Another drawback of our model was that it was trained solely on simulated Gaussian noise. Hence, it was valid only for those segments of LIGO detector data that can be approximated as stationary. Since LIGO data can be considered locally stationary over short time intervals, we used only 4-second segments of the dataset for predictions. This model could be improved by training it on LIGO detector data embedded with NR signals or approximate waveform templates, and then applying the LSTM autoencoder to predict the waveforms. We believe these corrections would significantly enhance the model's predictive performance.

We also proposed several QNN models that implement the quantum analogues of NN models. These models achieved considerably low MSE and may serve as a showcase for the application of QNNs in the field of GW astronomy. As QML is a nascent research area, there exist various methods to improve model accuracy. One straightforward approach is to increase the number of parameters in the ansatz, for instance, by increasing the number of repetitions in the ansatz to better parameterize the system. Another method is to modify the encoded quantum state using multiple repetitions of the feature map. Furthermore, custom feature maps and ansatz can be defined based on specific output requirements. These are some of the strategies that can be employed to enhance model accuracy.

Furthermore, QLSTMs can be utilized for learning time series sequences using a comparatively smaller number of parameters than classical LSTMs. They offer the potential to learn and model noise patterns present in LIGO interferometric data, thereby enabling the detection of gravitational wave signals embedded within noisy datasets. Implementing QLSTM-based autoencoders presents a novel approach for denoising gravitational wave data from LIGO. A quantum counterpart of the classical LSTM autoencoder can be developed to enhance accuracy and efficiency in signal modeling with reduced parameter complexity.

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